SURVIVING THE FUTURE

Crisis, Climate, and Human Resilience

A Data-Driven, Academically Grounded Guide to Living Through the Crises Caused by Global Warming

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PREFACE: WHY FUTURE GLOBAL WARMING IS CERTAIN

Global warming is not a hypothesis; it is an observed, measured, and physically explained reality. The certainty of future warming rests on three pillars: (1) the physics of the greenhouse effect, (2) the observed accumulation of greenhouse gases in the atmosphere, and (3) the inertia of the climate system. Even if all emissions stopped today, committed warming from past emissions would continue for decades. This book is written to be debated in academic spaces: every major claim is tied to peer-reviewed literature, data, and expert statements.

The Three Pillars of Certainty

- **Physics:** Greenhouse gases absorb outgoing infrared radiation. This is established physics, tested for over a century (Arrhenius, 1896; modern radiative transfer models).
- **Observation:** Atmospheric CO2 has risen from ~280 ppm (pre-industrial) to ~420 ppm (2024). Global mean temperature has risen ~1.1-1.2 C. These are direct measurements.
- **Inertia:** The ocean stores heat; ice sheets respond slowly. Even with zero emissions, warming continues for decades (committed warming).

Data underpinning the discussion is summarized as follows:

Table P.1 Key global climate indicators (2024)

Indicator	Value	Source
Global mean temperature rise since 1850-1900	~1.1-1.2 C	IPCC (2023)
CO2 concentration (2024)	~420 ppm	NOAA (2024)
Sea level rise since 1900	~20-25 cm	IPCC (2023)
Arctic sea ice decline	~13% per decade	NASA
People affected by climate disasters (annual)	~100 million+	IFRC
Economic damage from climate disasters (annual)	~$200-300 billion	World Bank

The expert consensus underpinning this book is summarized as follows.

Dr. James Hansen, former NASA GISS Director (2013 op-ed/2016 paper): 'Sea level rise of several meters could be expected with 2°C warming — 2°C would have consequences that can be described as disastrous.'

Prof. Katharine Hayhoe, Texas Tech Chief Scientist (2014, post-IPCC AR5): 'Climate change is here and now, and not in some distant time or place — the choices we make today will have a significant impact on our future.'

Dr. Michael Mann, Penn State climatologist (2004/2021): 'There is about as much scientific consensus about human-caused climate change as there is about gravity — multiple independent lines of evidence converge.'

Dr. Friederike Otto, Imperial College / World Weather Attribution co-founder (2021): 'The Pacific Northwest 2021 heatwave would have been virtually impossible without human-caused climate change.'

Prof. Johan Rockström, Potsdam Institute & Stockholm Resilience Centre (2009 Nature; 2023 update): 'We define planetary boundaries within which humanity can operate safely — six of nine boundaries are now transgressed, well outside the safe operating space.'

Preface (Extended)

This extended preface provides additional context for the reader. The book is organized to move from foundational science to practical action, ensuring that every reader can find relevant material. The structure reflects the complexity of the climate crisis and the need for multi-level responses.

Additional context: The climate crisis demands sustained attention and coordinated action across all sectors of society. This book aims to equip readers with the knowledge and tools needed to contribute meaningfully to that effort.

CHAPTER 1

INTRODUCTION: THE WARMING WORLD

Theme: the global crisis of warming and why survival begins with understanding it

In the summer of 2022, a monsoon system carried moisture from the Arabian Sea across Pakistan for eight consecutive weeks. By the end of August, one-third of the country was under water, thirty-three million people had been affected, and the national highway network had been cut in more than a hundred places. The flood was not a single event but a sequence: heat that dried the soil, then rain that could not infiltrate it, then reservoir releases that arrived after the rivers had already filled. Six months later, the same country was negotiating with the International Monetary Fund while its farmers reported that two million hectares of cropland would not be planted that season.

A year later, in 2023, Canada burned. More than eighteen million hectares of forest burned, an area larger than England, and the smoke crossed the Atlantic and turned the skies of New York orange. In September of the same year, two dams above the Libyan city of Derna failed after a storm that had been forecast days in advance; the water killed more than eleven thousand people in a city that had been warned but could not act. In October 2024, a year's worth of rain fell on the Spanish province of Valencia in a few hours and killed more than two hundred people in towns that had never been mapped as floodplains.

These events are usually described as natural disasters. They are not. Each of them required a physical trigger — rain, wind, heat — but the scale of the loss was determined by conditions that people created: where houses were built, how rivers were managed, whether warnings reached the people who needed them, whether the soil had been left compacted and bare, whether the population was old, poor, or living in informal settlements. The trigger is weather. The disaster is a failure of preparation, and preparation is something human beings can change.

This book is an attempt to describe that failure and the ways out of it, in both directions: backward, into the record of what has already happened, and forward, into the range of futures that the physical system has made possible. It is not a book of prophecy. It is a book about the conditions under which communities continue to function, and the conditions under which they do not.

1.1 The Decade That Changed the Baseline

The 2020s began with a global pandemic and ended their first half with the four hottest years ever measured. The World Meteorological Organization confirmed that 2024 was the first calendar year to average more than 1.5°C above the 1850–1900 baseline, at about 1.55°C, with 2023 close behind at 1.45°C. The two years were lifted by a strong El Niño, but the underlying trend was not: the previous ten years were the ten warmest in the instrumental record, and every one of them exceeded the average of the decade before it.

The warming has not been evenly distributed in time or space. The Arctic has warmed three to four times faster than the global mean; the Mediterranean and the Middle East have warmed roughly twice as fast; the North Atlantic has cooled in some patches even as the ocean as a whole has gained heat, because melting ice and shifting winds have altered circulation. A global average of 1.5°C therefore describes a world in which some regions have already experienced 2.5°C or 3°C of local warming, and in which the statistics of extreme heat have shifted far more than the mean.

The decade also changed the character of the evidence. Attribution studies that once took a year now take a month. Satellite records that began in the 1970s now span half a century. Insurance losses, crop models, hospital admissions and migration data have accumulated to the point where the effect of climate on society can be measured rather than inferred. The result is a body of evidence that is no longer primarily about the future: it is about what has already happened, to identifiable people, in identifiable places, at a cost that can be counted.

1.2 The Anatomy of a Global Crisis

A global crisis is not simply a large disaster. It is a set of stresses that arrive together, reinforce one another, and exceed the capacity of institutions to respond. Climate change qualifies because it acts on every sector at once: it raises food prices by reducing yields, raises insurance costs by increasing losses, raises migration by making some places uninhabitable, and raises the cost of borrowing by increasing uncertainty. Each effect is modest in isolation. Together they consume the fiscal and administrative space that governments need to cope with the next shock.

The reinforcement is the part that is usually missing from public discussion. A drought reduces the harvest; the reduced harvest raises food prices; higher food prices increase poverty; poverty increases vulnerability to the next flood; the flood destroys the assets that the household would have used to recover. The sequence can turn a single bad season into a decade of lost development, and it explains why the same hazard produces a few dozen deaths in one country and thousands in another.

This is also why the language of 'natural disaster' is misleading in a second sense. It suggests a boundary between the hazard and the society exposed to it, when in reality the boundary is the thing that policy controls. A levee, a building code, a crop insurance scheme, a heat action plan and a functioning weather service are all ways of moving that boundary. The chapters that follow examine each of them, but the logic is the same throughout: the physical hazard sets the pressure, and the social system determines whether the pressure becomes a catastrophe.

1.3 From Local Hazard to Systemic Failure

Most climate impacts begin as local events, but their consequences travel through markets, supply chains and migration routes. The 2011 floods in Thailand are the standard example: a sequence of monsoon rains inundated industrial estates north of Bangkok, global hard-drive production fell by about a quarter, and computer prices rose around the world. The flood was national; the shortage was global. The same mechanism now operates through semiconductor production in Taiwan, wheat exports

from the Black Sea, and container shipping through the Red Sea.

Systems fail when a shock exceeds redundancy. A power grid with spare capacity survives a heatwave; a grid without it cascades. A port with alternative routes survives a blockage; a port that is the only route does not. The 2021 Texas winter storm, the 2022 European gas crisis, and the 2023 Panama Canal drought all demonstrated the same principle in different sectors: the failure was not caused by the weather alone but by the absence of slack that would have absorbed it. Climate change is, among other things, a systematic reduction in slack.

For households, the translation is direct. A family that depends on one income, one water source, one road and one food market is exposed to every shock that touches any of them. Diversification — of income, of water storage, of food sources, of social contacts — is the household equivalent of redundancy in a grid. It is also the reason preparedness is not a single purchase but a set of overlapping capabilities, each of which buys time when another fails.

1.4 The Human Cost: What the Record Shows

The human cost of climate change is usually reported through disasters, and the disaster statistics are stark enough. The Centre for Research on the Epidemiology of Disasters recorded more than 7,000 major disasters between 2000 and 2019, affecting 4.2 billion people and killing 1.2 million. Weather and climate events accounted for the large majority of them. In the 2020s the pattern has continued: the 2022 Pakistan floods affected thirty-three million people; the 2023 Turkey–Syria earthquakes — geological rather than climatic, but a reminder of compound vulnerability — killed more than fifty thousand; the 2024 Valencia floods killed more than two hundred in a high-income country with a modern warning system.

But disasters capture only the visible part of the cost. The World Health Organization attributes about 150,000 deaths a year to climate change through heat, disease, malnutrition and displacement, a figure that most epidemiologists consider conservative because it excludes the long tail of chronic illness and lost livelihood. Heat alone is estimated to cause more than 500,000 deaths a year globally, with the European summer of 2022 accounting for about 61,700 of them. Air pollution, much of it from the same fuels that drive warming, kills about seven million people a year.

The economic record is equally uneven. The World Bank estimates that climate-related losses in developing countries already exceed US$200 billion a year, and that the figure understates the true cost because it counts damaged assets rather than lost development. A child who is malnourished during a drought does not recover the cognitive development that was lost; a farmer who sells livestock to survive a season does not rebuild the herd at the same rate. These effects are rarely counted in disaster statistics, but they are the mechanism by which a temporary shock becomes a permanent reduction in welfare.

1.5 Why the Worst Outcomes Are Not Yet Fixed

The physical system has inertia, and that inertia is usually presented as bad news: the warming already committed will continue for decades, sea level will rise for centuries, and some changes are irreversible. All of that is true. But the same inertia means that the future is not yet determined, because the emissions that would produce the worst outcomes have not yet happened. The difference between 1.8°C and 4.4°C at the end of the century is not a difference in physics; it is a difference in cumulative carbon, and cumulative carbon is still being decided.

The remaining carbon budget makes the arithmetic explicit. To have a fifty per cent chance of holding warming to 1.5°C, the world can emit roughly 250 billion tonnes of carbon dioxide more — about six years at current rates. For 2°C the budget is roughly 1,150 billion tonnes. These numbers are uncertain at the margins, but their order of magnitude is not, and it implies that every year of delay consumes a substantial share of the remaining budget. It also implies that the choice is not between action and no action but between action now and much larger, more expensive action later.

There is a second reason for qualified optimism. The cost of the technologies that reduce emissions has fallen faster than almost anyone predicted. Solar photovoltaic costs have fallen by roughly ninety per cent since 2010, onshore wind by about seventy per cent, and lithium-ion battery packs by about eighty-five per cent. In 2023 the world added more renewable generating capacity than fossil capacity for the first time, and the International Energy Agency now expects renewables to account for the majority of new power capacity through 2030. The obstacle is no longer primarily engineering; it is the speed at which grids, permits and political coalitions can change.

1.6 What 'Surviving' Actually Means

The word survival suggests a single moment: the storm passes, the flood recedes, the family is alive. In practice, survival is a process that extends long after the water leaves. A household that survives the flood but loses its documents, its savings and its access to credit may not survive the next two years. A city that survives the heatwave but loses its power grid for a week may lose thousands of people to hospitals that cannot run. The relevant question is not whether a system survives the event but whether it returns to a level of functioning from which it can absorb the next one.

This book therefore treats survival at three levels. Personal survival is about water, food, shelter, health and the knowledge to use them. Community survival is about mutual aid, local institutions and the physical infrastructure that connects households to services. Societal survival is about the fiscal capacity, the legitimacy and the administrative competence that allow a state to keep functioning when several crises arrive at once. The levels are nested: a household cannot be secure in a failed state, and a state cannot be stable when its households are desperate.

The nesting also explains why purely individual preparation is insufficient. A family that stores water and food is better off than one that does not, but it still depends on a functioning water utility, a supply chain, a hospital and a road. Preparedness that ignores the collective is a bet that everyone else will continue to provide the services on which the prepared household depends. That bet may pay off in a short emergency; it fails in a prolonged crisis, which is precisely the scenario that climate change makes

more likely.

1.7 Knowledge, Preparedness and Adaptive Capacity

Adaptive capacity is the ability of a system to adjust to change while remaining functional. It is not the same as wealth, though wealth helps; it is the combination of information, resources, institutions and social trust that allows a community to anticipate, absorb and recover. Bangladesh has far less wealth than the United States, but its cyclone preparedness is widely regarded as superior, because it has invested in shelters, volunteers, early warnings and drills that reach the village level. Adaptive capacity is therefore a choice about where to invest, not simply a by-product of income.

Knowledge is the cheapest component and the one most often neglected. A household that knows which road floods first, which neighbour has a generator, and which clinic stays open can make better decisions in the first six hours than one that has to discover those facts in the middle of the emergency. A city that knows which neighbourhoods have the oldest residents and the least tree cover can target cooling measures before the heatwave arrives. The information is rarely secret; it is simply not assembled, not practised, and not shared.

The chapters that follow treat knowledge as infrastructure. That means writing down the plan, testing it, updating it, and making sure the people who need it can find it in a language and format they can use. It also means distinguishing between knowledge that is stable — the physics of heat, the geography of a floodplain — and knowledge that changes, such as the location of a shelter or the phone number of a clinic. A plan that does not distinguish between the two will either become obsolete or become too vague to act on.

1.8 The Argument of This Book

The argument can be stated in four propositions. First, the climate has already changed enough to produce measurable harm, and the harm will increase because of warming that is already committed. Second, the harm is not distributed evenly: it falls hardest on people with the fewest resources, in places with the weakest institutions, and on hazards — heat, drought, flood — that are least insured. Third, the capacity to reduce harm exists and is largely a matter of preparation, governance and investment rather than of inventing new technology. Fourth, the difference between a manageable future and a catastrophic one is still a political choice, made year by year, in decisions about energy, land use and finance.

The book is organised around those propositions. The first part establishes the physical and observational basis: how warming works, how it is measured, and why the record is trustworthy. The second part examines the main channels through which harm arrives — heat, water, food, ecosystems, health, economies, migration and conflict. The third part turns to response: infrastructure, education, information, household preparedness, emergency health, social capital, technology and governance. The fourth part presents case studies and closes with action plans that a household, a community or a city can use.

Each chapter follows the same discipline. It begins with field conditions — what has actually happened, where, and to whom. It then brings in the research literature to explain why those conditions produced those outcomes, and it distinguishes between what is well established, what is contested, and what is not yet known. It ends with a set of future scenarios built from historical data, present observations and explicit assumptions. The aim is not to persuade the reader of a conclusion but to give the reader the materials to reach one.

1.9 How the Chapters Fit Together

The first six chapters are diagnostic. Chapters 1 to 3 establish the physical science and the observation systems; chapters 4 to 6 examine heat, water and flooding as the three hazards that account for most direct losses. Chapters 7 to 12 follow the consequences through food, ecosystems, health, economies, migration and conflict. These chapters are ordered by the sequence in which a shock typically propagates: heat and water arrive first, food and health follow within weeks, economies and migration follow within months, and conflict, where it occurs, emerges from the interaction of all of them.

Chapters 13 to 21 are about response. They cover infrastructure, education, data and misinformation, household preparedness, emergency health, social capital, renewable energy, psychology and governance. The order is deliberate: physical infrastructure is the most visible response and the slowest to build; education and information are the cheapest and the most durable; social capital is the resource that determines whether the other responses reach the people who need them. Governance appears last because it determines whether any of the others can be sustained.

Chapters 22 to 24 are applied. Two chapters of case studies examine what happened in specific places — heatwaves, fires, floods, droughts, food crises and security emergencies — and what distinguished the responses that worked from those that did not. The final chapter provides action plans at three scales: the household, the community and the city. Readers who need immediate guidance can begin there and return to the analytical chapters later; readers who want to understand the reasoning can read in order.

1.10 Reading in an Emergency

A book is a poor instrument in an emergency, and this one is designed with that limitation in mind. If a hazard is imminent, the priority is to act on the guidance of local authorities, not to consult a chapter. If the hazard has passed and the household is safe, the chapter on household preparedness provides a checklist for the next seventy-two hours and the next three months. If the reader is a planner or a teacher, the chapters on infrastructure, education and governance are the relevant starting points.

The book is also designed to be updated. Climate science, disaster statistics and technology costs change faster than books can be reprinted, and a reader who relies on a number without checking its date will eventually be misled. Each chapter therefore identifies the kind of evidence that would change its conclusions, and the final chapter lists the data sources that are updated most frequently. The habit of checking the date of

a claim is itself a survival skill, because misinformation often consists of true statements that were true once.

Finally, the book is written for readers who are not specialists. Technical terms are explained when they first appear; numbers are given with their units and their dates; and uncertainty is stated rather than concealed. The aim is not to make the reader a climate scientist but to make the reader capable of following an argument, detecting an exaggeration, and deciding what to do. That capacity — not the memorisation of facts — is what allows a community to adapt when the facts change.

1.11 Case Study: The 2022 Pakistan Floods

The floods began in June 2022 with monsoon rain that was heavy but not unprecedented. What made them catastrophic was the sequence. A spring heatwave had melted snow and glaciers early and baked the soil into a crust; when the rain arrived, it ran off instead of infiltrating. The Indus and its tributaries rose together, and the dams and barrages that had been built to control them released water downstream to protect their own structures. By late August, the government had declared a national emergency, and satellite imagery showed a lake stretching across Sindh.

The numbers are a case study in how climate impacts are counted. About 33 million people were affected, 1,700 died, and 2.2 million houses were damaged or destroyed. The Post-Disaster Needs Assessment estimated damage at US$14.9 billion and economic losses at US$15.2 billion, together about 4.8 per cent of GDP. Agriculture lost most of a season: two million hectares of crops were destroyed, and the cotton harvest fell by about a third. The damage was concentrated in Sindh and Balochistan, the two poorest provinces, which is a pattern repeated in disaster after disaster.

Attribution research found that climate change made the extreme rainfall more likely, though the magnitude of the effect varied between studies and depended on the definition of the event (Otto et al., 2023). But attribution explains only the trigger. The vulnerability had been built over decades: irrigation systems that had silted up, embankments that had not been maintained, housing in floodplains, and a population whose livelihoods depended on a single crop season. The flood was a climate event; the disaster was a development event.

The response revealed both the strengths and the limits of international aid. Donor pledges eventually exceeded US$10 billion, but they arrived over months, while the most urgent needs were in the first weeks. The Pakistani military, which has the country's largest logistics capacity, led the relief effort; provincial governments coordinated shelters; and local charities, particularly religious foundations, distributed food faster than the formal system. The pattern — rapid local response, slow international finance — is typical, and it is why the chapters on governance and social capital treat local institutions as part of the infrastructure.

Recovery has been uneven. By 2024, many households had rebuilt with the same materials in the same places, because they had no alternative and because reconstruction assistance covered only a fraction of the cost. The lesson is not that Pakistan should have prevented the rain. It is that recovery which restores the

pre-existing vulnerability is not recovery; it is the preparation of the next disaster. The same lesson appears in the case studies of every chapter that follows.

For readers outside Pakistan, the case has three implications. It shows that a climate hazard can erase years of development in a single season. It shows that vulnerability is produced by decisions about land, water and housing that are made long before the hazard arrives. And it shows that the difference between a disaster and a catastrophe is often the speed and reach of the response — which is a matter of institutions, not of meteorology.

1.12 Future Scenarios and Predictions: What the Next Two Decades Hold

The historical record provides the starting point. Since 1970, global surface temperature has risen by about 0.18°C per decade, sea level by about 3.4 millimetres per year, and the frequency of extreme heat has increased by a factor of roughly three in many regions. The number of weather-related disasters reported to EM-DAT has risen from about 100 a year in the 1970s to more than 300 a year in the 2010s, though part of that increase reflects better reporting. The losses, adjusted for inflation, have grown by about 5 to 7 per cent a year, faster than exposure alone would predict.

The present position is that the world is on a trajectory consistent with about 2.7°C of warming by 2100 if current policies are maintained, according to the International Energy Agency's assessment of announced pledges and implemented measures. That is below the 4.4°C of the highest scenario but well above the 1.5°C target. The near-term trajectory is largely locked in by emissions already released and by infrastructure already built: even if every country met its net-zero pledge, the next two decades would still warm by about 0.4°C above the current level.

The scenarios below are therefore not predictions of what will happen but conditional projections of what could happen under stated assumptions. They differ mainly in the speed of decarbonisation and in the response of slow variables such as ice sheets and the overturning circulation. Each is anchored to observed data: the 2024 concentration of 424 parts per million, the 1.55°C annual anomaly, the 4.5 millimetres per year of sea-level rise, and the current rate of energy transition.

Table 1.1 Three futures for the next two decades

Scenario	Assumption	2035	2050	Signature impacts
Accelerated transition	Emissions peak before 2025, halve by 2035, net zero by 2060	~1.7°C	~1.8°C	Heat extremes stabilise; sea-level rise ~0.3 m by 2050; coral reefs still decline
Current policies	Implemented measures only, slow tightening after 2030	~1.9°C	~2.1°C	Severe heatwaves annual in South Asia; crop losses 5–10%; 200 million displaced by 2050
Delayed transition	Emissions plateau until 2040, rapid catch-up afterwards	~2.1°C	~2.4°C	Arctic summers ice-free; reef collapse; 400 million in coastal flood zones

The difference between the first and third rows is about half a degree by 2050 and nearly a degree by 2100, but the difference in consequences is much larger than the temperature numbers suggest. Half a degree is the difference between a heatwave that kills hundreds and one that kills tens of thousands; between a harvest that fails once a decade and one that fails every other year; between a reef that bleaches and recovers and one that dissolves. The reason is that impacts are non-linear: they depend on thresholds, and thresholds are crossed at particular temperatures rather than spread evenly across the range.

What would change these projections? A faster collapse in the cost of storage would make the first row more likely; a sustained failure of international cooperation would make the third more likely. A large release of methane from permafrost or a faster loss of Antarctic ice would push all three rows upward. The one thing that would not change them is better forecasting: the projections already incorporate the best available science, and the uncertainty that remains is dominated by choices, not by ignorance.

CHAPTER 2

THE PHYSICAL SCIENCE OF GLOBAL WARMING

Theme: the physical science of global warming and its history

In March 1958, a thirty-year-old geochemist named Charles David Keeling hauled a gas analyser up the northern slope of Mauna Loa, a volcano in Hawai'i, and began measuring the carbon dioxide content of clean Pacific air. The first readings came in at about 315 parts per million. Keeling had chosen the site for the same reason meteorologists choose mountain tops: the air there has travelled thousands of kilometres over open ocean and carries no local smoke, no city traffic, no factory exhaust. What it carries is the background composition of the atmosphere. Within two years the measurements showed something no laboratory experiment could show — the background itself was rising, year after year.

That single instrument turned a nineteenth-century physics argument into an accounting problem. Fourier, Foote, Tyndall and Arrhenius had established that carbon dioxide absorbs infrared radiation and that more of it should warm the surface. Keeling established the other half of the ledger: how fast humans were adding it. By 2024 the same series stood at roughly 424 parts per million, about fifty per cent above the pre-industrial value, with the increase still accelerating (NOAA Global Monitoring Laboratory, 2025). No other number in this book is measured so directly, over so long a period, by so few instruments.

The physics matters for the rest of the book because it sets the boundary conditions for everything else. A heatwave in Karachi, a harvest failure in the Brazilian Cerrado, an insurance withdrawal in Florida and a flood in Emilia-Romagna are different events with different proximate causes, but they share a common denominator: an atmosphere holding more energy and more water vapour than it did a century ago. This chapter establishes that denominator. It explains the mechanism that warms the surface, the gases that drive it, the evidence that the warming is real, the accounting that attributes it, and the range of futures consistent with what is already in the pipeline.

It also shows where the certainty stops. The direction of change is not in serious dispute; the magnitude of the response to a given concentration still is, within limits that matter enormously for policy. The difference between a climate that warms by 1.8°C and one that warms by 4.4°C by 2100 is not a matter of scientific taste. It is the difference between a world in which adaptation is expensive and a world in which parts of it are impossible, and it turns on assumptions about emissions that no physical model can settle on its own.

2.1 The Greenhouse Effect

The argument began with a puzzle about temperature. In 1824 Joseph Fourier calculated that a body at the Earth's distance from the Sun should be colder than the Earth actually is, and proposed that the atmosphere slows the escape of heat — a blanket, in his image, though he did not know what the blanket was made of. Thirty-two years later, Eunice Foote ran a simple experiment with glass cylinders in sunlight and found that a cylinder filled with carbon dioxide heated more than one filled with

ordinary air, writing that an atmosphere of that gas 'would give to our earth a high temperature'. In 1859 John Tyndall built an apparatus sensitive enough to measure which gases absorb infrared radiation and found that water vapour was the strongest absorber, with carbon dioxide and ozone close behind (Tyndall, 1861).

The mechanism those experiments revealed is straightforward. Sunlight arrives mostly as visible light, to which the atmosphere is largely transparent. The surface absorbs it and re-emits the energy as infrared radiation, and it is at those wavelengths that greenhouse gases intercept the outgoing beam. They re-radiate in all directions, including downward, so the effective altitude at which the planet finally loses heat to space rises. Because the air at higher altitude is colder, it radiates less efficiently, and the surface must warm until the outgoing energy again balances the incoming sunlight. In 1896 Svante Arrhenius carried out the first quantitative calculation of this balance, estimating that a doubling of carbon dioxide would raise global temperature by roughly five degrees Celsius — an estimate strikingly close to modern values, reached with pencil, paper and a great deal of patience.

The most common objection to this account is that carbon dioxide is a trace gas, four hundredths of one per cent of the atmosphere, and that something so scarce cannot govern the temperature of a planet. The objection misunderstands how absorption works. A gas does not need to be abundant to be opaque at particular wavelengths; it needs only to be present in sufficient quantity to absorb the radiation that tries to pass. At the centre of the carbon dioxide bands the atmosphere is already nearly opaque, so added molecules matter most at the edges of those bands, where absorption is incomplete. That is why the warming effect grows roughly with the logarithm of concentration rather than in direct proportion: each doubling adds about the same increment of forcing as the one before, regardless of how high the starting point already is.

2.2 Greenhouse Gases and Their Sources

Four gases do almost all of the work. Carbon dioxide is the largest single contributor and the one that accumulates; methane is far more potent per molecule but short-lived; nitrous oxide is potent and long-lived in roughly equal measure; and a family of fluorinated industrial gases, present in minute quantities, traps heat thousands of times more effectively than CO2. In 2024 the global average concentration of carbon dioxide reached about 424 parts per million, methane about 1,934 parts per billion and nitrous oxide about 336 parts per billion — values that have no precedent in the ice-core record of the last 800,000 years (NOAA, 2025; IPCC, 2021).

The sources are known with unusual precision because they are counted twice: once from the bottom up, by adding up fuel sales, land-use change and industrial processes, and once from the top down, by measuring the changing composition of the atmosphere and the ocean. The two accounts now agree to within a few per cent. Fossil fuel combustion and industry released about 37.4 billion tonnes of CO2 in 2024, land-use change perhaps four billion more (Friedlingstein et al., 2024). Energy use accounts for roughly three-quarters of total greenhouse gas emissions; agriculture, forestry and land use for about a fifth; the rest comes from industry and waste (IPCC,

2022c).

The difference in lifetimes between the gases is the most policy-relevant fact in this section. A molecule of methane survives in the atmosphere for about a decade before it is oxidised, so cutting methane emissions reduces warming within years rather than centuries. Carbon dioxide is different: perhaps a third of what is emitted today will still be in the atmosphere in a thousand years. That asymmetry explains why climate strategy pursues two tracks at once — rapid methane cuts to slow the rate of warming in the near term, and deep CO2 cuts to determine where the warming finally stops. Confusing the two tracks is a common source of bad policy, because a methane success can look like progress on the problem while the cumulative CO2 ledger continues to grow.

2.3 Evidence of Warming

The evidence for warming does not rest on a single thermometer or a single group. Surface stations, weather balloons, satellites, ocean buoys, glacier surveys, tide gauges, boreholes, the timing of spring, the poleward movement of fish stocks and the retreat of mountain glaciers all point the same way. Each system has its own errors and its own blind spots; what makes the conclusion robust is that the errors are different and the signal is common. When independent records built from different instruments by different groups agree, the probability that all of them are wrong in the same direction falls sharply (IPCC, 2021, Chapter 2).

The last two years illustrate both the strength and the subtlety of the evidence. The World Meteorological Organization estimated 2023 at about 1.45°C above the 1850–1900 average and 2024 at about 1.55°C, the warmest year in the instrumental record (WMO, 2025). Neither figure should be read as 'the climate has reached 1.5°C'; a single year bounces above and below the trend because of El Niño, volcanic aerosols, solar variability and the weather of the ocean. The trend itself, averaged over a decade, is currently rising at roughly 0.2°C per decade — about twenty times faster than the average rate of change since the last ice age (IPCC, 2021).

A related dispute concerns the satellites. Microwave sounding units have measured the temperature of the lower troposphere since 1979, and early analyses appeared to show less warming than the surface record. The discrepancy was traced to the slow drift of satellite orbits, which changed the time of day at which measurements were taken, and to the decay of the instruments themselves. Once those effects were corrected, the satellite record agreed with the surface stations within their respective uncertainties, and both agreed with model simulations driven by observed forcings (IPCC, 2021). The episode is worth remembering because it illustrates a general rule: a disagreement between two measurement systems is usually a clue about the instruments, not a refutation of the phenomenon they are both trying to measure.

2.4 Temperature Records and Trends

A global temperature record is not a single measurement but a statistical construction, and it is worth knowing how it is built because the construction is where most public argument about it goes wrong. Stations move, instruments change, cities

grow around once-rural sites, and ships once measured sea temperature by dipping buckets and now measure it at the engine intake. Each of these changes introduces a bias of a few tenths of a degree, and each bias has to be estimated and removed. Four independent groups — HadCRUT in Britain, NASA GISS in the United States, Berkeley Earth and NOAA — do this work with different methods, and their global series agree to within about 0.05°C (Morice et al., 2021; Lenssen et al., 2019; Rohde and Hausfather, 2020).

What the corrected record shows is a rise of about 1.2°C since the late nineteenth century, unevenly distributed: land has warmed faster than ocean, the Arctic three to four times faster than the global average, and nights faster than days. The record also shows that the warming has not been smooth. There are decades of little change, sharp El Niño years, and pauses that lasted a decade; all of them sit on an upward slope that has steepened since the 1970s. The apparent contradiction between short pauses and long trends is the single most exploited feature of the record in public debate, and it is resolved by the same rule that governs financial markets: look at the trend, not the last quarter.

The exceptional warmth of 2023 and 2024 has a partial explanation that is worth separating from the trend. An El Niño event transferred heat from the tropical Pacific to the atmosphere; the eruption of Hunga Tonga in 2022 injected water vapour, itself a greenhouse gas, into the stratosphere; a sharp reduction in sulphur emissions from shipping removed a reflective aerosol that had been masking part of the warming; and the solar cycle was near its maximum. Each of these contributed a fraction of a degree. Taken together they can explain why the two years sat above the trend line, but not why the trend line itself is where it is. That distinction — between the noise on the signal and the signal — is the difference between a weather story and a climate one.

2.5 Industrial Revolution and Emissions

For most of human history the carbon content of the atmosphere was remarkably stable, varying by only a few parts per million between ice ages and warm periods. That stability ended with coal. The concentration had been about 280 parts per million for thousands of years; by 1900 it had reached roughly 296, and the curve began to bend upward sharply after 1950, when oil, gas and industrial agriculture were added to coal. Historians of the environment call this inflection the Great Acceleration: in a single generation after the Second World War, population, fertiliser use, water consumption, vehicle ownership and carbon emissions all rose together, faster than at any previous point in human history (Steffen et al., 2015).

The arithmetic of that rise is usually summarised by an identity rather than a story. Emissions equal population multiplied by income per person, multiplied by the energy needed for each unit of income, multiplied by the carbon emitted for each unit of energy. Improvements in the last two terms have been real and substantial; they have simply been outpaced by growth in the first two. The same identity explains why emissions are so unequally distributed. The richest tenth of the world's population accounts for roughly half of consumption-based emissions, while the poorest half accounts for about twelve per cent (Chancel, 2022). That distribution is not a moral

footnote; it is the reason historical responsibility remains the most contested question in climate negotiations.

The geography of emissions has shifted faster than the geography of responsibility. China is now the largest annual emitter, responsible for roughly thirty per cent of current fossil CO2, followed by the United States at about eleven, the European Union at about seven and India at about seven (Friedlingstein et al., 2024). Measured cumulatively since 1850, the order is different: the United States and Europe together account for roughly forty per cent of historical emissions, China for about fourteen and India for about four. Per person, the gap is wider still: an average American emits several times more than an average Indian or African. Which accounting is used determines which obligations follow, and that is why the choice of metric is itself a political act.

2.6 IPCC Assessments

The Intergovernmental Panel on Climate Change was created in 1988 by two United Nations bodies, the World Meteorological Organization and the UN Environment Programme, in response to a simple political problem: governments needed an authoritative account of a scientific literature that none of them could read in full. Its method is unusual. Hundreds of scientists volunteer to assess published research, they produce a technical report running to thousands of pages, and then governments negotiate the summary for policymakers line by line. The process is slow, conservative and occasionally painful to read, but it has one great virtue: no single government, industry or scientist can determine its conclusions.

The sixth assessment cycle, published between 2021 and 2023, is the most important single source in this book. Its first working group concluded that it is 'unequivocal that human influence has warmed the atmosphere, ocean and land', and that the best estimate of the warming already experienced is about 1.1°C above 1850–1900 (IPCC, 2021). The second group found that roughly 3.3 to 3.6 billion people live in contexts that are highly vulnerable to climate change (IPCC, 2022b). The third found that the policies in place at the time of writing would produce warming of about 2.7°C by the end of the century (IPCC, 2022c). Those three sentences, taken together, are the empirical spine of the argument that follows.

Reading an IPCC summary requires attention to its grammar. The panel uses a calibrated vocabulary in which 'likely' means a probability between 66 and 100 per cent, 'very likely' between 90 and 100, and 'virtually certain' above 99. It also distinguishes between an assessed range, which reflects the judgement of the author team across all available evidence, and the spread of model results, which reflects only the models. Statements such as 'warming of 1.5°C will be reached in the early 2030s' are assessments, not model outputs, and they carry the authority and the caution of that distinction. Readers who quote the panel without that grammar tend to overstate either its certainty or its doubt.

2.7 Ocean Heat and Sea Level

The ocean has absorbed about ninety per cent of the excess heat trapped by greenhouse gases, and roughly a quarter of the carbon dioxide. That absorption is why the surface has not warmed faster, and it is also why the consequences arrive late. Water has an enormous heat capacity, so the ocean integrates warming over decades and then releases it slowly. Ocean heat content has set a record every year for the past decade; the increase in the upper two kilometres between 2022 and 2023 alone was larger than the total energy used by humanity in a year (Cheng et al., 2024). Marine heatwaves have roughly doubled in frequency since the 1980s, and the fourth global coral bleaching event was declared in 2024.

Sea level is the slowest and most consequential of these responses. Global mean sea level rose about 21 centimetres between 1900 and 2018, and the rate has accelerated from roughly 1.3 millimetres a year in the early twentieth century to about 4.5 millimetres a year today (IPCC, 2021; NASA, 2024). Roughly forty per cent of that rise comes from thermal expansion, a quarter from glaciers and a quarter from the ice sheets, with the remainder from changes in land water storage. Because the processes that drive it respond slowly, sea level will continue to rise for centuries even if emissions fall to zero. The practical question is therefore not whether to adapt, but how much and how soon — and the answer differs by an order of magnitude between a low-emission and a high-emission trajectory.

The coast is where the global average becomes a local emergency. Global mean sea level is an abstraction; what floods a street is relative sea level, which combines the global rise with local subsidence, sediment compaction and changing storminess. Jakarta has been sinking at rates of several centimetres a year in some districts, so its relative sea level rises far faster than the global average; Semarang and parts of Manila face similar combinations. About 1.1 billion people live in low-elevation coastal zones and about 900 million of them in Asia (Neumann et al., 2015). For those populations, the difference between 0.4 and 0.8 metres of rise by 2100 is not a scientific interval; it is the difference between a managed retreat and an unmanaged one.

2.8 Ice Loss and Cryosphere

The cryosphere is where the climate system keeps its slowest and least reversible commitments. Arctic sea ice has declined by roughly 12 to 13 per cent per decade in September since satellite records began in 1979, and the remaining summer ice is thinner and younger than it was a generation ago (Notz and Stroeve, 2016). The Greenland ice sheet has been losing mass at about 260 billion tonnes a year, Antarctica at roughly 150 billion tonnes, and the world's glaciers at about 270 billion tonnes a year outside the ice sheets (Shepherd et al., 2020; IMBIE, 2018; Zemp et al., 2019). Each of those numbers is small relative to the total ice stored on the planet and large relative to anything human infrastructure has had to absorb before.

Three consequences follow. The first is a feedback: white ice reflects sunlight that dark ocean absorbs, so lost ice accelerates local warming. The second is freshwater forcing: meltwater changes the density of the North Atlantic and can slow the

circulation that carries heat northward, with consequences for European weather that remain actively debated (Caesar et al., 2021; Ditlevsen and Ditlevsen, 2023). The third is commitment: ice sheets respond to temperature over centuries, so the sea level implied by today's warming will continue to arrive long after the emissions that caused it have stopped. This is why the difference between 1.5°C and 2°C matters far more for coasts than the difference between two decades of weather.

The Arctic case also illustrates how definitions carry consequences. 'Ice-free' does not mean an ocean without ice; it means less than one million square kilometres of ice at the September minimum, a threshold chosen because it is roughly the area at which the remaining ice becomes fragmented and mobile. Under all plausible emission trajectories the Arctic is expected to reach that state at least once before 2050, and under high emissions it becomes the normal summer condition (Notz and Stroeve, 2016). The consequences extend beyond the region: darker water absorbs more sunlight, the jet stream responds to the changed temperature gradient, and new shipping routes open across a sea that was once a barrier.

2.9 Carbon Cycle and Feedbacks

Of every tonne of carbon dioxide released by human activity, roughly half stays in the atmosphere. The other half is taken up in approximately equal measure by the ocean and by land ecosystems, and that service is the reason atmospheric concentrations have risen more slowly than emissions. The service is not guaranteed. Measurements suggest that the fraction absorbed by land and ocean sinks has been declining as a share of emissions, and that the land sink in particular is becoming more variable as droughts and fires reduce uptake (Friedlingstein et al., 2024). A weakening sink means that the same emissions produce a larger atmospheric increase — a feedback that is already visible in the data.

The ocean also absorbs carbon dioxide chemically, and the resulting change in seawater chemistry is direct and measurable: surface pH has fallen by about 0.1 units since the pre-industrial period, equivalent to a roughly thirty per cent increase in hydrogen ion concentration (IPCC, 2021). Organisms that build shells from calcium carbonate are sensitive to that change, which is why acidification is treated in this book as a parallel crisis rather than a footnote to warming. The same arithmetic that makes the ocean a buffer makes it a slow victim: it has absorbed the heat and the carbon, and it will pay for both for centuries.

The carbon budget is the bridge between physics and policy, and its arithmetic is brutally simple. Every tonne of CO2 emitted adds roughly the same amount of warming, regardless of when it is emitted; the total that can be emitted while holding warming below a given threshold is therefore a finite quantity. The best estimate for a fifty per cent chance of staying below 1.5°C was about 500 billion tonnes of CO2 from 2020, and continued emissions have since reduced it to roughly 200 billion — about five years at current rates (IPCC, 2021; Friedlingstein et al., 2024). The budget does not care about pledges, announcements or targets; it responds only to the tonnage that actually leaves the ground.

2.10 Attribution of Warming

Attribution is the discipline of asking not whether the climate has changed but why. Its central tool is the fingerprint: climate models are run with all known forcings, then with human influences removed, and the resulting spatial patterns are compared with observations. The pattern of greenhouse warming — tropospheric warming with stratospheric cooling, more warming over land than ocean, more at high northern latitudes — is distinctive, and no combination of solar and volcanic variability reproduces it (Hegerl et al., 1996; IPCC, 2021). This is why the sixth assessment could describe human influence as unequivocal rather than merely likely.

A younger branch of the field applies the same logic to individual events. Event attribution compares the probability or intensity of a heatwave, flood or drought in a world with and without human influence, using large ensembles of model simulations constrained by observations. The method has produced conclusions that are now widely cited: the 2021 Pacific Northwest heat dome was judged virtually impossible without climate change, and the 2023 Derna flood in Libya was made far more likely by warming sea surface temperatures. The method has limits — it says little about vulnerability, and single-event results carry wide uncertainty — but it has changed the terms of the compensation debate, because it allows a specific loss to be connected to a specific cause.

Attribution also has a geography of evidence. Events in wealthy countries tend to be studied quickly because they are insured, instrumented and politically salient; events in poorer countries are studied later, if at all, and often with fewer observations. The result is an asymmetry in what can be said, not in what is happening. Researchers call this the attribution gap, and it matters for compensation: a country that cannot demonstrate attribution finds it harder to claim loss and damage, even when the physical case is strong. Closing the gap is therefore a matter of both science and justice.

2.11 Case Study: The Keeling Curve

The most consequential measurement in climate science began as a side project. Charles David Keeling, a chemist by training, had developed a technique for measuring carbon dioxide with far greater precision than his contemporaries, and in 1956 he took a set of flasks to the Olympic Peninsula in Washington State and to the mountains of California to test it. The results surprised him: background concentrations were higher than the accepted value, and they varied with the season in a way nobody had documented. When he presented the finding, Roger Revelle and Hans Suess at the Scripps Institution of Oceanography invited him to continue the work as part of the International Geophysical Year.

The location he chose — Mauna Loa Observatory, 3,400 metres above the Pacific — gave the record its authority. Air arriving there has spent days or weeks over open ocean, so it samples the global background rather than a local plume. Keeling insisted on calibrating his instrument against a set of reference gases, on continuous operation rather than occasional sampling, and on publishing the raw data alongside the analysis. Those choices, which seemed pedantic at the time, are the reason the curve has

survived decades of scrutiny without revision of its central finding.

The data themselves tell a story in two layers. The first is a saw-tooth: carbon dioxide rises through the northern winter and falls through the northern summer as the forests of Siberia, Canada and Europe take up carbon during the growing season. That annual pulse is the breathing of the biosphere, and its amplitude has grown as the land has warmed. The second is the trend beneath it: a steady climb from about 315 parts per million in 1958 to roughly 424 in 2024, with no year below the previous decade's average since the record began.

The curve is also a record of what one instrument cannot see. It measures the atmosphere at one latitude and one altitude; it says nothing directly about the ocean, which absorbs most of the added heat, or about the land, which absorbs a quarter of the carbon. Those measurements came later, from ships, buoys, aircraft and satellites, and they are consistent with the mountain-top series. The Keeling record functions as a keystone rather than a complete arch: remove it and the rest still stands, but the structure becomes much harder to date and to defend.

The project was nearly cancelled twice. In the 1960s the funding agency cut the budget on the grounds that the work was routine monitoring rather than research, and Keeling had to fight to keep the instruments running; a decade later similar pressures returned. Each time the record was preserved by a small group of scientists who understood that a time series is only valuable if it is unbroken. The episode is instructive for anyone designing a monitoring system today: continuity is a scientific asset that is easy to destroy and impossible to reconstruct.

What the curve did politically was to make the abstract concrete. Before Mauna Loa, the greenhouse effect was a calculation; after it, it was a graph that a parliamentarian could hold up. The same function is served today by the satellite record of sea level and by the global temperature series, and the lesson generalises beyond climate: a long, clean, public measurement is one of the most powerful instruments a society can build, because it converts a contested claim into a shared reference point.

There is a final, quieter lesson in the project's design. Keeling published his data in full, including the seasonal wiggles that his colleagues found distracting, and he insisted that the calibration gases be exchanged between laboratories so that independent groups could compare results. That practice — open data, shared standards, documented methods — became the template for every subsequent global observing system. It is also the reason the curve has survived political attack: a record that anyone can check is much harder to dismiss than a conclusion that must be trusted.

2.12 Future Scenarios and Predictions: The Physical Science

Any statement about the physical future of the climate is conditional, and it is worth being explicit about the condition. The atmosphere responds to the total accumulated carbon dioxide, not to the annual rate at which it is emitted. That single fact determines the shape of every projection in this chapter: two trajectories that emit the same cumulative total produce roughly the same peak warming, whether they reach that total quickly or slowly. What differs is the damage along the way, the options available for adaptation, and the risk of crossing thresholds that cannot be reversed on

human timescales.

The historical record supplies the baseline. Concentrations were stable at about 280 parts per million for six thousand years before 1750; they reached 315 in 1958, about 370 in 2000 and roughly 424 in 2024, with cumulative emissions since 1850 now close to 2,500 billion tonnes of CO2 (Friedlingstein et al., 2024). Global mean temperature has risen about 1.2°C over the same period and is currently rising at roughly 0.2°C per decade. Sea level, which lagged the temperature rise for most of the twentieth century, is now accelerating.

The present position can be stated with unusual precision. For a fifty per cent chance of holding warming below 1.5°C, the remaining carbon budget is roughly 200 billion tonnes of CO2 — about five years of current emissions. For 2°C the budget is larger but still finite, and the policies in place in 2025 imply warming of about 2.7°C by 2100 rather than the 1.5°C target governments have endorsed (IPCC, 2022c; Friedlingstein et al., 2024). Three trajectories bracket the plausible range, and each rests on a different assumption about policy, technology and behaviour.

The first assumes rapid decarbonisation: emissions fall by about half by 2030 and reach net zero around mid-century, with carbon removal compensating for residual sources. Warming peaks near 1.8°C and declines slowly thereafter. The second assumes that current policies continue and are strengthened gradually, producing net zero late in the century; warming reaches about 2.7°C. The third assumes that decarbonisation stalls and that coal, oil and gas remain dominant; warming reaches roughly 4.4°C, with the largest increases over land and at high northern latitudes (IPCC, 2021, Table 4.5).

A useful thought experiment clarifies what is already committed. If every emission stopped tomorrow, the atmosphere would stop accumulating CO2 within a few years, but global temperature would stabilise rather than fall, and sea level would continue to rise for centuries as the ocean and the ice sheets finished responding to the warming that has already occurred. The commitment is not a reason for fatalism; it is a reason to distinguish between the warming that cannot now be avoided and the warming that can still be prevented. The first determines how much adaptation is necessary, the second determines whether adaptation remains possible.

Regionally, the three trajectories diverge far more than the global averages suggest. Under the high trajectory, the Mediterranean and the Middle East approach conditions in which summer heat becomes a chronic hazard rather than an episodic one; South Asia faces combinations of heat and humidity that exceed the physiological limits of outdoor work for part of the year; and the Arctic warms by several degrees more than the global mean. Under the low trajectory those outcomes are attenuated but not eliminated. The difference between the scenarios is therefore not a difference in global average temperature; it is a difference in which regions remain habitable in the way their inhabitants currently understand the term.

Table 2.1 Three physical trajectories to 2100 (relative to 1850–1900)

Trajectory	Policy assumption	2050	2100	Observable markers
Rapid decarbonisation (SSP1-2.6)	Emissions halve by 2030, net zero around 2060	~1.6°C	~1.8°C (1.3–2.4)	Atmospheric CO2 peaks below 450 ppm; sea-level rise near 0.4 m
Current policies strengthened (SSP2-4.5)	Slow tightening, net zero late century	~2.0°C	~2.7°C (2.1–3.5)	CO2 near 500 ppm by 2060; Arctic summers ice-free before 2050
Stalled decarbonisation (SSP5-8.5)	Fossil dominance continues to 2100	~2.6°C	~4.4°C (3.3–5.7)	CO2 above 700 ppm; sea-level rise above 0.8 m and accelerating

What would change these numbers is worth stating as plainly as the numbers themselves. A larger climate sensitivity than the central estimate, a faster loss of the land carbon sink, or an earlier collapse of Arctic summer sea ice would all push the outcomes toward the upper end. A faster decline in methane, a stronger aerosol effect from air-quality policy, or large-scale carbon removal would push them down. None of these possibilities can be resolved by argument; they will be resolved by measurement, and the measurements that matter are the ones this chapter has described — concentration, temperature, ocean heat, ice mass and sea level.

For the reader, the practical implication is a shift in attention. The question is no longer whether the physics is settled; it is which of three trajectories is being followed, and how quickly the answer becomes visible. That question is answered not by climate models but by the energy system, which is the subject of the chapters that follow.

CHAPTER 3

CLIMATE SYSTEMS, FEEDBACKS AND MONITORING

Theme: climate systems, feedback mechanisms and monitoring

A climate is not a temperature. It is a machine with several moving parts — an atmosphere, an ocean, a cryosphere, a biosphere and a land surface — coupled so tightly that a change in one is felt in all the others, usually after a delay. The atmosphere responds in days, the ocean in decades, the ice sheets in centuries. This difference in response time is the single most important fact for anyone trying to understand why the consequences of emissions arrive long after the emissions themselves, and why some of those consequences cannot be reversed on any timetable that matters to the people experiencing them.

The couplings also explain why the climate does not warm smoothly. Heat stored in the ocean can be released years later; a shift in wind patterns can cool a continent for a decade while the planet as a whole warms; a volcanic eruption can suppress global temperature for two or three years. These internal variations are not noise in the sense of meaningless — they are the weather of the climate system, and they determine whether a particular harvest fails or survives, whether a particular city floods this year or next.

Feedback is the mechanism that turns a small push into a large response, or a large push into a small one. Warming increases atmospheric water vapour, which is itself a greenhouse gas, which produces more warming; that is a positive feedback, and it roughly doubles the effect of carbon dioxide alone. Warming melts reflective ice, exposing dark ocean that absorbs more sunlight; another positive feedback. But warming also increases the radiation the planet emits to space, which is a negative feedback and the reason the system has not run away. The balance between these effects, not the strength of any one of them, determines how sensitive the climate is.

This chapter is about that balance, about the thresholds where it can change abruptly, and about the instruments that allow us to watch it happen. It closes with the least glamorous and most consequential subject in climate science: data quality, and the question of what a measurement can and cannot tell you.

3.1 The Climate System

The Earth receives about 340 watts of solar energy for every square metre of its surface, averaged over day and night and over the seasons. Roughly thirty per cent of that is reflected back to space by clouds, ice and bright surfaces — the albedo — and the remainder is absorbed by the atmosphere, the ocean and the land. Because the planet is neither heating nor cooling catastrophically on human timescales, the energy leaving must roughly equal the energy arriving. The greenhouse effect works by slowing the departure, not by adding energy; the system compensates by warming until the outgoing radiation again balances the incoming sunlight (Trenberth, Fasullo and Kiehl, 2009).

What makes the system complicated is that its components have different memories. The atmosphere forgets a temperature anomaly within weeks; the surface ocean remembers it for months to years; the deep ocean for centuries; an ice sheet for

millennia. A pulse of heat injected into the atmosphere is therefore partitioned: some escapes to space immediately, some is absorbed by the ocean and returned decades later, and some is consumed melting ice. The same is true of carbon dioxide, which is exchanged continuously among the atmosphere, the biosphere and the ocean, with a small fraction remaining airborne for thousands of years.

The practical consequence is that the climate responds to a change in forcing with a lag that depends on the component being observed. Surface temperature responds within a few years, sea level over centuries, and ice sheets over millennia. Policymakers who expect a rapid visible response to emission cuts are therefore looking at the wrong variable; the response is real but it arrives first in the ocean and only later at the coast.

3.2 Atmosphere and Ocean Circulation

The atmosphere moves heat from the tropics toward the poles through three great cells in each hemisphere: the Hadley cell, which lifts moist air at the equator and brings it down over the subtropics; the Ferrel cell, which drives the mid-latitude westerlies; and the polar cell. The boundaries between these cells are where the jet streams run, and the position of the jets determines whether a winter storm reaches Lisbon or stalls over the North Sea. As the tropics warm faster than the poles, the temperature gradient weakens and the jets tend to shift poleward, though the effect is uneven and remains an active research question (IPCC, 2021, Chapter 4).

The ocean moves heat in parallel, and more slowly. The Atlantic Meridional Overturning Circulation carries warm surface water northward, cools it in the Nordic seas, and returns it southward as cold, dense deep water; the Gulf Stream is its most visible surface expression. The circulation transports roughly eighteen million cubic metres of water per second — about a hundred times the flow of the Amazon — and delivers heat to Europe that helps keep its winters milder than those of comparable latitudes in Canada. It is driven by differences in temperature and salinity, which is why added freshwater from Greenland's melting ice is watched so closely.

Superimposed on these steady circulations are oscillations: the El Niño–Southern Oscillation, which shifts heat across the tropical Pacific every two to seven years and alters rainfall from Peru to Indonesia; the North Atlantic Oscillation, which governs European winter weather; the Indian Ocean Dipole; the Southern Annular Mode. These patterns do not create the long-term trend, but they determine which years are hot, which are wet, and where the damage falls.

3.3 Positive and Negative Feedback Loops

The largest feedback is also the least visible. Warmer air holds more water vapour — about seven per cent more for each degree of warming, following the Clausius–Clapeyron relation — and water vapour is a powerful greenhouse gas. The increase in water vapour roughly doubles the warming that carbon dioxide would produce on its own, which is why the climate sensitivity to a doubling of CO2 is about three degrees rather than one and a half. This is not a speculative mechanism; it is measured, and it is the reason the same physics that gives a modest direct effect gives a large total one (IPCC, 2021, Chapter 7).

Ice and snow provide a second positive feedback. White surfaces reflect sunlight; dark ocean and dark land absorb it. When ice retreats, the newly exposed surface absorbs more energy, which warms the region, which melts more ice. The feedback is strongest in the Arctic, where it contributes to the three- to fourfold amplification of warming relative to the global mean. A third comes from the carbon cycle: warming increases the respiration of soil microbes and the frequency of fire, releasing carbon that had been stored, which produces further warming.

Against these stand the negative feedbacks that keep the system stable. The planet radiates more energy as it warms, following the Stefan–Boltzmann law; the lapse rate — the rate at which temperature falls with altitude — adjusts in a way that damps the response in the tropics; and increased weathering of silicate rocks removes carbon dioxide from the atmosphere over geological time. Clouds are the great uncertainty: low clouds reflect sunlight and cool, high clouds trap heat and warm, and how their balance changes with warming remains the largest single source of spread among climate models.

3.4 Tipping Points and Abrupt Change

Some parts of the climate system do not respond gradually. They behave like a see-saw held just past its balance point: a small additional push produces a large, rapid and often irreversible change. Researchers call these tipping elements, and a 2022 assessment identified sixteen of them, including the Greenland and West Antarctic ice sheets, the Atlantic overturning circulation, the Amazon rainforest, boreal permafrost and warm-water coral reefs (Armstrong McKay et al., 2022). The assessment found that several would be at risk at warming levels between 1.5 and 2°C, which is to say within the range that current policies are likely to produce.

The concept is easy to misunderstand. A tipping point is not a cliff edge at a precise temperature; it is a region of the parameter space in which the system loses stability, and the exact location is uncertain because the systems are complex and poorly observed. Nor does crossing a threshold always produce an instantaneous change: the Greenland ice sheet may take centuries to disappear once committed, and the Amazon may degrade over decades through fire and drought. What changes at the threshold is not the speed of the change but its reversibility.

The policy implication is asymmetric. If the system were linear, overshooting a target and returning later would be an acceptable strategy. If it contains thresholds, overshoot becomes a gamble with a permanent loss on the other side. That asymmetry, rather than any single temperature number, is the strongest argument for treating 1.5°C as a limit to be approached with caution rather than a target to be traded against other priorities.

3.5 Temperature and Satellite Monitoring

The global temperature record is assembled from about thirty thousand land stations, several thousand ships and buoys, and a handful of satellite instruments, all processed by four independent groups using different methods. Satellites have measured the temperature of the lower troposphere since 1979, first with microwave sounders and

later with infrared interferometers; they provide global coverage but suffer from orbital drift, calibration changes and the difficulty of separating the signal of the lower atmosphere from that of the surface. Reanalysis products, which combine observations with a weather model, fill the gaps and provide consistent fields of temperature, wind and humidity over the whole planet (Hersbach et al., 2020).

Each system has a characteristic error. Surface stations are precise but unevenly distributed, with dense coverage in Europe and North America and sparse coverage in parts of Africa and South America. Satellites are uniform but indirect, measuring radiance rather than temperature and requiring a radiative transfer model to convert one into the other. Reanalyses are complete but inherit the biases of their models. The skill of the system comes from the disagreement: when independent methods converge, the result is robust; when they diverge, the divergence points to a problem worth investigating.

The architecture of that system is itself a policy achievement. The Global Climate Observing System, established in 1992, sets standards for instruments, data exchange and continuity, and the Paris Agreement's transparency framework depends on it. When a country reports its emissions or its progress, it relies on measurements that are only credible because they are comparable across borders. Monitoring, in other words, is not a technical appendix to climate policy; it is the infrastructure that makes policy verifiable.

3.6 Ocean and Ice Monitoring

The ocean was, until recently, the least observed part of the climate system — a gap that mattered because it stores more than ninety per cent of the excess heat. That changed with Argo, an international programme that began deploying autonomous profiling floats in 2000 and now maintains roughly four thousand of them. Each float sinks to two thousand metres, measures temperature and salinity as it rises, transmits its data by satellite and repeats the cycle every ten days. The result is a three-dimensional picture of the ocean's heat content, updated continuously, available to anyone with an internet connection.

Ice is observed by a different combination of instruments. Satellite altimetry, which has measured sea surface height since 1992, also measures the height of ice sheets; the GRACE mission, launched in 2002, weighs them by detecting tiny changes in the Earth's gravity field. Radar and laser altimeters measure thinning, and microwave sensors map the extent of sea ice every day. Each method has blind spots — altimetry struggles at the steep margins of ice sheets, gravity missions have coarse spatial resolution — but together they have produced a consistent picture of accelerating loss (Shepherd et al., 2020).

The data from these systems are open, which is unusual for expensive infrastructure and essential for credibility. A government that disputes a sea-level projection can download the altimetry, the tide-gauge record and the gravity data and check the arithmetic itself. That openness does not settle political disputes, but it narrows the space in which they can be conducted by assertion rather than evidence.

3.7 Extreme Event Attribution

Attribution began with a question about a single heatwave. After the European summer of 2003 killed tens of thousands of people, a team led by Peter Stott asked whether the probability of such an event had changed because of human influence, and answered by comparing model simulations with and without greenhouse gases (Stott, Stone and Allen, 2004). The answer — that the risk had roughly doubled — established a method that has since been applied to hundreds of events, from the Russian heatwave of 2010 to the floods of 2022 in Pakistan.

The method has become faster and more standardised. The World Weather Attribution group, founded in 2014, now produces rapid analyses within weeks of an event, using peer-reviewed protocols and large ensembles of model runs. Its findings are stated as changes in probability or intensity rather than as certainties: the 2021 Pacific Northwest heat dome was judged to be virtually impossible without climate change; the 2022 Pakistan floods were made more likely by warming, though vulnerability and land management determined the scale of the disaster; the 2023 Derna flood in Libya was linked to warmer-than-usual Mediterranean waters.

Attribution has limits that are easy to forget. It is more reliable for heat than for rainfall, more reliable at continental than at local scales, and weakest where observations are sparse — which is usually where the most vulnerable populations live. It also says nothing about who is responsible or who should pay; those are political questions that attribution can inform but cannot answer. Used carefully, it converts a diffuse sense of injustice into a defensible claim; used carelessly, it invites the expectation that every disaster has a single attributable cause.

3.8 Early Warning Systems

Early warning is the most cost-effective adaptation measure available, and it is still missing where it is needed most. The United Nations' Early Warnings for All initiative, launched in 2022, estimated that about half of the world's countries lack a multi-hazard early warning system, and that where systems exist they often fail at the last mile — the warning reaches a capital city but not a village, or arrives in a language and format that the people at risk cannot use (UNDRR and WMO, 2023). The same reports estimate that a twenty-four-hour warning of a coming storm can reduce damage by about thirty per cent.

Bangladesh is the standard example of what is possible. After the Bhola cyclone of 1970 killed an estimated three hundred thousand people, the country built a system of cyclone shelters, volunteer networks and mobile-phone alerts that has cut fatalities dramatically: Cyclone Amphan in 2020, a storm of comparable strength, killed roughly a hundred people. The improvement was not primarily technological. It came from shelters placed where people live, warnings issued in local languages, and drills that made evacuation routine.

Heat is the harder case, because heat kills quietly and without a visible landfall. Ahmedabad's heat action plan, adopted after a 2010 heatwave killed more than a thousand people, combines colour-coded forecasts, cooling centres, altered working

hours and outreach to vulnerable groups; evaluations suggest it has reduced mortality substantially. The lesson generalises: a warning is only an early warning if it arrives with the capacity to act on it.

3.9 Data Quality and Uncertainty

Uncertainty is not the same as ignorance, and the distinction matters for how scientific statements should be read. A measurement can be uncertain in a way that is precisely quantified — plus or minus a tenth of a degree, with a stated confidence interval — and a projection can be uncertain in a way that reflects genuine disagreement among plausible models. The first kind of uncertainty shrinks with better instruments; the second shrinks with better understanding, and sometimes not at all. Confusing the two leads either to false confidence or to unwarranted paralysis.

Climate science distinguishes between internal variability, which is the unpredictable component generated by the system itself; model uncertainty, which is the spread produced by different representations of the same physics; and scenario uncertainty, which reflects the fact that future emissions depend on choices not yet made. Only the first is reduced by more data, and only the third is reduced by policy. When a projection is presented as a range, the width of that range is usually dominated by scenario uncertainty at the end of the century and by model uncertainty in the near term.

There is also a discipline problem that the field has only recently begun to address. A dataset that is used to make a political argument will be scrutinised in a way that a dataset used only for research is not, and the scrutiny often targets the processing rather than the physics. The defence is transparency: open code, documented adjustments, published uncertainty ranges, and an explicit account of what would change the conclusion. Those practices are now standard in the major temperature and sea-level products, and they are the reason the records have survived decades of political attack.

3.10 Using Data for Decisions

The final step is the one most often skipped: converting a probabilistic projection into a decision. The standard approach in engineering is to design for a return period — a flood defence built for the one-in-a-hundred-year event — but that approach assumes a stationary climate, which is precisely what no longer holds. A defence designed for the statistics of the twentieth century will be overtopped more often than its designers intended, not because it was badly built but because the statistics it was built against have moved.

The alternative is robust decision-making: rather than optimising for a single best estimate, identify the options that perform acceptably across a wide range of futures, and prefer those that can be adjusted as information arrives. The Netherlands' Delta Programme, which plans for sea-level rise up to several metres while building in stages that can be accelerated or paused, is the clearest example. The Thames Barrier, designed in the 1970s for a scenario that now looks optimistic, is another: its closure rate has risen from a few times a year to several times a year, and the decision to raise it

further is taken on the basis of monitoring rather than prediction.

What these examples share is a willingness to treat uncertainty as a design parameter rather than an obstacle. That requires institutions capable of revisiting decisions, budgets that can be reallocated, and a public that understands why a plan might change. The technical tools are available; the constraint is almost always institutional, which is why the chapters that follow devote as much attention to governance as to physics.

3.11 Case Study: Argo and the Ocean That Was Hiding

Before 2000, the ocean's interior was sampled the way a continent might be mapped by a few dozen travellers: ships lowered instruments at chosen points, took a reading, and moved on. The coverage was sparse, biased toward shipping lanes, and concentrated in the northern hemisphere. Estimates of ocean heat content therefore carried uncertainties large enough to obscure the trend for a decade at a time, which meant that the largest reservoir of the climate system was also the least measured.

Argo changed that with a design of remarkable simplicity. A float is dropped from a ship; it sinks to a programmed depth, drifts with the current, then rises while measuring temperature and salinity. At the surface it transmits its data and its position by satellite, then sinks again. Each cycle takes about ten days, each float lasts about four years, and the international programme maintains roughly four thousand of them, distributed across every ocean basin. The data are released within a day and are free to use.

The scientific consequences were immediate. Ocean heat content became measurable with an uncertainty small enough to detect the annual increase; the seasonal cycle of the upper ocean became visible; and the deep ocean's role in absorbing heat could be quantified rather than inferred. When the record was extended backward with older ship data and forward with Argo, it showed a steady accumulation of heat that has continued every year since, with the last decade the warmest in the record (Cheng et al., 2024).

Argo also demonstrates a governance lesson. The programme works because countries contribute floats, share data and accept common standards without a central authority that can compel them. It is coordinated by a small international secretariat and sustained by the self-interest of the participants, each of whom gets access to the whole dataset in exchange for contributing a share. That model — voluntary, standardised, open — has proved more durable than many formal treaties.

The gaps that remain are instructive. Ice-covered seas, marginal seas and the deep ocean below two thousand metres are still poorly sampled, and the floats themselves are biased toward regions where deployment is easy. The next generation of instruments, including deep Argo and under-ice floats, is being deployed to close those gaps. The lesson for other domains is that a monitoring system is never finished; it is a commitment to keep looking, renewed each time the technology changes.

3.12 Future Scenarios and Predictions: Systems, Feedbacks and Thresholds

The future of the climate system is not a single number but a set of conditional paths, and the condition is the same one that governs the previous chapter: the cumulative carbon emitted. What this chapter adds is the possibility of abruptness. The historical record shows that the system has changed gradually for most of the last ten thousand years, and that it has also changed abruptly in the more distant past — the Younger Dryas cold snap, the collapse of ice sheets at the end of the last glaciation — when a threshold was crossed. The question for the next century is whether the thresholds now being approached will be crossed slowly or quickly.

The present position is that several of the slower variables have already begun to move. The Atlantic overturning circulation is weaker than at any time in the last millennium according to proxy reconstructions, though the size and significance of the decline remain contested (Caesar et al., 2021). Arctic sea ice is on a trajectory to be essentially absent in September before mid-century under all but the lowest emission scenarios. Warm-water coral reefs have already suffered four global bleaching events, and the interval between them has shortened.

The assumptions that drive the scenarios are therefore not only about emissions but about the response of these slow variables. If the overturning circulation weakens moderately, European winters cool slightly while the tropics warm faster; if it collapses, the regional consequences are severe and the global consequences are still debated. If the Greenland ice sheet is committed to long-term loss, sea level rises for centuries regardless of what happens to emissions after mid-century. These are not alternative theories; they are different positions on a continuum of probability, and the monitoring systems described above are what allow the probabilities to be updated.

Table 3.1 System thresholds and what to watch for

Element	Threshold of concern	Time to observe change	Signal to monitor
Atlantic overturning	Weakening beyond ~15% of baseline	Decades	Florida Strait transport; Labrador Sea density
Arctic summer sea ice	Below 1 million km² in September	Years	NSIDC extent; ice thickness from CryoSat
Greenland ice sheet	Committed loss above ~1.5°C	Centuries	GRACE gravity; surface melt days
Amazon rainforest	Dieback beyond ~20–25% forest loss	Decades	Fire counts; dry-season length; carbon flux
Coral reefs	Frequent bleaching at 1.5°C; collapse at 2°C	Years	Degree-heating weeks; reef surveys

Two conclusions follow for decisions. The first is that monitoring is not a scientific luxury but an early-warning system for the planet: the sooner a threshold is detected, the more options remain. The second is that the standard tools of cost-benefit analysis are poorly suited to a system with thresholds, because they average over outcomes that are not interchangeable. A small probability of an irreversible change is not equivalent to a larger probability of a recoverable one, and any appraisal that treats them as equivalent will systematically underinvest in prevention.

What would change these conclusions? A clear, sustained weakening of the overturning circulation would raise the urgency of every adaptation decision in Europe and West Africa. A decade without further Arctic sea-ice decline would suggest that the system is more resilient than the models assume. Neither is likely to be decisive on its own, and that is the nature of a system observed at the pace of decades: the evidence arrives slowly, and the decisions cannot wait for it.

CHAPTER 4

EXTREME HEAT AND URBAN HEALTH

Theme: extreme heat, urban heat islands and the physiology of heat death

Heat is the deadliest climate hazard in most years, and the least visible. A flood leaves debris, a fire leaves ash, a hurricane leaves a coastline that can be photographed from space. A heatwave leaves bodies in apartments, morgues that fill quietly, and death certificates that list cardiac arrest rather than exposure. The number of people killed by heat is therefore almost always discovered later, by epidemiologists comparing the deaths that occurred with the deaths that would have been expected, rather than by officials counting at the scene.

The physical basis is simple. The human body produces heat continuously and must lose it to the environment. It does so by radiation, convection, evaporation of sweat and, to a small extent, by breathing. When air temperature approaches skin temperature, radiation and convection stop working and evaporation becomes the only route. When humidity is also high, evaporation slows, and the body's core temperature begins to rise. At a wet-bulb temperature near 35°C — the theoretical limit at which sweat can no longer cool the body — even a healthy adult at rest will overheat within hours.

Cities make this worse in ways that are measurable. Concrete and asphalt absorb solar radiation during the day and release it at night, so the urban heat island is strongest after sunset, when the body most needs relief. Buildings block wind, traffic adds waste heat, and the loss of trees removes shade and transpiration. The result is that a heatwave of the same meteorological magnitude can be survivable in a village and lethal in a city, and that the difference is concentrated in the neighbourhoods with the least green cover and the oldest housing.

This chapter examines heat as a physical hazard, as a physiological stress, as an urban design problem and as a public health emergency. It pays particular attention to the question that determines whether a heatwave becomes a disaster: who is exposed, who is warned, and who has somewhere cooler to go.

4.1 Measuring Heat: Temperature, Humidity and Wet-Bulb

A thermometer in the shade measures air temperature, but the body responds to a combination of temperature, humidity, wind and radiation. The heat index, familiar from weather forecasts, approximates the apparent temperature by combining temperature and relative humidity. Wet-bulb temperature goes further: it is the lowest temperature to which a surface can be cooled by evaporation, and it is the variable most closely tied to the body's ability to shed heat. A dry 45°C is dangerous; a humid 35°C wet-bulb is close to unsurvivable for more than a few hours.

The theoretical limit of 35°C wet-bulb was proposed by Sherwood and Huber (2010) on the basis of a resting metabolic rate. Laboratory work since then has found that the critical wet-bulb temperature for healthy young adults at rest is closer to 31°C, and lower still during light activity (Vecellio, Wolf, Cottle and Kenney, 2022). Older people, people with cardiovascular disease, pregnant women and children reach their

limits earlier. The practical implication is that conditions once considered rare are already occurring in South Asia and the Persian Gulf, and that the margin between discomfort and death is narrower than the 35°C figure suggests.

Heat is also cumulative. A single hot day is survivable; a week of hot nights is not, because sleep is when the body recovers. Public health agencies therefore increasingly report heat in terms of consecutive days above a threshold, or degree-days, rather than daily maxima. The distinction matters for warnings: a forecast of 40°C for one afternoon is a different emergency from a forecast of 35°C for seven consecutive days with nights above 28°C.

4.2 The Physiology of Heat Illness

Heat illness progresses along a continuum. Heat cramps are painful muscle spasms caused by salt loss; heat exhaustion produces weakness, dizziness, headache and nausea as blood pools in the skin and the heart struggles to maintain blood pressure; heatstroke occurs when the core temperature exceeds about 40°C and the body's thermoregulation fails. Heatstroke is a medical emergency with a mortality rate that can exceed 50 per cent if cooling is delayed, and it can leave survivors with permanent neurological damage.

The pathway to death is often cardiovascular rather than thermal. During a heatwave, the heart works harder to pump blood to the skin for cooling, while sweating reduces blood volume. People with narrowed coronary arteries, heart failure or kidney disease can die of a heart attack or kidney failure without ever reaching the temperature that defines heatstroke. This is why excess mortality during heatwaves is dominated by cardiovascular and respiratory causes, and why death certificates rarely mention heat.

Certain medications increase risk. Diuretics reduce blood volume; beta-blockers limit the heart's ability to increase output; anticholinergics reduce sweating; some antidepressants impair the perception of thirst. People who live alone may not notice the onset of symptoms, and people with dementia may not be able to seek help. The physiological and the social therefore combine: the same temperature that causes discomfort in a healthy adult can be lethal to an isolated elderly person on the top floor of a building without shade.

4.3 The Urban Heat Island

The urban heat island is one of the best-documented effects of human settlement. Cities are typically 3 to 7°C warmer than their rural surroundings during the day and can be more than 10°C warmer at night. The causes are well understood: dark surfaces absorb more solar radiation, buildings have a high thermal mass and release heat slowly, the geometry of streets traps radiation, vehicles and air-conditioning units release waste heat, and vegetation that would otherwise cool the air through transpiration is scarce.

The intensity varies enormously within a single city. Satellite measurements of land surface temperature in Jakarta, Delhi, Manila and São Paulo show differences of 5 to 10°C between neighbourhoods a few kilometres apart, with the hottest areas corresponding to dense informal settlements, industrial zones and bare ground, and the

coolest to parks, tree-lined streets and water bodies. Because these differences track income, the urban heat island is also a map of inequality: the people with the least capacity to buy cooling live in the hottest places.

Urban form determines how the heat island will change. A city that grows through sprawl with dark roofs and wide asphalt roads will intensify it; a city that protects green space, plants street trees, uses reflective materials and preserves ventilation corridors can keep it in check even as the climate warms. The decisions are made in building codes, zoning rules and investment plans, which is why heat policy is inseparable from urban planning.

4.4 Who Dies in a Heatwave

The epidemiology of heat mortality is remarkably consistent across countries. The elderly account for the majority of deaths, particularly those over 75 and those living alone. People with pre-existing cardiovascular, respiratory, renal or psychiatric conditions are at elevated risk. Outdoor workers, construction labourers, agricultural workers and delivery riders are exposed to the highest temperatures for the longest periods. Homeless people, prisoners, refugees and people in informal settlements are exposed both to heat and to the absence of cooling.

The Chicago heatwave of July 1995 is the canonical study. More than 700 people died in five days, most of them in their homes, and the distribution was not explained by temperature alone. Eric Klinenberg's social autopsy found that neighbourhoods with similar demographics and similar temperatures had very different death rates, and that the difference was associated with social isolation, fear of crime, and the quality of the local social infrastructure — whether people knew their neighbours, whether they trusted the city, whether they were willing to leave their apartments (Klinenberg, 2002).

The European heatwave of August 2003 killed an estimated 70,000 people, with France alone accounting for nearly 15,000. The deaths were concentrated among the elderly, many of whom were left in apartments without air conditioning while their families were on holiday, and the crisis exposed the absence of a public health plan for heat. The 2022 European summer, by contrast, killed about 61,700 people despite a much better warning system, which suggests that warnings alone are insufficient when the underlying vulnerability remains (Ballester et al., 2023).

4.5 Counting the Dead: Excess Mortality

Heat deaths are counted by comparing the number of deaths observed during a heatwave with the number expected from historical averages for the same period. The difference is excess mortality. The method is robust to the misclassification of individual death certificates, because it does not require the certifier to recognise heat as a cause; it only requires that the total number of deaths be counted accurately. This is why epidemiological estimates of heat deaths are usually much higher than official counts.

The method has limitations. Heat can displace deaths that would have occurred within weeks — a phenomenon known as mortality displacement, or harvesting — which means that a heatwave may accelerate deaths rather than cause them. Studies generally find that displacement accounts for a minority of excess deaths, especially for severe heatwaves, but the size of the effect is contested. Excess mortality also captures deaths from other causes that rise during heatwaves, such as heart attacks and kidney failure, which may not be recorded as heat-related.

The global burden is large. The Lancet Countdown estimates that heat-related mortality among people over 65 has increased by about 70 per cent since the early 1990s, and that more than 500,000 deaths a year are associated with heat. The figure is an estimate, not a count, and it depends on assumptions about temperature-mortality relationships that vary by location. But the direction is unambiguous, and the trend is one of the clearest signals that climate change is already affecting human health.

4.6 Heat Action Plans and Early Warning

Ahmedabad, India, adopted one of the first heat action plans in South Asia after a 2010 heatwave killed more than 1,300 people. The plan combines a colour-coded forecast system, cooling centres in public buildings, adjusted working hours for outdoor labourers, water distribution, and outreach to slum communities through community health workers. An evaluation published in 2018 estimated that the plan had prevented about 1,190 deaths a year, though the authors cautioned that the estimate depended on assumptions about the counterfactual (Hess et al., 2018).

The components of an effective plan are now fairly standard. A warning system that triggers specific actions at specific thresholds; a communication strategy that reaches people in their own languages through channels they trust; a registry of vulnerable people, particularly the elderly living alone; designated cooling spaces that are open, accessible and free; and a protocol for hospitals and emergency services to prepare for surges. The weakest link is usually the last mile: the warning exists, but the person at risk does not hear it, cannot act on it, or has nowhere to go.

Plans also fail when they are designed for the wrong hazard. A plan built around daily maximum temperature will miss a heatwave with moderate daytime temperatures and very hot nights, which is precisely the pattern that kills in humid cities. Plans built around a single city will miss rural workers and informal settlements outside the administrative boundary. And plans that are not rehearsed will not be implemented when the emergency arrives, because the agencies involved will be learning their roles in real time.

4.7 Heat at Work: Productivity and Labour

Heat affects economies through labour as well as through death. The International Labour Organization estimates that heat stress already reduces global working hours by about 2.2 per cent, equivalent to 80 million full-time jobs, and that the loss could rise to 2.4 per cent by 2030 under a high-emission scenario. Agriculture accounts for about 60 per cent of the lost hours, followed by construction and industry. The losses are concentrated in South Asia, Sub-Saharan Africa and Southeast Asia,

where outdoor work is common and cooling is scarce.

The mechanism is physiological. Above about 26°C wet-bulb, the body must divert blood to the skin to dissipate heat, which reduces the blood available to muscles and the brain. Physical capacity declines, error rates rise, and the risk of accidents increases. Workers adapt by taking breaks, shifting hours to early morning or night, and drinking more water, but these adaptations have limits, and they are often unavailable to workers paid by output rather than by time.

The economic consequence is not only lost output but lost income. A construction worker who cannot work during the hottest hours loses a fraction of daily wages; a farmer whose crop fails loses a season. Because informal workers rarely have heat insurance or sick leave, the cost is transferred to the household. Heat policy is therefore also labour policy, and the most effective interventions often involve collective bargaining, enforceable rest breaks and access to shade and water on site.

4.8 Cooling, Energy and Equity

Air conditioning saves lives, and its spread is one of the reasons heat mortality has fallen in high-income countries even as heatwaves have become more severe. The International Energy Agency counted about 1.6 billion air-conditioning units in 2016 and projected that the number could reach 5.6 billion by 2050, with most of the growth in developing countries. Cooling already accounts for about a tenth of global electricity consumption, and demand peaks during exactly the hours when the grid is most stressed.

The equity problem is that the people who most need cooling are least able to afford it. In many cities, electricity is unreliable, the upfront cost of an air conditioner is prohibitive, and landlords do not pay for efficiency improvements. Passive cooling — shade, insulation, reflective roofs, cross-ventilation, courtyards, trees — is cheaper and does not increase emissions, but it requires design decisions that are locked in for decades and is difficult to retrofit into informal housing.

There is also a feedback loop. Air conditioning releases waste heat into the street, raising the outdoor temperature and increasing the demand for cooling in neighbouring buildings. The effect is small relative to the heat island but measurable in dense districts. The solution is not to discourage cooling but to combine it with efficiency standards, building codes, urban greening and protection for people who cannot pay — a package that treats cooling as infrastructure rather than as a consumer good.

4.9 Designing Cooler Cities

Urban heat can be reduced by design, and the interventions are well understood. Street trees are the most effective single measure: a mature canopy can lower surface temperature by 10 to 15°C and air temperature by 2 to 4°C within its shade. Reflective roofs and pavements raise albedo and reduce heat absorption. Green roofs and walls insulate buildings and cool the air through transpiration. Water features, parks and ventilation corridors allow cool air to move through the city.

The evidence base has improved. A study of 601 European cities estimated that increasing tree cover to 30 per cent could have prevented about 2,600 heat deaths a year, roughly a third of the total, and that the benefit was largest in southern and eastern Europe (Iungman et al., 2023). Modelling of US cities found that reflective roofs and street trees could reduce afternoon temperatures by 1.5 to 3°C in dense neighbourhoods. The interventions pay for themselves through reduced cooling demand, improved air quality and lower mortality.

The obstacle is not knowledge but implementation. Trees take decades to mature, compete for space with utilities and parking, and require maintenance that municipal budgets often lack. Reflective roofs need building codes that are enforced. Ventilation corridors need planning rules that resist the pressure to build on every open space. The cities that succeed tend to have a dedicated heat office, a long-term budget and a political constituency that treats heat as a public health issue rather than an inconvenience.

4.10 The Limits of Adaptation

Adaptation to heat has real limits. Once wet-bulb temperatures approach the physiological threshold, no amount of shading, cooling or behaviour change can make outdoor work safe, and even indoor spaces without mechanical cooling become lethal. The regions most exposed — the Indus Valley, the Persian Gulf, the North China Plain, parts of West Africa — are also regions with large populations, limited air conditioning and fragile grids. In those places, adaptation buys time but does not remove the hazard.

The second limit is economic. Cooling for everyone would require a massive expansion of electricity generation, much of which would need to be low-carbon to avoid worsening the problem it addresses. The International Energy Agency estimates that efficient cooling could halve the growth in demand relative to a business-as-usual path, but efficiency requires standards, financing and enforcement that many countries lack. The third limit is behavioural: people underestimate their own vulnerability, especially if they have survived previous heatwaves, and warnings that are repeated without visible consequences lose credibility.

These limits do not make adaptation pointless. They define the boundary between what can be managed and what must be prevented. A city that plants trees, insulates homes, protects outdoor workers and opens cooling centres can prevent most heat deaths at 1.5°C or 2°C of warming. At 3°C or 4°C, the same city will face conditions that no amount of planning can fully offset. That is why heat policy must be paired with emission reduction: adaptation and mitigation are not alternatives but two halves of the same decision.

4.11 Case Study: The European Summer of 2022

The summer of 2022 was the hottest in Europe since records began, and it was preceded by a spring drought that had already lowered rivers and dried soils. Temperatures exceeded 40°C in Britain for the first time, reached 47°C in Portugal, and persisted across the Mediterranean for weeks. Wildfires burned more than 800,000 hectares across the continent, the largest area in the European record, and the combined

effect of heat and drought reduced maize and sunflower harvests by double-digit percentages.

The mortality estimate published in Nature Medicine attributed about 61,700 deaths to the summer, with the highest numbers in Italy, Spain, Germany and France (Ballester et al., 2023). The figure was lower than the 70,000 of 2003 in absolute terms, but the summer was hotter, which suggests that adaptation had reduced the death rate per degree even as the absolute burden remained enormous. The authors noted that the estimate depended on the choice of temperature-mortality model and that the true figure could be higher.

The pattern of deaths revealed the limits of adaptation. Most deaths occurred among people over 75, many with cardiovascular or respiratory disease, and many in households without air conditioning. Southern European countries with older housing stock and less air conditioning had higher death rates than northern countries with cooler climates but better building standards. The lesson is that vulnerability is not simply a function of latitude: it is a function of housing, income, age structure and the presence of a plan.

The policy response varied. France, which had built a national heat plan after 2003, activated a colour-coded alert system, opened cooling centres and sent text messages to vulnerable households. Spain and Italy introduced restrictions on outdoor work during the hottest hours. Britain, which had no comparable plan, declared a national emergency for the first time and cancelled appointments at hospitals that were already under pressure. The differences in outcome are difficult to isolate because many factors changed at once, but the comparison suggests that plans reduce deaths when they are specific, funded and rehearsed.

The summer also exposed the interaction between heat and other systems. Railways buckled in Britain, nuclear plants in France reduced output because river water was too warm for cooling, and hospitals cancelled surgeries because operating theatres could not be kept cool. These failures were not caused by the temperature itself but by infrastructure designed for a cooler climate. They illustrate the central point of this chapter: heat becomes a disaster when it encounters systems that were not built to tolerate it.

For planners, the case suggests three priorities. The first is to treat heat as a health emergency with a dedicated budget, not as a weather inconvenience. The second is to identify the people at risk before the heatwave, through registries of elderly and isolated residents, rather than trying to find them during it. The third is to protect the infrastructure that heat disables — power, water, transport, hospitals — because a heatwave that shuts down the grid becomes a different and much larger emergency.

4.12 Future Scenarios and Predictions: Heat in a Warmer World

The historical record shows a clear trend. The number of days with extreme heat has increased on every continent, the frequency of heatwaves has risen by a factor of roughly three since the 1950s, and the hottest year on record has been exceeded repeatedly in the past decade. Heat mortality has risen with it: the Lancet Countdown estimates that heat deaths among people over 65 have increased by about 70 per cent

since the 1990s, and that the trend is accelerating in Europe and South Asia.

Present conditions vary enormously by place. In Delhi, the summer of 2024 produced weeks of temperatures above 45°C and a record electricity demand; in Jakarta, the combination of high humidity and a strong urban heat island produces wet-bulb conditions that are already dangerous for outdoor workers; in Karachi, the 2015 heatwave killed more than 1,200 people in a city where many neighbourhoods lack reliable water and power. These are not future scenarios. They are the baseline from which the projections begin.

The assumptions that drive the projections are therefore about adaptation as much as about emissions. If cities invest in shade, cooling, warning systems and housing, mortality can fall even as temperatures rise. If they do not, mortality will rise faster than the temperature because exposure is growing with urbanisation and ageing. The scenarios below combine the two: they show what happens to heat risk under different emission paths and different levels of adaptive investment.

Table 4.1 Heat exposure in selected cities, present and 2050

City	Days above 35°C now	Days above 35°C in 2050 (SSP2-4.5)	Wet-bulb risk	Critical adaptation
Delhi	~100	~140	High in monsoon	Cooling centres; outdoor work bans
Karachi	~60	~100	Very high	Water supply; power reliability
Jakarta	~30	~70	Very high	Shade; drainage; social housing
Lagos	~20	~60	High	Grid; informal settlement upgrading
Athens	~30	~55	Moderate	Building codes; tree canopy
Phoenix	~110	~150	Low but extreme	Cooling for homeless; grid capacity

Two conclusions follow. The first is that heat risk is not determined by temperature alone: the same number of hot days produces very different mortality depending on housing, water, power and social protection. The second is that the largest gains come from interventions that are boring and cheap — shade, water, warnings, check-ins on elderly neighbours — rather than from technologies that are expensive and unproven. A city that does the boring things well can save thousands of lives at 1.5°C and still save many at 2.5°C, though the margin narrows.

What would change these conclusions? A breakthrough in cheap, efficient cooling would reduce mortality in hot countries but would increase electricity demand and emissions unless it were paired with clean generation. A collapse of the power grid during a heatwave would turn a manageable event into a catastrophe, as happened in Texas in 2021 and in India in 2012. And a wet-bulb event above the physiological threshold would create a situation in which adaptation is impossible and only evacuation remains.

CHAPTER 5

WATER SCARCITY AND DROUGHT

Theme: water scarcity, drought and the politics of a finite resource

In June 2019, the reservoirs that supply Chennai, a city of about nine million people, fell to one per cent of their capacity. The city government declared a water emergency, trucks delivered water to neighbourhoods that had none, and trains carried water from the Cauvery river more than three hundred kilometres to the city. Offices asked employees to work from home. Restaurants closed. The monsoon arrived in October, and within weeks the reservoirs were full again — but the crisis had lasted four months, and it had been predicted for more than a decade.

Chennai is not an arid city. It receives about 1,400 millimetres of rain a year, more than London or Paris. The crisis was not caused by a lack of rainfall in the long term but by a sequence of weak monsoons, the loss of lakes and wetlands to construction, the depletion of groundwater, and a supply system that had not kept pace with growth. The city had built its prosperity on the assumption that water would always be available somewhere nearby, and the assumption failed.

The same assumption is failing across much of the world. Fresh water is a small fraction of the planet's water — about 2.5 per cent — and most of that is locked in glaciers or buried deep underground. The water that is accessible, in rivers, lakes and shallow aquifers, is renewed by rainfall and snowmelt at a rate that is essentially fixed. Humanity uses about half of that renewable supply, and in many regions it uses more, mining groundwater that took thousands of years to accumulate.

This chapter examines water scarcity as a physical, economic and political problem. It distinguishes between drought, which is a temporary shortage of rain, and scarcity, which is a persistent imbalance between demand and supply. It examines the agricultural, urban and industrial demands that drive the imbalance, and the adaptations — storage, efficiency, reuse, desalination, pricing and institutions — that can restore it. It closes with the question that will shape the next century: what happens when a shared river no longer carries enough water for the people who depend on it.

5.1 The Physics and Geography of Fresh Water

The global water cycle moves about 500,000 cubic kilometres of water through evaporation and precipitation each year, but only about 40,000 cubic kilometres falls on land, and only a fraction of that is accessible to humans at a reasonable cost. Rivers and lakes hold about 90,000 cubic kilometres; groundwater holds about 10 million cubic kilometres, but most of it is deep, saline or unrenewable. The renewable freshwater supply that humanity can use is therefore a thin and unevenly distributed layer of the planet's water.

Distribution is as important as quantity. Nine countries hold about half of the world's renewable freshwater, while forty countries have less than 1,000 cubic metres per person per year — the threshold commonly used to define water scarcity. The Middle East and North Africa are the most water-scarce regions, but scarcity is also severe in parts of India, Pakistan, Mexico and South Africa, where growing populations

and economies have outgrown the supply that rainfall provides.

Climate change alters the cycle in three ways. It increases evaporation as temperatures rise, which reduces the water available in soils and reservoirs. It changes precipitation patterns, making rain more concentrated in intense events and less reliable as a steady supply. And it reduces snowpack and glacier storage, which acts as a natural reservoir that releases water slowly through the dry season. The third effect is especially important in the Himalayas, the Andes and the Alps, where millions of people depend on meltwater.

These physical facts do not by themselves produce scarcity. Scarcity is produced by the interaction between supply and the institutions that allocate it. A country can have abundant water and still experience shortages if its infrastructure leaks, its agriculture is inefficient, its cities grow without planning, or its water rights are allocated to uses that no longer make sense. Understanding that distinction is the first step toward treating scarcity as a problem that can be governed rather than as a fate that must be endured.

5.2 Drought: Four Definitions

Drought is usually defined in four ways. Meteorological drought is a prolonged period of below-average rainfall. Agricultural drought is a shortage of soil moisture during the growing season, which can occur even when rainfall is normal if the rain falls at the wrong time or runs off compacted ground. Hydrological drought is a depletion of rivers, reservoirs and groundwater, which usually develops more slowly than the other two. Socioeconomic drought occurs when the shortage begins to affect the supply of goods and services, including food, energy and water.

The four definitions explain why drought is often declared late. A government may see normal rainfall and conclude that there is no drought, while farmers are already losing a season because the rain came in one week instead of eight. Hydrological drought may persist for years after the rain returns, because aquifers and reservoirs recover slowly. The 2022 Horn of Africa drought was declared after five consecutive failed rainy seasons, by which time more than twenty million people were acutely food insecure.

Drought also has a long tail. The economic losses extend beyond the harvest: livestock die, rural incomes collapse, migration accelerates, and children drop out of school. A study of the 2011–2017 drought in the Horn of Africa found that the effects on nutrition and education persisted for years after the rains returned. Drought is therefore not an event that ends when the rain begins; it is a shock whose consequences are transmitted through households and institutions.

5.3 Groundwater: The Invisible Reserve

Groundwater is the largest accessible store of fresh water on Earth and the source of drinking water for about half the world's population. It is also the most vulnerable to overuse, because it is invisible: a farmer who pumps a well cannot see the aquifer declining, and a city that licenses new wells cannot easily measure the cumulative effect. The result is a global pattern of depletion that has been called a

slow-moving crisis.

The Ogallala aquifer under the US Great Plains is the canonical example. It supplies water to about 30 per cent of US irrigated land, and parts of it have lost more than half their saturated thickness. In some areas of Kansas and Texas, the water table has fallen by more than 50 metres, and farmers have abandoned irrigation. The aquifer recharges at a rate of a few millimetres a year, while withdrawals are measured in metres, which means that the resource is being mined rather than used.

Similar patterns appear in the Indo-Gangetic Plain, the North China Plain, Mexico's central plateau, and the Arabian Peninsula. In Jakarta, groundwater extraction has caused land subsidence of up to 25 centimetres a year in some districts, which increases flood risk and has contributed to the decision to move the national capital. The common feature is that the water is being used faster than it is replenished, and that the users have no direct incentive to stop because the cost of depletion falls on someone else, often on future generations.

There are success stories. Arizona's Groundwater Management Act of 1980 required new developments to demonstrate a hundred-year water supply, and it slowed depletion in the most populated areas, though it did not eliminate it. Israel's water economy combines drip irrigation, recycled wastewater and desalination to support agriculture and a growing population with less water per capita than any other developed country. The lesson is that groundwater can be governed, but only with measurement, enforcement and a legal framework that treats the aquifer as a shared resource.

5.4 Rivers Under Stress

The great rivers of the world are the clearest visible measure of water stress. The Colorado River no longer reaches the sea in most years; its flow has declined by about 20 per cent since the early twentieth century, and the two largest reservoirs, Lake Mead and Lake Powell, fell below 30 per cent of capacity in 2022. The river supplies water to about 40 million people and 1.2 million hectares of farmland across seven US states and northern Mexico, under an allocation system designed in 1922 that assumed a flow larger than the river has delivered.

The Yellow River in China has been fully allocated since the 1990s and sometimes fails to reach the sea; the Indus supplies a basin that is home to more than 250 million people and is dependent on snow and glacier melt that is projected to decline; the Murray–Darling in Australia has experienced severe droughts and ecosystem collapse in its lower reaches. In each case, the river is both a water supply and an ecosystem, and the two uses compete for the same flow.

The politics of rivers are upstream-downstream politics. A dam built upstream reduces the flow available downstream; a city that withdraws water for its residents reduces the water available for farmers; a country that claims a share based on historical use may be claiming water that other countries believe is theirs. International river basins cover about half the land surface of the planet and are home to about 40 per cent of its population, which is why water is often described as a source of both cooperation and conflict.

5.5 Water Quality and the Hidden Scarcity

Water can be physically present and still unusable. Contamination by pathogens, nitrogen and phosphorus from agriculture, heavy metals, arsenic and industrial chemicals reduces the supply of safe water as effectively as a drought. The World Health Organization estimates that about 2.2 billion people lack access to safely managed drinking water, and that contaminated water is associated with hundreds of thousands of diarrhoeal deaths a year, most of them among children under five.

Arsenic is a particularly insidious example. Wells drilled in Bangladesh and West Bengal in the 1970s to provide microbiologically safe water inadvertently tapped aquifers contaminated with naturally occurring arsenic. Tens of millions of people were exposed, and the health consequences — skin lesions, cancers, cardiovascular disease — appeared decades later. The episode is a reminder that a technological solution can create a new problem while solving an old one, and that water quality must be monitored as carefully as water quantity.

Salinity is another hidden form of scarcity. In coastal aquifers, over-extraction draws seawater inland; in irrigated regions, rising water tables bring salts to the surface, where evaporation concentrates them. Salinisation has degraded hundreds of millions of hectares of farmland and continues to spread in the Indus basin, the Tigris–Euphrates plain and parts of Australia. Unlike a drought, salinity does not end when the rain returns; it can make land unusable for decades.

5.6 Agriculture: The Largest User

Agriculture accounts for about 70 per cent of global freshwater withdrawals, and in some developing countries the share is closer to 90 per cent. Irrigation underpins about 20 per cent of cultivated land but produces about 40 per cent of global food, which means that any constraint on irrigation has an outsized effect on food security. The efficiency of that use is often low: flood irrigation loses large amounts to evaporation and runoff, and poorly maintained canals leak before the water reaches the field.

The gains from improving efficiency are large but not automatic. Drip irrigation can reduce water use by 30 to 50 per cent compared with flood irrigation while maintaining or increasing yields, but it requires capital, maintenance and a reliable supply of clean water. In India, the world's largest user of groundwater, subsidised electricity for pumping has encouraged over-extraction, and the response has been a mix of subsidies for drip systems, restrictions on new wells and, in some states, direct payments to farmers who switch to less water-intensive crops.

The deeper problem is that water is usually priced below its cost. A farmer who pays nothing for water has no incentive to economise, and a city that charges less than the cost of supply cannot maintain its pipes. Pricing reform is politically difficult because water is a basic need and because farmers are organised. The alternatives — quotas, tradable rights, tiered tariffs that protect basic consumption while charging more for luxury use — are all attempts to impose a price signal without denying people the water they need to survive.

5.7 Urban Water Systems

Cities concentrate demand and lose water before it reaches the tap. Non-revenue water — the difference between what a utility produces and what it bills — averages about 30 per cent globally and exceeds 50 per cent in some cities. The losses come from leaking pipes, illegal connections, metering errors and unbilled public use. In a city with a shortage, repairing the distribution network can be cheaper than building a new reservoir.

Urban growth also consumes the landscapes that once stored water. Chennai's lakes and wetlands were built over as the city expanded; Jakarta's groundwater was pumped because the piped network did not reach most neighbourhoods; Mexico City's aquifers were drained faster than they could recharge. The result is that the city becomes dependent on distant sources, which increases cost, vulnerability and political tension with the regions that supply it.

The responses that work combine supply, demand and storage. Singapore's water strategy — local catchment, imported water, desalination and recycled water, known as the four national taps — is the most cited example, and it has allowed a city-state with no natural aquifer to achieve water security. Windhoek, Namibia, has been recycling wastewater directly into the drinking supply since 1968. These cases show that scarcity is not simply a function of rainfall; it is a function of how deliberately a city manages the cycle.

5.8 Drought, Food and Prices

Drought transmits to the economy through food prices. When a major producing region loses a harvest, global prices rise, and the effect is largest for poor households that spend a large share of their income on food. The 2010–2011 drought in Russia and the subsequent export ban contributed to the global food price spike that year; the 2022 drought in Europe reduced maize and sunflower output and added to the price pressures already caused by the war in Ukraine.

The transmission is not automatic, because trade, storage and substitution can absorb part of the shock. But those buffers are thinner than they appear. Global grain stocks are held by a small number of countries, and exporters sometimes restrict shipments in a crisis, which amplifies the price increase. Importing countries respond by buying more, which amplifies it further. The result is that a regional drought can become a global price event through the interaction of policy responses rather than through the physical shortage alone.

For households, the consequence is a change in diet. When prices rise, families substitute cheaper staples, reduce protein and fresh produce, and sometimes skip meals. The nutritional effect persists long after prices fall, particularly for children, whose growth and cognitive development can be permanently affected. Drought is therefore a food security problem as much as a water problem, and the two cannot be addressed separately.

5.9 Water Conflict and Cooperation

Water has been called a source of war, but the historical record is more complex. The Pacific Institute's Water Conflict Chronology records thousands of water-related tensions, most of them local, and far more instances of cooperation than of armed conflict over water. The Nile, the Indus, the Mekong and the Rhine all have long histories of negotiation, and some of the most durable treaties in the world are water treaties.

That does not mean water is not a source of conflict. It means that the conflict usually takes the form of displacement, dispossession and violence within countries rather than war between them. Dam construction displaces communities; irrigation withdrawals deprive downstream farmers; pollution destroys fisheries; water tariffs provoke protests. The risk of interstate war over water is real but rare, and it is highest where water scarcity interacts with existing political hostility and weak institutions.

The cooperative record offers lessons. The Indus Waters Treaty of 1960 survived three wars between India and Pakistan and has been remarkably resilient, because it created a joint commission with technical authority and a dispute-resolution mechanism. The Nile Basin Initiative, the Mekong River Commission and the Senegal River Organisation show that shared basins can be governed when the parties have an interest in the information and the rules that the institution provides. Water is more often a reason to talk than a reason to fight.

5.10 Adaptation: Storage, Efficiency and Reuse

The portfolio of water adaptations is well known. Storage — dams, reservoirs, aquifer recharge, rainwater harvesting — bridges the gap between wet and dry seasons. Efficiency — drip irrigation, leak repair, industrial recycling — produces more output per cubic metre. Reuse — treating wastewater for irrigation or industry — turns a waste stream into a resource. Desalination — removing salt from seawater or brackish groundwater — creates new supply at a cost in energy and money.

Each option has limits. Dams displace people and silt up; the world has built most of the good sites. Efficiency can be offset by rebound effects, in which farmers use the saved water to expand irrigated area. Desalination is expensive, energy-intensive and produces brine that must be disposed of carefully; it is affordable for wealthy coastal cities but not for inland agriculture. Reuse requires treatment, monitoring and public acceptance, which is often the hardest part.

The most important adaptation is institutional. Water is managed through rights, prices, quotas and basin agreements, and the difference between a well-managed basin and a poorly managed one is often larger than the difference between a wet year and a dry one. Australia's Murray–Darling Basin Plan, which caps total withdrawals and allows water to be traded, is an example of an attempt to manage a basin as a system rather than as a collection of individual claims. It has been criticised for the environmental outcomes it has not delivered, but it demonstrates that the institutional design is at least as important as the engineering.

5.11 Case Study: Chennai's Day Zero

Chennai's water crisis did not arrive without warning. The city depends on four reservoirs and on groundwater, and the reservoirs had been declining for years as urban expansion covered the lakes that once captured monsoon rain. In 2018 the northeast monsoon failed, and by June 2019 the reservoirs held about one per cent of their capacity. The city government declared a water emergency and began to truck water to neighbourhoods, while the state arranged for trains to carry water from the Cauvery.

The crisis affected different residents differently. Households with private wells and storage tanks managed; households in informal settlements, which depended on public taps and tanker deliveries, queued for hours. Schools closed, restaurants reduced service, and information technology companies asked staff to work from home. The economic cost was estimated in the hundreds of millions of dollars, and the political cost was higher: the crisis became a symbol of the failure of urban water management in a rapidly growing city.

The response included short-term measures — tankers, trains, restrictions on non-essential use — and longer-term investments in desalination, sewage treatment and rainwater harvesting. The city also began to restore some of the lakes and wetlands that had been built over, recognising that the cheapest storage was the landscape itself. By 2024, the reservoirs had recovered, but the underlying imbalance had not been resolved: the city still depends on a monsoon that is becoming less reliable and on groundwater that is being depleted.

The case illustrates a pattern that appears in cities around the world. Water crises are rarely caused by a single year of low rainfall; they are caused by the accumulation of decisions that reduced storage, increased demand and deferred maintenance. The crisis is the moment when the accumulated deficit becomes visible. Because the deficit is institutional, the solution must be institutional: a plan that combines supply, demand, storage and governance, and that is implemented before the reservoirs empty.

For other cities, the lessons are specific. First, measure and publicly report the water balance, including losses in the distribution network and the rate of groundwater depletion. Second, protect the landscapes that store water — lakes, wetlands, floodplains and recharge areas — before they are built over. Third, diversify supply, including reuse and desalination where affordable, and price water in a way that funds maintenance. Fourth, identify the households that will be hit first and build a delivery system that reaches them, because the crisis will expose the difference between a city that has water and a city that has access to it.

5.12 Future Scenarios and Predictions: Water in 2050

The historical data show a consistent trend. Global water withdrawals have roughly tripled since 1950, driven by population growth, irrigation and industrial demand. Groundwater depletion has accelerated in the past two decades, and the area of land affected by drought has increased. The World Bank estimates that water scarcity could reduce GDP growth in some regions by up to 6 per cent by 2050, and the United Nations Convention to Combat Desertification estimates that drought could displace

700 million people by 2030.

Present conditions give the baseline. About 4 billion people experience severe water scarcity for at least one month a year, and 2.2 billion lack safely managed drinking water. Agriculture uses about 70 per cent of withdrawals, and in many basins the allocation of water exceeds the renewable supply. The gap is filled by groundwater mining, which cannot continue indefinitely. The question is not whether the deficit will be closed but how: by efficiency, by reallocation, by new supply, or by crisis.

The assumptions that drive the projections are population, diet, technology and governance. A world that converges toward efficient irrigation, recycles more water and limits groundwater withdrawal can meet demand with less water than it uses today. A world that continues current trends will face a shortfall of about 40 per cent between available supply and projected demand by 2030, according to the 2030 Water Resources Group. The scenarios below show what the main basins might look like under different assumptions.

Table 5.1 Water stress in major basins, present and 2050

Basin	Population (million)	Current stress	2050 gap (SSP2-4.5)	Main adaptation
Indus	~300	Extreme	30–40%	Drip irrigation; groundwater regulation
Colorado	~40	Extreme	20–30%	Demand caps; reuse; fallowing
Nile	~280	High	25–35%	Storage; efficiency; basin agreement
Citarum (Java)	~25	Very high	20–30%	Wastewater treatment; reforestation
Murray–Darling	~2.4	High	10–20%	Water trading; environmental flows
Tigris–Euphrates	~60	Very high	30–40%	Dam coordination; irrigation reform

Two conclusions follow. The first is that the gap is largest in basins where agriculture dominates and groundwater is unregulated, which means that the most important interventions are agricultural rather than urban. The second is that the gap can be closed with existing technology; the obstacle is the political cost of reallocating water from one user to another. A basin that caps withdrawals will produce less in the short term, and the farmers who lose water will resist.

What would change these conclusions? A breakthrough in cheap desalination would expand supply for coastal cities but not for inland agriculture. A change in diet toward less water-intensive food would reduce demand substantially, since meat and dairy require far more water per calorie than grains. And a sustained increase in precipitation in some basins, which some climate projections suggest, would relieve pressure in those places — but the same projections show declines in others, and the uncertainty is largest where the stakes are highest.

CHAPTER 6

FLOODS AND SEA LEVEL RISE

Theme: floods, coastal risk and the slow emergency of rising seas

On 29 October 2024, a line of thunderstorms stalled over the province of Valencia in eastern Spain and released more rain in eight hours than the region normally receives in a year. The Turia and Magro river systems filled within minutes. Cars were swept out of underpasses, bridges collapsed, and more than two hundred people died, most of them in towns that had been built on floodplains but had not experienced a flood of that magnitude in living memory. The regional government's alert reached mobile phones after the water had already risen.

Floods are the most frequent and most costly of climate disasters. Between 1995 and 2015, floods accounted for about 47 per cent of all weather-related disasters and affected 2.3 billion people, according to the United Nations Office for Disaster Risk Reduction. They kill in two ways: by drowning and trauma during the event, and by disease, displacement and loss of livelihood in the months that follow. The World Health Organization estimates that floods affect more people than any other natural hazard.

The physical mechanism is simple and getting stronger. Warmer air holds more water vapour, about seven per cent more for each degree of warming, following the Clausius–Clapeyron relation. When that moisture condenses, it falls in heavier bursts. At the same time, land surfaces have been made less absorbent by compaction, paving and the loss of wetlands, so a larger fraction of the rain becomes runoff. The combination means that a storm of the same meteorological magnitude delivers more water to rivers and streets than it did a century ago.

Sea level rise adds a second, slower dimension. The ocean has risen about 20 to 25 centimetres since 1880, and the rate has increased to about 4.5 millimetres a year. That sounds small until it is combined with storm surge, high tide and subsidence: a city that has sunk 20 centimetres and faces a sea 20 centimetres higher experiences a 40-centimetre change in the effective height of its defences. This chapter examines both dimensions, because in most coastal cities they arrive together.

6.1 The Physics of Flooding

Flooding occurs when the volume of water arriving at a place exceeds the capacity of the channel, the soil or the drainage system to carry it away. The volume is determined by rainfall intensity, the area that drains into the channel, the moisture already in the soil, and the speed with which water moves across the surface. A catchment that is dry and vegetated can absorb a storm that would overwhelm the same catchment after a wet month or after the vegetation has been removed.

The intensification of rainfall is one of the most robust findings in climate science. Observations show that extreme precipitation has increased in most regions, and that the increase is consistent with the Clausius–Clapeyron relationship. A warmer atmosphere can hold more water, and when the conditions for condensation are met, more of it falls at once. The effect is not uniform: some regions are becoming drier, and

in those regions the same mechanism can produce shorter, more violent storms separated by longer droughts.

Land use determines how much of that rain becomes flood. Urbanisation typically increases peak runoff by a factor of two to five, because pavement and roofs replace soil and vegetation. Deforestation in upland catchments increases erosion and reduces infiltration. Drainage systems designed for the rainfall of 1970 cannot carry the rainfall of 2024. The result is that floods are produced by the interaction of the atmosphere and the landscape, and that engineering a channel without managing the landscape upstream often moves the problem downstream.

6.2 River Floods and the Great Deltas

The largest concentration of flood risk is in the deltas of Asia. The Ganges–Brahmaputra, the Mekong, the Irrawaddy, the Indus and the Yangtze carry enormous volumes of water and sediment into low-lying plains that are home to hundreds of millions of people. The deltas are fertile because rivers deposit sediment there; they are vulnerable because the same sediment is subsiding, and because the channels that carry the water are constrained by embankments that prevent the floodplain from absorbing the flood.

The Mekong Delta is the clearest example. It supports about 17 million people and produces more than half of Vietnam's rice, but it is sinking as groundwater is pumped and as sediment is trapped behind dams upstream. Saltwater intrusion has advanced tens of kilometres inland during dry seasons, damaging rice fields and forcing farmers to switch to shrimp or to leave. The combination of subsidence, reduced sediment and sea level rise means that the delta is losing height relative to the sea even while the sea itself is rising.

The Indus delta illustrates a different mechanism. Upstream dams and barrages have reduced the flow reaching the delta, which has caused seawater to intrude and mangroves to retreat. The 2022 floods demonstrated that the river still carries enormous volumes in extreme years, but the infrastructure built to control those volumes — embankments, canals, barrages — had been designed for a different pattern of flow and failed in ways that concentrated the damage in particular districts. The lesson is that flood control infrastructure is only as good as the assumptions about flow on which it was designed.

6.3 Coastal Flooding and Sea Level Rise

Coastal flooding has three components: mean sea level, tides and storm surge. Mean sea level is rising because ocean water expands as it warms and because land ice is melting. Tides are predictable, which is why the timing of a storm matters: a surge that arrives at high tide can overwhelm defences that would hold at low tide. Storm surge is the temporary rise in sea level caused by wind and low pressure, and it is the component that does most of the damage in tropical cyclones.

Subsidence is the fourth component and often the largest. Many coastal cities are built on deltas or on reclaimed land, and they sink as groundwater is extracted and as the sediment beneath them compacts. Jakarta has subsided by up to four metres in some

districts over the past three decades; Bangkok, Manila, Ho Chi Minh City and Shanghai have all subsided by tens of centimetres. In these cities, the effective rate of sea level rise is the global rate plus the local rate of sinking, which can be ten times larger.

The consequences are already visible in the frequency of nuisance flooding. Miami Beach now experiences tidal flooding on sunny days several times a year, a phenomenon that barely occurred a generation ago. Jakarta experiences routine flooding during the monsoon and during high tides, even without rain. These events are not catastrophic individually, but they impose costs on transport, sanitation and property values, and they signal that the defences are being overtaken by the water.

6.4 Jakarta: The Sinking Capital

Jakarta is the most studied example of a city losing its race with the sea. About 40 per cent of the city lies below sea level, and some northern districts have subsided by more than four metres since the 1970s. The subsidence is caused mainly by groundwater extraction: only about half the population is connected to the piped water network, and the rest pumps from aquifers. The city has 13 rivers and a network of canals, and it receives about 1,700 millimetres of rain a year, concentrated in a wet season that regularly produces floods.

The 2020 New Year flood displaced about 400,000 people and killed more than 60. The 2013 and 2007 floods were comparable or worse. The city has responded with a sea wall, pumping stations, dredging and a giant offshore dyke project, but the project has been delayed by land acquisition, funding and disputes about its design. In 2022, the government passed a law to move the national capital to Nusantara in East Kalimantan, citing congestion, inequality and environmental vulnerability, including flooding.

The capital move does not solve Jakarta's problem; it relocates the state's administrative functions while leaving about 30 million people in the metropolitan area. The lesson is not that relocation is wrong but that it is not a substitute for managing subsidence. Stopping groundwater extraction, providing piped water to the whole city, restoring natural drainage and building defences are all necessary, and all are slow, expensive and politically difficult. Jakarta is therefore a test case for whether a large city can adapt faster than the ground beneath it sinks.

6.5 Flash Floods and Urban Drainage

Flash floods are the most lethal type of flood because they arrive with little warning. They occur when intense rainfall falls on steep terrain or on urban surfaces that cannot absorb it, and the water can rise within minutes. The Valencia flood of 2024, the Kuala Lumpur flash floods of 2021 and the Seoul floods of 2022 all followed this pattern: a convective storm, a drainage system designed for a milder climate, and a population that did not know which roads had become rivers.

Urban drainage design uses return periods: a pipe or canal is sized for a storm expected once every 5, 10 or 50 years. The assumption is that the climate is stationary and that the return period is stable. Both assumptions have failed. A system designed for a 10-year storm in 1990 may now experience that storm every three or four years, and the design storm itself has become more intense. The result is that cities are being

flooded by storms that their engineers reasonably expected them to survive.

The responses are known but slow. Separating stormwater from sewage reduces contamination and increases capacity. Green infrastructure — permeable pavements, rain gardens, retention ponds, constructed wetlands — absorbs and slows runoff. Zoning that keeps floodplains free of construction reduces exposure. Early warning systems that target drivers and pedestrians, rather than only river levels, save lives in flash floods. The common obstacle is that these measures require coordination across agencies and budgets that are usually organised by sector.

6.6 Who Dies in Floods

Flood mortality is concentrated in a few well-understood groups. People who are poor, elderly, disabled or living in informal settlements are at highest risk; so are people who attempt to cross flooded roads, which is the leading cause of death in flash floods in high-income countries. In low-income countries, drowning in homes and on agricultural land is more common, and women are often at higher risk because of restrictions on mobility and because they are less likely to have been taught to swim.

The 1970 Bhola cyclone in Bangladesh killed an estimated 300,000 to 500,000 people, making it one of the deadliest disasters in history. The 1991 cyclone killed about 138,000. In both cases, the deaths were concentrated among the poor who lived on low-lying islands and had no shelters. The subsequent investment in cyclone shelters, early warning and volunteer networks reduced deaths in later storms by orders of magnitude, which demonstrates that flood mortality is a function of preparedness rather than of the hazard alone.

The indirect mortality is larger and less visible. Floods contaminate water supplies, destroy crops, disrupt health services and displace populations, and these effects produce deaths from diarrhoeal disease, malnutrition and untreated chronic illness in the weeks and months after the water recedes. The 2010 Pakistan floods, which affected about 20 million people, were followed by a surge in waterborne disease and child malnutrition. Counting only the bodies found during the flood therefore understates the human cost.

6.7 Levees, Dams and Polders

Engineering has protected many cities from flooding, and it has also created new risk. Levees and embankments keep water out of the protected area, but they raise the water level in the channel and increase the consequences if they fail. The Netherlands has built the most sophisticated flood defence system in the world — the Delta Works, a network of dams, storm surge barriers, dikes and pumps that protects about a quarter of the country below sea level — and it maintains that system with a national programme and a statutory budget.

New Orleans is the counter-example. The city's levees failed during Hurricane Katrina in 2005, killing about 1,800 people and causing more than US$125 billion in damage. Investigations found that the failure was not caused by the storm exceeding the design height but by defective construction, inadequate maintenance and a design that had not accounted for the subsidence of the land and the loss of coastal wetlands that

had once absorbed surge. The lesson is that a defence is a system, and that its reliability depends on maintenance and on the condition of the landscape around it.

Polders — areas enclosed by dikes and drained by pumps — are a model for coastal cities with the resources to maintain them. They require continuous pumping, regular inspection and a political commitment that outlasts electoral cycles. Tokyo's underground reservoirs, Rotterdam's water plazas and Singapore's ABC waterways show a different approach: instead of keeping water out, they create space for it to go during extreme events. The choice between resistance and accommodation depends on the local geography and on how much risk a society is willing to accept.

6.8 Insurance, Retreat and Managed Realignment

Insurance is the financial technology that allows a society to absorb flood losses, and it is retreating from the most exposed places. In the United States, the National Flood Insurance Program has been insolvent for years and has been forced to raise premiums, which has made some properties unaffordable. Private insurers have withdrawn from parts of California and Florida, where wildfire and hurricane risk have increased faster than premiums could adjust.

When insurance becomes unavailable, the market begins to price the risk, and property values fall. That process is already visible in some coastal communities, where homes are sold at a discount or abandoned. The alternative to a disorderly retreat is a managed one: buyouts of the most exposed properties, restoration of floodplains, and relocation assistance for households that cannot afford to move on their own. The Netherlands has used managed realignment for decades, returning some polders to the rivers to reduce flood levels elsewhere.

Retreat is politically difficult because it is seen as giving up. But in many places it is the only option that is economically rational, and delaying it increases the eventual cost. The communities that have managed retreat best are those that began early, compensated households fairly, and combined relocation with investment in the places people moved to. Those that have managed it worst waited until a disaster forced the issue, which is the most expensive and least equitable way to adapt.

6.9 Ecosystem-Based Protection

Natural systems protect coasts at a fraction of the cost of engineering. Mangrove forests reduce wave height and attenuate storm surge; coral reefs break waves before they reach the shore; sand dunes and beaches absorb energy; wetlands store floodwater and release it slowly. A study of 59 coastal defence projects found that mangrove belts can reduce wave heights by 60 to 70 per cent over a few hundred metres, and that the benefit is greatest for small storms, which are the most frequent.

The protection is not absolute. Mangroves cannot stop a tsunami or a category 5 storm surge, and they require space and suitable conditions to survive. Coral reefs are themselves dying from warming and acidification, so the protection they provide is declining at the moment it is most needed. Dunes migrate and require space to migrate; seawalls prevent that migration and cause the beach to erode. Ecosystem-based protection works best as part of a layered strategy, not as a replacement for all

engineering.

The economics are favourable. Restoring mangroves is cheaper than building seawalls, and it provides additional benefits: fisheries nursery habitat, carbon storage, timber and tourism. Indonesia has the largest mangrove area in the world and has committed to restoring hundreds of thousands of hectares, though the record of restoration projects has been mixed because planting has sometimes occurred in places where mangroves cannot survive. The lesson is that ecosystem-based adaptation requires the same rigour as engineering: site selection, monitoring and maintenance.

6.10 Warning, Evacuation and Recovery

Most flood deaths are preventable with warning and evacuation, and the countries that have invested in both have seen mortality fall dramatically. Bangladesh's cyclone preparedness programme, built after the disasters of 1970 and 1991, combines forecasts, shelters, volunteers and drills, and it has reduced deaths in comparable storms from hundreds of thousands to dozens. The programme works because it is embedded in communities, not because it relies on technology alone.

Effective warning requires four elements: a forecast that is accurate enough to justify action; a message that reaches the people at risk in a language and format they understand; a population that knows what to do and has somewhere to go; and a decision-making system that is willing to act before the event is certain. The failure of any one element can turn a warning into a false alarm or a missed evacuation. The Valencia flood of 2024 failed at the second and third elements: the warning arrived late and the population did not know how quickly the water would rise.

Recovery determines whether a flood becomes a long-term disaster. Households that receive timely compensation, temporary housing and support to rebuild recover faster; those that do not may remain displaced for years. The 2004 Indian Ocean tsunami and the 2011 Japan earthquake both showed that recovery is fastest when reconstruction is combined with livelihood support and when local institutions are strengthened rather than bypassed. Floods are events; recovery is a process, and the process determines the final cost.

6.11 Case Study: Jakarta and the Sea

Jakarta's flood problem is the product of three histories. The first is geological: the city sits on a delta of thirteen rivers, built on soft sediment that compacts under the weight of construction. The second is hydraulic: colonial and post-colonial engineers built canals to drain the city, but the canals were designed for a smaller population and a cooler climate. The third is political: the city grew rapidly in the second half of the twentieth century, and much of that growth occurred in areas that were already floodplains or that were reclaimed from swamps.

The result is a city that sinks. Groundwater extraction, driven by the absence of a reliable piped supply, has lowered the water table and compacted the aquifers. In northern Jakarta, land has subsided by up to four metres since the 1970s, and the subsidence continues at rates of 1 to 15 centimetres a year. Because the sea is also rising, the relative rate of sea level rise in the city is several times the global average.

The 2020 New Year flood, the 2013 flood and the 2007 flood each displaced hundreds of thousands of people.

The response has been a combination of engineering and retreat. The city has built pumping stations, dredged canals, raised embankments and begun a sea wall. A giant offshore dyke, the National Capital Integrated Coastal Development project, was designed to enclose the bay and create a reservoir for water supply, but it has been delayed by land acquisition, financing and concerns about its environmental impact on fishing communities. In 2022, the government began moving the capital to Nusantara, citing Jakarta's congestion, inequality and environmental vulnerability.

The capital move is the largest planned relocation in Indonesia's history, and it is not a solution to Jakarta's flooding. It moves the administrative apparatus, not the 30 million people in the metropolitan area, and it may reduce the political attention that the city needs. The government has paired the move with investment in Jakarta's water supply and flood defences, but the scale of the investment is uncertain and the subsidence continues.

The case illustrates a general problem: cities can adapt to sea level rise if they have resources, institutions and time, but the three are rarely available together. Jakarta has resources, but its institutions are fragmented between the national government, the provincial government and the surrounding municipalities. It has time, but less than it thinks, because subsidence is faster than the global rate. And it has a plan, but the plan requires a level of coordination that has not yet been achieved.

For other coastal cities, the lessons are clear. Stop groundwater extraction before the land sinks; provide piped water to everyone, because otherwise people will pump. Protect and restore mangroves and wetlands, which cost less than seawalls and provide additional benefits. Build defences in stages that can be raised as the sea rises. And plan for retreat from the most exposed areas, because some of them cannot be defended at any acceptable cost.

6.12 Future Scenarios and Predictions: Coasts in 2100

The historical record is unambiguous. Global mean sea level has risen by about 20 to 25 centimetres since 1880, and the rate has accelerated from about 1.4 millimetres per year in the early twentieth century to about 4.5 millimetres per year today. The frequency of coastal flooding has increased even where the sea level rise is modest, because flooding depends on the combination of sea level, tide, surge and waves, and because subsidence and coastal development have increased exposure.

Present conditions show the scale of the exposure. About 600 million people live in low-elevation coastal zones less than 10 metres above sea level, and about 1 billion live within 100 kilometres of a coast. The value of assets in these zones is enormous: in the United States alone, coastal property worth trillions of dollars is exposed to storm surge. In developing countries, the exposure is concentrated in informal settlements that have no insurance and little public protection.

The assumptions that drive the projections are the emission path, the response of ice sheets, and the rate of subsidence. Under the low-emission scenario, sea level is projected to rise by about 0.3 to 0.4 metres by 2100; under the high-emission scenario,

by 0.6 to 0.8 metres; and if the West Antarctic ice sheet becomes unstable, by more than a metre. The difference matters less for the global average than for the frequency of extreme events: a 0.5-metre rise can turn a once-in-a-century flood into a once-in-a-decade flood, and in some places into an annual one.

Table 6.1 Coastal exposure in selected cities

City	Subsidence (cm/yr)	People exposed to 1-in-100 flood	2050 sea level (SSP2-4.5)	Adaptation
Jakarta	1–15	~2.5 million	+0.15 m	Sea wall; water supply; relocation
Manila	1–5	~1.5 million	+0.17 m	Drainage; mangrove restoration
Bangkok	1–3	~1 million	+0.16 m	Polders; groundwater control
Ho Chi Minh City	1–3	~1 million	+0.16 m	Embankments; delta management
Miami	0.2–0.5	~0.4 million	+0.18 m	Pumps; elevation; managed retreat
Alexandria	0.3–0.6	~0.5 million	+0.17 m	Sea wall; beach nourishment

Two conclusions follow. First, the dominant variable in many cities is not global sea level rise but local subsidence, which is caused by human decisions and can be slowed or stopped. Second, the cost of adaptation rises faster than the sea, because each increment of rise requires a larger and more expensive defence, and because the probability of failure increases as the design threshold is approached.

What would change these conclusions? A collapse of the West Antarctic ice sheet would make the high-end projections much more likely and would shift the policy focus from protection to retreat. A breakthrough in cheap, large-scale pumping or in floating construction could allow some cities to remain in place longer. And a reduction in emissions would slow the rise, but it would not stop it, because the ocean and the ice sheets respond over centuries. For coasts, mitigation determines the speed of the emergency; adaptation determines whether the city survives it.

CHAPTER 7

FOOD SECURITY AND AGRICULTURE

Theme: food security, crop physiology and the fragility of the global harvest

In May 2022, the government of India banned wheat exports with a single notification. The trigger was a heatwave in March that had arrived earlier and lasted longer than any recorded spring heat in the wheat belt, and had cut the harvest by an estimated five per cent. The ban was small in global terms — India accounts for a modest share of world wheat trade — but it arrived when the war in Ukraine had already removed a large part of the Black Sea supply, and prices that had been rising for a year rose further.

In July 2023, India restricted rice exports, and the effect was larger. India accounts for about forty per cent of the global rice trade, and the restriction, combined with an El Niño that was reducing production in other Asian countries, pushed rice prices to their highest level since 2008. Panic buying in the Philippines, Indonesia and West Africa amplified the increase. The rice was not physically scarce in most importing countries; the expectation of scarcity was enough to empty shelves.

These episodes show how food security fails. It is not only a matter of how much the world grows; it is a matter of how the harvest is distributed, stored, traded and priced. A heatwave in one region becomes a price shock in another through export restrictions, speculative buying and the thinness of global stocks. The poorest households, which spend half their income on food, absorb the shock first, and they absorb it by eating less and eating worse.

This chapter examines the physical effects of climate on crops, livestock and fisheries, and the economic and political mechanisms that turn those effects into hunger. It pays particular attention to the crops that feed most of humanity — wheat, maize and rice — and to the smallholders who grow much of the food consumed in low-income countries. It closes with the adaptations that are known to work and the reasons they are not spreading fast enough.

7.1 How Heat Affects Crops

Plants grow within a narrow temperature range, and the range differs by crop and by stage of development. Wheat and rice perform best around 20 to 25°C; maize around 25 to 30°C. Above the optimum, photosynthesis slows, respiration increases, and the plant partitions less energy into grain. The effect is not linear: a few days above 35°C during flowering can reduce grain set by a large fraction, even if the rest of the season is favourable.

Night temperature matters as much as daytime heat. Cool nights allow the plant to recover from the day's water loss and to complete the metabolic processes that fill grain. When nights remain above 25°C, respiration continues through the night, consuming the carbohydrates that would otherwise become yield. Studies in the Indo-Gangetic Plain have found that night temperature is a stronger predictor of rice yield than daytime maximum, which is one reason climate projections that report only daytime averages underestimate the risk.

The combined effect of warming and CO2 is complex. Higher CO2 increases photosynthesis in C3 crops such as wheat, rice and soy, an effect known as CO2 fertilisation, which can offset part of the heat penalty. But the effect is smaller in field conditions than in greenhouses, because nutrients and water become limiting, and it reduces the concentration of protein, zinc and iron in the grain. A crop that yields slightly more but is less nutritious is not necessarily an improvement for the people who eat it.

7.2 Water, Soil and the Limits of Irrigation

Agriculture consumes about 70 per cent of global freshwater withdrawals, and irrigation is the largest single use. In regions where rainfall is unreliable, irrigation is what makes the difference between one harvest and two, or between survival and failure. But irrigation depends on water that is increasingly contested: aquifers are being depleted, rivers are fully allocated, and snowmelt that feeds many systems is declining.

Soil is the other invisible input. A healthy soil holds water, cycles nutrients and supports the microbial communities that make them available to plants. Erosion, compaction and the loss of organic matter reduce that capacity, which makes crops more vulnerable to drought even when rainfall is unchanged. The global loss of topsoil is estimated at 24 billion tonnes a year, and the degradation of arable land is one of the reasons yields have stagnated in some regions despite improved varieties.

These constraints interact. A field with depleted soil and no irrigation cannot benefit from a heat-tolerant variety; a farmer who cannot afford fertiliser cannot compensate for nutrient dilution by adding more. The result is that the same climate shock produces very different losses on different farms, and that adaptation must address soil, water and seeds together rather than one at a time.

7.3 Pests, Pathogens and Pollinators

Warmer temperatures expand the range of many agricultural pests and change their life cycles. The fall armyworm, a moth native to the Americas, spread across Africa and then Asia in the 2010s and has become a major pest of maize; its expansion was aided by trade and by warmer winters that allowed it to survive in new regions. Aphids, whiteflies and locusts respond to temperature and rainfall in ways that are difficult to predict but increasingly favourable to outbreaks.

Plant diseases follow the same pattern. Fungal pathogens that were once limited to warm regions are moving poleward and upward; coffee leaf rust, which devastated Central American plantations in 2012–2013, was associated with higher temperatures and more variable rainfall. The response is often more pesticide, which raises costs and creates resistance, or the abandonment of susceptible crops, which reduces diversity.

Pollinators are a separate vulnerability. About three-quarters of the world's leading food crops depend at least partly on animal pollination, and pollinators are themselves sensitive to heat, drought and the loss of habitat. Studies in tropical regions have found that bee diversity declines with warming, and that the decline reduces yields of crops such as coffee, cocoa and many fruits and vegetables. The effect is easy to overlook because it is gradual, but it accumulates in the same way that soil loss does.

7.4 Extreme Events and Harvest Failure

Most of the year-to-year variability in global production comes from extreme events rather than from gradual warming. A drought during the grain-filling period, a flood at planting, a heatwave during flowering or a frost at the wrong moment can remove a large share of a national harvest. Because global stocks are thin relative to consumption, a series of such events in different regions can produce a global crisis.

The 2007–2008 food price crisis is the classic example. Drought in Australia, poor harvests in Europe and the United States, and the diversion of maize to biofuel combined to draw down stocks, and when major exporters restricted shipments, prices doubled. The result was riots in more than thirty countries and a reversal of poverty reduction in many others. The physical shortfall was modest; the policy response turned it into a catastrophe.

The 2022–2023 period repeated the pattern with different triggers. Heat in India, drought in the Horn of Africa, the war in Ukraine and an El Niño that was forecast to reduce rice output in Asia combined to raise prices and reduce availability in importing countries. Each event was within the range of natural variability; together they exceeded the capacity of the trade system to buffer, which is exactly what climate change is expected to make more frequent.

7.5 Livestock and Pastoralism

Livestock keep about 1.3 billion people employed and provide protein, traction and savings for pastoral communities. Heat affects animals directly: dairy cows reduce milk production when the temperature-humidity index exceeds about 72, poultry suffer mortality in hot weather, and grazing animals lose condition when rangelands dry out. The 2022 drought in the Horn of Africa killed millions of animals, destroying the savings of pastoral households.

Pastoralism is a system adapted to variability. Herders move with the rains, keep mixed herds of different species, and use social networks to share animals and information. Climate change stresses that system by making rainfall less predictable, reducing the number of good seasons, and increasing the frequency of droughts that occur before herds have recovered from the previous one. The result is not a single disaster but a slow erosion of the asset base.

The livestock sector is also a major source of emissions, contributing about 12 per cent of global greenhouse gases through enteric fermentation, feed production and land-use change. That creates a tension in policy: reducing livestock numbers would reduce emissions and free land for other uses, but it would also affect the livelihoods and nutrition of the poorest households. The solution is not a single global target but a set of context-specific changes: better feed, better breeds, better grazing management and a shift in high-income countries toward less meat-intensive diets.

7.6 Fisheries and Aquaculture

Fisheries provide about 17 per cent of the animal protein consumed globally, and more than three billion people depend on fish as a significant source of protein and micronutrients. Climate change affects fish through warming, acidification, oxygen loss and the movement of species toward cooler water. The result is that stocks are shifting away from the tropical regions where dependence is highest, and toward the poles where fishing fleets are less able to follow.

Coral reefs are the most vulnerable fishery habitat. Reef fish depend on the structure and productivity of live coral, and when reefs bleach and die, the fish community changes: herbivorous species take over, larger predatory fish decline, and catch rates fall. In Southeast Asia and the Pacific, where reef fisheries support millions of coastal households, the loss is both nutritional and economic. Warming of 2°C is projected to eliminate most of the world's reefs.

Aquaculture is the fastest-growing food sector and can offset part of the decline in capture fisheries, but it has its own vulnerabilities. Coastal ponds are exposed to sea level rise, storm surge and saline intrusion; the feed supply depends on wild fish and on crops that are themselves affected by climate; and disease outbreaks spread more easily in warm water. Aquaculture can be part of the adaptation, but it is not a substitute for protecting the ecosystems that underpin wild stocks.

7.7 Prices, Trade and Stocks

The global food trade buffers local production shocks, but it also transmits them. A country that experiences a drought imports more, which raises prices for everyone else; an exporting country that restricts shipments to protect its own consumers amplifies the shortage. The result is that the price effect of a climate shock can be larger than the physical effect, and that the poorest countries, which are most dependent on imports, pay the highest price.

Stocks are the first line of defence, and they are thinner than they appear. Global grain stocks are held by a small number of countries and by private traders who have no obligation to release them in a crisis. Official stocks are sometimes low because holding them is expensive and because governments prefer to intervene in markets when prices rise. The 2007–2008 crisis revealed that the stocks meant to stabilise the market were insufficient once exporters began to hoard.

Trade policy is therefore climate policy. A world that maintains open markets, limits export restrictions and invests in storage can absorb a series of regional shocks without a global crisis. A world that responds to every shortage with export bans and panic buying converts regional harvest failures into global price spikes, and the spikes fall hardest on the households with the least capacity to absorb them.

7.8 Smallholders and the Poorest Consumers

Smallholder farms produce about a third of the world's food and a much larger share in low-income countries, but they manage only a small fraction of the investment and irrigation. They are the most exposed to climate shocks and the least able to absorb them, because they lack insurance, credit, storage and access to markets. A single failed season can force a household to sell assets, withdraw children from school and borrow

at high interest, which reduces its capacity to cope with the next season.

Women are disproportionately affected. In many countries they are responsible for water, fuel and food processing, and they often have weaker land rights and less access to extension services. When a drought increases the time needed to collect water or fuel, girls are often the first to be withdrawn from school. The result is that a climate shock does not simply reduce income; it changes the trajectory of a household, sometimes permanently.

Social protection is the most effective instrument for preventing that trajectory. Cash transfers, school feeding, employment guarantee schemes and crop insurance can keep households from selling productive assets during a shock. Ethiopia's Productive Safety Net Programme, India's Mahatma Gandhi National Rural Employment Guarantee Act and Mexico's Progresa-Oportunidades have all been shown to reduce the impact of droughts on consumption, and they are often more cost-effective than emergency assistance after the crisis.

7.9 Adaptation: Seeds, Soil, Water and Insurance

The portfolio of agricultural adaptation is well established. Heat- and drought-tolerant varieties, earlier planting, crop diversification, improved irrigation, soil conservation and agroforestry can reduce losses under moderate warming. Climate-smart agriculture, a term that covers many of these practices, has been adopted widely in development policy, though the evidence for its effectiveness varies by context and by how it is measured.

Genetic improvement is the most powerful tool and the slowest to deploy. Breeding a new variety takes eight to twelve years, and the climate that the variety will face when it is released is not the climate for which it was selected. Gene banks hold the diversity that breeders need, but they are underfunded, and the traits that matter — heat tolerance at flowering, deep roots, resistance to new pests — are often found in landraces that are being lost as farmers adopt modern varieties.

Insurance and finance are the enabling conditions. A farmer who cannot borrow cannot invest in irrigation or improved seed; a farmer without insurance may avoid risk by planting low-yielding but reliable crops. Index-based insurance, which pays out when rainfall or temperature crosses a threshold, has been promoted as a solution, but its uptake has been limited by basis risk, cost and trust. The most successful programmes combine insurance with credit, extension and access to markets, because insurance alone does not solve the underlying problem of low productivity.

7.10 The Food System as a Whole

Food security is not only a question of production. It depends on storage, transport, processing, retail and waste. About a third of the food produced is lost or wasted, and reducing that waste is equivalent to increasing production without using more land or water. It depends on diets: about a third of crop calories are fed to animals, and a shift toward less meat-intensive diets in high-income countries would free land and water for human food.

It depends on the political economy of food. Subsidies that encourage water-intensive crops in dry regions, tariffs that protect inefficient producers, and land tenure systems that exclude smallholders all reduce the system's capacity to adapt. The barriers are rarely technical; they are institutional and distributional, which is why the chapters on governance and justice are as relevant to food as the chapters on agronomy.

The final point is that the food system and the climate system are coupled. Agriculture, forestry and land use account for about a fifth of global greenhouse gas emissions, and food loss and waste add more. A transformation that reduces emissions from agriculture — through better fertiliser management, less deforestation, healthier diets and less waste — would also make the food system more resilient, because it would reduce the pressure on land and water that makes it fragile.

7.11 Case Study: The 2023 Rice Shock

Rice is the most important food crop in the world by the number of people who depend on it. About 3.5 billion people obtain a fifth or more of their calories from rice, and most of them live in Asia, where the crop is often produced by smallholders and consumed close to where it is grown. The global rice trade is thin relative to production — only about ten per cent of output crosses a border — which means that a small disruption in a major exporter can move prices for everyone.

In 2023, that is what happened. India, which accounts for about forty per cent of global rice exports, restricted shipments after a poor monsoon and rising domestic prices. The restriction coincided with an El Niño that was reducing production in Thailand, Vietnam and the Philippines, and with a global market that was already nervous after the war in Ukraine. Rice prices rose by about 30 per cent in a few months, and importing countries responded by buying more, which pushed prices higher.

The effects were felt far from the harvest. In West Africa, where rice is a staple and imports account for a large share of consumption, governments subsidised purchases and released stocks. In the Philippines, the government capped rice prices, which reduced supply and created queues. In Indonesia, the state logistics agency increased imports and distributed rice through social assistance. None of these countries had experienced a harvest failure; they were experiencing the transmission of a policy decision through a thin market.

The case shows that food security is a systems problem. The physical shock was a weak monsoon, which is within the range of natural variability. The crisis was produced by the interaction of that shock with export restrictions, thin stocks, panic buying and the concentration of trade in a few exporters. Climate change increases the probability of the physical shock, but the severity of the crisis depends on how the trade system responds.

The lessons are specific. First, importing countries should hold strategic reserves and diversify their suppliers, rather than relying on the cheapest source. Second, exporting countries should coordinate restrictions rather than impose them unilaterally, because unilateral restrictions invite retaliation and panic. Third, international institutions should monitor stocks and prices in real time, so that governments can distinguish a physical shortage from a speculative one. Fourth, social protection should

be in place before the crisis, because the households that are pushed into hunger by a price spike cannot wait for a new programme to be designed.

7.12 Future Scenarios and Predictions: Feeding the World in 2050

The historical evidence is that warming has already reduced yields relative to what they would have been without it. A meta-analysis of field experiments found that each degree of warming reduces global wheat yields by about 6 per cent, maize by 7.4 per cent, rice by 3.2 per cent and soybeans by 3.1 per cent, before adaptation (Zhao et al., 2017). The losses are larger in tropical regions, where crops are already close to their heat limits, and smaller in temperate regions, where longer growing seasons can partly offset the heat.

Present conditions give the baseline. Global food production has kept pace with population growth for most of the past half-century, but the rate of yield improvement has slowed, and about 735 million people were undernourished in 2022. The food system is also responsible for about a third of global greenhouse gas emissions, which means that it must reduce emissions while increasing production — a combination that is difficult but not impossible.

The assumptions that drive the projections are population growth, diet, technology and trade. A world that stabilises population, reduces food waste, shifts toward healthier diets and invests in adaptation can feed a larger population with less land and water. A world that continues current trends will face rising prices, increasing dependence on imports and a growing number of climate-driven food emergencies. The scenarios below show the range.

Table 7.1 Crop yield sensitivity and 2050 projections

Crop	Yield change per °C	2050 change (SSP2-4.5)	Main adaptation	Key region
Wheat	−6%	−10 to −15%	Heat-tolerant varieties; earlier sowing	South Asia; Europe
Maize	−7.4%	−15 to −20%	Drought tolerance; diversification	Sub-Saharan Africa; US
Rice	−3.2%	−5 to −10%	Flood-tolerant varieties; irrigation	South and Southeast Asia
Soybean	−3.1%	−5 to −8%	Crop rotation; heat tolerance	Brazil; US; Argentina
Coffee	−10 to −20%	−25 to −50%	Shade; altitude shift; new varieties	Ethiopia; Colombia; Vietnam
Livestock	−5 to −10% (milk)	−10 to −20%	Shade; breeds; feed quality	South Asia; Sahel

Two conclusions follow. First, the crops that feed the poorest people are the most sensitive to heat, and the regions where they are grown are the ones where adaptation is least funded. Second, the projected losses are not catastrophic in the global aggregate, but they are severe in particular places, and they interact with poverty, conflict and weak institutions to produce famine rather than shortage.

What would change these conclusions? A breakthrough in heat-tolerant varieties, or in the efficiency of photosynthesis, would reduce the losses, but it would take a decade or more to reach farmers. A shift toward plant-based diets in high-income countries would reduce demand for feed grains and free land for human food. And a sustained increase in investment in irrigation, storage and rural roads would reduce the post-harvest losses that currently account for a large share of the gap. The most important variable, however, is emissions: the difference between 1.5°C and 3°C is the difference between a food system that can adapt and one that cannot.

CHAPTER 8

ECOSYSTEMS AND BIODIVERSITY LOSS

Theme: ecosystem collapse, extinction risk and the services that nature provides

In the first half of 2024, the Great Barrier Reef experienced its fifth mass bleaching event in eight years. Surveys by the Australian Institute of Marine Science found that about 73 per cent of surveyed reefs had bleached, with the highest levels in the southern section, which had previously escaped the worst heat. Bleaching is not death: a bleached coral has expelled the algae that feed it and can recover if the water cools quickly. But the interval between events has become shorter than the time a reef needs to recover, and that is what turns a recurring disturbance into a permanent decline.

The reef is the most visible example of a general pattern. Warming is reorganising the living world: species are moving toward the poles and up mountains, the timing of seasons is shifting, and the disturbances that ecosystems evolved with — fire, drought, storms — are becoming more frequent and more severe. The result is not a uniform decline but a reshuffling, in which some species thrive, many decline, and the relationships among them change in ways that are difficult to predict.

The consequences for people are not sentimental. Ecosystems provide food, fibre, clean water, pollination, flood protection, carbon storage and medicine. They regulate the climate and the water cycle, and they underpin the livelihoods of billions of people. When a reef dies, the fish that depend on it decline, and the coastal communities that depend on the fish lose protein and income. When a forest dries out, it stops absorbing carbon and starts releasing it, which accelerates the warming that caused the drying.

This chapter examines the main mechanisms by which climate change affects ecosystems, the thresholds beyond which the changes become irreversible, and the services that are at stake. It pays particular attention to three systems that are both highly vulnerable and highly consequential: coral reefs, tropical forests and permafrost. It closes with the conservation and restoration strategies that can buy time, and with an honest assessment of what they cannot save.

8.1 The Biodiversity Baseline

The living world is already under pressure from land conversion, overexploitation, pollution and invasive species, and climate change is arriving on top of those pressures. The Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services estimated in 2019 that about one million species are threatened with extinction, and that the average abundance of native species in most major terrestrial biomes has fallen by at least 20 per cent since 1900. The Living Planet Index, which tracks monitored vertebrate populations, recorded an average decline of 69 per cent between 1970 and 2018 (WWF, 2022).

These numbers describe a system already in deficit. Climate change adds a new set of pressures: heat stress, altered rainfall, ocean acidification, rising sea level and more frequent extreme events. The interaction is what makes the situation dangerous. A

forest fragment that is already degraded by logging has less capacity to survive a drought; a coral reef that is already polluted has less capacity to recover from bleaching; a species whose range has been reduced by land clearing has fewer places to move when its remaining habitat becomes too warm.

The baseline also matters for what can be saved. Ecosystems that retain their structure and diversity are more resilient than those that have been simplified, and the difference can be large. A intact mangrove forest attenuates storm surge; a cleared one does not. A diverse grassland recovers from drought; a monoculture may not. Protecting what remains is therefore as important as restoring what has been lost, and it is usually cheaper.

8.2 Range Shifts and Phenology

Species respond to warming by moving. On land, they move up mountains and toward the poles; in the ocean, they move toward cooler water and greater depths. A meta-analysis of observed shifts found that terrestrial species have moved to higher elevations at an average of about 11 metres per decade and toward the poles at about 17 kilometres per decade, while marine species have shifted at about 72 kilometres per decade (Chen et al., 2011). The rates are fast in ecological terms and slow in comparison with the rate at which the climate is changing.

The problem is that many species cannot move fast enough, and some have nowhere to go. A species that lives on a mountain top has no higher ground; a species that lives in a lake has no other lake; a species that lives on a low island has no landward refuge if the coast is developed. The result is that range shifts benefit mobile generalists and penalise specialists, which changes the composition of communities even when total biodiversity is maintained.

Phenology — the timing of seasonal events — is shifting even faster. Earlier springs advance the emergence of insects, the flowering of plants and the arrival of migratory birds, but the shifts are not synchronised across species. A bird that arrives at its breeding ground on its traditional date may find that the insects it feeds on have already peaked. These mismatches are difficult to observe and can reduce breeding success without any obvious change in the landscape.

8.3 Coral Reefs

Coral reefs occupy less than one per cent of the ocean floor but support about a quarter of marine species and provide food, income and coastal protection for hundreds of millions of people. They are also among the most sensitive ecosystems to warming, because the symbiotic algae that live in coral tissue are expelled when the water exceeds their thermal tolerance. The result is bleaching, and if the heat persists, the coral dies.

The frequency of bleaching has increased sharply. Before the 1980s, mass bleaching was rare; since 1998 there have been four global bleaching events, and the interval between them has shortened from about twelve years to about six. The Great Barrier Reef has bleached five times since 2016, and the 2024 event affected reefs that had previously been considered refuges because they lived in cooler water. Global coral cover has declined by about half since the 1950s, and the decline has accelerated.

The projections are stark. At 1.5°C of warming, about 70 to 90 per cent of warm-water reefs are projected to decline; at 2°C, more than 99 per cent. These are not gradual declines; they are the loss of the physical structure that supports the ecosystem, followed by the collapse of the fish community. Some corals may survive in cooler refuges, and some may adapt, but the reef as a functioning system is unlikely to persist in the form that has existed for thousands of years.

8.4 Tropical Forests and the Amazon

Tropical forests store about half of the carbon in terrestrial vegetation and support a large share of the world's biodiversity. They also influence rainfall over vast regions: the Amazon recycles moisture that falls as rain across South America, and the Congo basin does the same for central Africa. When forests are cleared or degraded, both the carbon and the rainfall are lost, and the two losses reinforce each other.

The Amazon is the most studied case. About 17 per cent of the forest has been cleared, and a further fraction is degraded by logging and fire. Climate models and observational studies suggest that beyond a threshold of about 20 to 25 per cent deforestation, the forest may lose its ability to recycle moisture and begin to convert to savannah in the eastern and southern parts. The 2023–2024 drought, the most severe in the instrumental record, lowered river levels to historic lows and created conditions for fires that burned millions of hectares.

The forest is also a carbon bomb. If it dies back, the carbon it stores will be released, accelerating warming and making the dieback more likely. The Amazon therefore contains a positive feedback that is not fully represented in most climate models, and the uncertainty is asymmetric: the risk of a rapid transition is small in any given decade but the consequences are irreversible on human timescales. Protecting the forest is therefore both a mitigation measure and an adaptation measure, and it is one of the cheapest available.

8.5 Permafrost and the Tundra

Permafrost — ground that remains frozen for at least two consecutive years — covers about a quarter of the Northern Hemisphere and stores an estimated 1,400 to 1,600 billion tonnes of carbon, roughly twice the carbon in the atmosphere. As the Arctic warms three to four times faster than the global average, the ground thaws, and microbes decompose the organic matter that has been frozen for millennia, releasing carbon dioxide and methane.

The release is already measurable. Arctic rivers carry increasing amounts of dissolved organic carbon; thaw slumps and thermokarst lakes are appearing in regions that were previously stable; and winter carbon dioxide emissions from northern soils have increased. The processes are slow relative to fossil fuel emissions, but they are large and difficult to reverse, and they are not included in the carbon budgets that governments use to plan emission reductions.

Thawing permafrost also damages infrastructure. Roads, pipelines, buildings and airports built on frozen ground deform when the ground subsides, and the cost of repair is already significant in Alaska, Canada and Russia. For communities that depend on ice

roads and frozen ground for transport, the loss of permafrost is a direct threat to access to food, fuel and medical care.

8.6 Freshwater Ecosystems

Freshwater ecosystems — rivers, lakes, wetlands and aquifers — are among the most degraded and the most important for people. They provide drinking water, irrigation, fisheries, transport and flood protection, and they support a disproportionate share of biodiversity. Climate change affects them through warming, changed flow, reduced snowmelt, increased evaporation and more frequent droughts and floods.

Warming lakes stratify earlier and more strongly, which reduces the mixing that brings nutrients from the bottom and oxygen from the surface. The result is low-oxygen zones that kill fish and release phosphorus and methane from sediments. Rivers fed by glaciers change from snowmelt-dominated to rain-dominated flow, which reduces summer flow precisely when demand is highest. Wetlands dry out, releasing the carbon they stored and losing their capacity to buffer floods.

Freshwater species are declining faster than terrestrial or marine species. The Living Planet Index for freshwater populations shows an average decline of about 83 per cent since 1970, driven by dams, pollution, water abstraction and invasive species, with climate change now adding to those pressures. The loss matters directly to people: inland fisheries provide protein for hundreds of millions of people, and the loss of river connectivity blocks the migration of fish that communities depend on.

8.7 Ocean Acidification and Deoxygenation

The ocean absorbs about a quarter of the carbon dioxide that humans emit, and when carbon dioxide dissolves in seawater it forms carbonic acid. The result is that surface ocean pH has fallen from about 8.2 to 8.1 since the pre-industrial period, a 30 per cent increase in hydrogen ion concentration. The change is large by geological standards, and it is occurring far faster than at any time in the past 65 million years.

Acidification affects organisms that build shells and skeletons from calcium carbonate. Corals, oysters, mussels, pteropods and some plankton are particularly sensitive, especially in the cold waters where carbonate is least available. Experiments show reduced calcification, thinner shells and slower growth, and field observations in the Southern Ocean and the Arctic already show effects. The consequences propagate through food webs, because the organisms affected are often at the base of the chain.

Deoxygenation is a related problem. Warmer water holds less dissolved oxygen, and nutrient runoff from agriculture increases the growth of algae that consume oxygen when they decompose. The result is an expansion of low-oxygen zones in coastal waters and the open ocean, which forces fish into shallower water and reduces the habitat available to species that need cold, oxygen-rich water. The combination of warming, acidification and deoxygenation is sometimes called the triple threat to marine life.

8.8 Extinction Risk and Climate Velocity

Extinction is the endpoint of a process that begins with range contraction, population decline and loss of genetic diversity. Climate change contributes to that process by reducing habitat, changing the timing of life cycles and increasing the frequency of extreme events. The species most at risk are those with small ranges, low mobility, specialised habitat requirements and long generation times — which includes many amphibians, mountain species, island endemics and large mammals.

The concept of climate velocity measures how fast a species would have to move to stay within its preferred temperature range. In flat terrain, the velocity is high because a small change in temperature corresponds to a large change in latitude; in mountainous terrain, it is lower because the species can move up a short distance. The concept helps explain why species in plains and on low islands are more vulnerable than species in mountains, and why connectivity between habitats is essential.

Projections of extinction risk depend on assumptions about dispersal and adaptation, and the range is wide. A widely cited analysis found that at 2°C of warming, about 18 per cent of insects, 16 per cent of plants and 8 per cent of vertebrates would lose more than half their geographic range, and that at 3.2°C the figures rise to 49 per cent, 44 per cent and 26 per cent (Warren et al., 2018). Range loss is not extinction, but it is the precursor, and it is much easier to measure.

8.9 Ecosystem Services and Human Dependence

The value of ecosystems to people is usually described as services: provisioning services such as food, fibre and fresh water; regulating services such as climate, flood and disease regulation; supporting services such as soil formation and nutrient cycling; and cultural services such as recreation and spiritual significance. The categories are a useful accounting device, but they understate the extent to which the services are interdependent and irreplaceable.

Pollination is a clear example. About three-quarters of the leading food crops depend at least partly on animal pollinators, and the value of that service is estimated in the hundreds of billions of dollars. If pollinators decline, the effect is not a uniform reduction in food supply but a shift toward crops that do not need them — wind-pollinated grains — which reduces dietary diversity and increases the cost of fruits, vegetables and nuts.

Coastal protection is another. Mangroves, reefs and dunes reduce wave energy and storm surge, and the loss of these systems increases the cost of coastal defence. A study of the 2004 Indian Ocean tsunami found that villages behind intact mangroves suffered fewer casualties and less damage than villages on exposed coasts, though the effect depended on the thickness of the belt and the angle of the waves. Ecosystem services are not a luxury; they are part of the infrastructure.

8.10 Conservation, Restoration and Their Limits

Conservation can buy time. Protecting large, connected areas allows species to move; reducing non-climate pressures such as pollution, overfishing and invasive species increases resilience; restoring degraded habitats can recover some of the services that have been lost. The Kunming–Montreal Global Biodiversity Framework,

adopted in 2022, set a target of protecting 30 per cent of land and sea by 2030, and many countries have committed to it.

Restoration is harder than protection. Planting trees is easy; growing a forest that stores carbon, supports biodiversity and provides livelihoods is not. Restored mangroves often fail because they are planted in the wrong place or because the hydrology that sustains them has been altered. Restored coral reefs are limited by the same warming that killed them. Restoration can recover structure within a decade or two, but recovery of biodiversity and function takes much longer, and some systems cannot be restored at all.

The limits are ultimately set by temperature. Conservation can protect species from land conversion, hunting and pollution, but it cannot protect them from a climate that exceeds their tolerance. That is why the biodiversity crisis and the climate crisis are the same crisis, and why the targets for one cannot be met without the targets for the other. Protecting 30 per cent of the planet is necessary but not sufficient; the other 70 per cent, and the atmosphere above it, determine whether the protected areas remain viable.

8.11 Case Study: The Great Barrier Reef

The Great Barrier Reef is the largest living structure on Earth, extending more than 2,300 kilometres along the Queensland coast and comprising about 3,000 individual reefs. It is a UNESCO World Heritage site, a source of income for tourism worth billions of dollars, and a cultural landscape for the Traditional Owners who have managed its waters for thousands of years. It is also the best-documented example of an ecosystem being pushed beyond its thermal limits.

The reef has bleached five times since 2016: in 2016, 2017, 2020, 2022 and 2024. The 2016 event killed about half the corals in the northern section; the 2017 event affected the central section; the 2020 event was the most widespread on record; and the 2024 event affected reefs that had previously escaped because they lived in cooler water. Between events, coral cover recovered in some places, particularly in the southern section, but the recovery has been repeatedly interrupted.

The reef is also affected by other pressures. Runoff from agriculture carries sediment, nitrogen and pesticides into coastal waters, which reduces water quality and encourages algal growth. Crown-of-thorns starfish outbreaks consume coral; cyclones break it; and fishing removes herbivorous fish that would otherwise control algae. Management has addressed some of these pressures — runoff targets, fishing closures, starfish control — but it cannot address the heat.

The Australian government has invested in adaptation: coral breeding programmes, assisted migration of heat-tolerant corals, shading and cooling experiments, and research on cloud brightening. These are small-scale and uncertain, and they cannot protect 2,300 kilometres of reef. The most effective adaptation is emission reduction, which is outside the jurisdiction of the reef's managers, and that is the central dilemma of coral conservation: the people with the authority to protect the reef do not control the variable that determines its future.

The case has wider lessons. First, repeated disturbances are more damaging than isolated ones, because recovery time is essential. Second, local pressures compound climate stress, so reducing them buys time even if it cannot stop the decline. Third, adaptation technologies are not a substitute for mitigation, and presenting them as one delays the action that matters. Fourth, the loss of a reef is not only an ecological event; it is an economic and cultural event for the communities that depend on it.

8.12 Future Scenarios and Predictions: The Living World in 2100

The historical record shows that ecosystems are already changing. The Living Planet Index recorded a 69 per cent average decline in monitored vertebrate populations between 1970 and 2018; global coral cover has halved since the 1950s; and the area of primary forest lost each year remains above 10 million hectares. These declines predate the worst of the warming, which means that the climate pressure is arriving on an already weakened system.

Present conditions are a mixture of loss and resilience. Some ecosystems retain the capacity to recover: the southern Great Barrier Reef recovered much of its coral cover after the 2016–2017 events, and some forests regrow after logging. Others have crossed thresholds: the Amazon is emitting more carbon than it absorbs in some regions, and parts of the Arctic are now net sources of carbon rather than sinks. The difference between the two groups is the key to understanding what can be saved.

The assumptions that drive the projections are the emission path, the rate of land-use change and the effectiveness of conservation. Under a low-emission path, many species and ecosystems can persist, though in altered form. Under a high-emission path, the projections converge on widespread loss: coral reefs as we know them disappear, tropical forests decline, and a large fraction of species face range loss. The scenarios below summarise the range.

Table 8.1 Ecosystem thresholds and projected change

Ecosystem	Threshold of concern	Change at 2°C	Change at 3°C	Service at risk
Coral reefs	1.5°C; frequent bleaching	70–90% decline	>99% decline	Fisheries; coastal protection
Amazon forest	20–25% deforestation + warming	Dieback in east/south	Large-scale conversion	Carbon storage; rainfall
Boreal forest	Fire and pest outbreaks	Increased burning	Shift to shrub/grass	Carbon; timber
Permafrost	Thaw onset	10–20% thaw	30–50% thaw	Carbon; infrastructure
Mangroves	Sea level > migration rate	10–30% loss	30–50% loss	Coastal defence; nurseries
Freshwater	Flow and temperature change	20–30% species loss	40–60% species loss	Water; inland fisheries

Two conclusions follow. First, the difference between 2°C and 3°C is the difference between a world where coral reefs survive in reduced form and one where they are largely gone, and between a forest that stores carbon and one that releases it. Second, the loss is not evenly distributed: tropical ecosystems, which hold the most biodiversity and support the most vulnerable people, are the most exposed.

What would change these conclusions? A rapid reduction in deforestation would buy time for the Amazon and reduce emissions. A breakthrough in coral restoration, or the discovery of heat-tolerant symbionts, could preserve some reefs, but it would not restore the physical structure at scale. And a change in the fire regime — through better management and less ignition — would reduce the conversion of forest to savannah. None of these substitutes for emission reduction, but together they can reduce the rate of loss while the climate policy is negotiated.

CHAPTER 9

HEALTH IMPACTS OF CLIMATE CHANGE

Theme: climate-sensitive disease, health systems and the epidemiology of a warming world

In 2024, the Pan American Health Organization recorded more than 12 million cases of dengue in the Americas, the highest number ever reported and roughly three times the previous record. Brazil alone reported more than six million suspected cases. The epidemic was driven by a combination of factors: warmer temperatures that shortened the incubation period of the virus inside the mosquito, heavier rainfall that created breeding sites, urbanisation that brought people and mosquitoes together, and the movement of people and goods that carried the virus across borders.

Dengue is one of the clearest examples of a disease that is climate-sensitive. The Aedes aegypti mosquito that transmits it thrives in warm, humid cities, and its range has expanded as temperatures have risen. The virus replicates faster in warmer mosquitoes, which means that a mosquito that would have been infectious after twelve days may become infectious after seven. The result is that a city which was too cool for sustained transmission a generation ago may now experience regular outbreaks.

The health effects of climate change are broader than infectious disease. Heat kills directly and indirectly; air pollution from wildfire smoke and fossil fuel combustion damages lungs and hearts; droughts and floods reduce food production and contaminate water; and extreme events disrupt the health systems that would otherwise treat the sick. The World Health Organization estimates that climate change will cause about 250,000 additional deaths per year between 2030 and 2050 from heat, malnutrition, malaria and diarrhoea alone, a figure that excludes many of the pathways described in this chapter.

This chapter examines those pathways, the populations they affect most, and the capacity of health systems to respond. It pays particular attention to the difference between direct effects, which are relatively easy to attribute to a heatwave or a flood, and indirect effects, which operate through ecosystems, food systems and infrastructure and are therefore harder to measure but often larger.

9.1 Direct Effects: Heat, Injury and Displacement

The direct effects of climate on health are the easiest to see. Heat causes heatstroke, cardiovascular collapse and kidney injury, and it exacerbates pre-existing conditions; the European summer of 2022 was associated with about 61,700 deaths. Floods cause drowning, trauma and electrocution, and they displace people into temporary shelters where disease spreads easily. Storms cause injuries during the event and infections in the aftermath, when wounds are exposed to contaminated water.

The direct effects are concentrated in specific moments, which makes them visible and therefore politically salient. A heatwave produces a spike in deaths that can be measured within weeks; a flood produces images that circulate globally. But the concentration also creates a bias in how the health burden is understood: the deaths that occur during an event are counted, while the deaths that occur months later, from the

disruption of health services, the loss of income or the deterioration of chronic disease, are not.

Displacement is the bridge between direct and indirect effects. People who lose their homes to a flood or a fire may live in crowded shelters, lose access to medication and treatment, and face months of uncertainty. The health consequences of displacement — infectious disease, mental distress, interruption of chronic care — often exceed the direct consequences of the event that caused it.

9.2 Vector-Borne Diseases

Malaria, dengue, chikungunya, Zika and yellow fever are all sensitive to temperature and rainfall. The mosquitoes that transmit them have temperature optima for survival, reproduction and virus development, and their ranges are bounded by cold and by dryness. Warming expands the suitable range, lengthens the transmission season and accelerates the development of the pathogen inside the vector.

Malaria is the most consequential. About 249 million cases and 608,000 deaths were reported in 2022, mostly in sub-Saharan Africa and mostly among children under five. The relationship between malaria and climate is complex: too much rain can wash away breeding sites, and too little can eliminate them; the effect depends on the species of mosquito and the local ecology. But the overall trend is toward a larger transmission zone, particularly in the highlands of East Africa and the Andes, where warmer temperatures allow the parasite to complete its cycle.

Dengue has expanded even faster. The Aedes aegypti mosquito has spread to more than 150 countries, and the virus now circulates in urban areas where it was previously absent. The 2024 epidemic in the Americas was the largest on record, and similar surges occurred in Southeast Asia and the Western Pacific. The expansion is driven by urbanisation and travel as well as by climate, which is why the response must combine vector control, surveillance and health system preparedness rather than climate policy alone.

9.3 Waterborne and Foodborne Disease

Floods and droughts both increase the risk of waterborne disease, through different mechanisms. Floods overwhelm sanitation systems, contaminate wells and carry pathogens into homes; droughts concentrate pathogens in scarce water and reduce the water available for hygiene. Cholera, typhoid, hepatitis A and diarrhoeal diseases all follow this pattern, and children under five are the most vulnerable.

The 2022 Pakistan floods produced a surge in diarrhoeal disease and malaria, and the 2010 Haiti earthquake and subsequent cholera outbreak showed how quickly a contaminated water supply can become a national emergency. In drought-affected regions, the problem is often not the absence of water but its quality: as sources dry up, people use whatever water is available, including water that is contaminated with sewage or agricultural runoff.

Foodborne disease is also climate-sensitive. Warmer temperatures increase the growth rate of bacteria such as Salmonella and Campylobacter in food, and flooding can contaminate crops with pathogens. The European Centre for Disease Prevention and

Control has documented increases in foodborne infections during heatwaves, and the risk is highest in countries with limited cold-chain infrastructure. The result is a health burden that is rarely attributed to climate but is nonetheless part of it.

9.4 Air Pollution and Wildfire Smoke

Air pollution is the largest environmental health risk in the world, causing about seven million deaths a year according to the World Health Organization. Much of that pollution comes from the same fossil fuels that drive climate change, which means that reducing emissions produces immediate health benefits — the co-benefit that is often the strongest argument for climate action in the short term.

Wildfire smoke is a growing component. The 2023 Canadian fires burned more than eighteen million hectares and sent smoke across North America and the Atlantic, exposing tens of millions of people to particulate matter concentrations that exceeded health guidelines. The 2019–2020 Australian fires exposed millions of people to smoke for weeks and were associated with thousands of hospital admissions and hundreds of deaths. Smoke is particularly dangerous because it contains fine particles that penetrate deep into the lungs and enter the bloodstream.

The health effects of air pollution are not limited to the respiratory system. Fine particles are associated with cardiovascular disease, stroke, diabetes and adverse birth outcomes. Ozone, which forms more readily at higher temperatures, damages lung tissue and reduces crop yields. The combination means that climate change increases air pollution at the same time that air pollution increases climate change, through the deposition of black carbon on snow and ice.

9.5 Nutrition and Child Development

The health effects of climate change on nutrition are indirect and profound. Higher temperatures reduce crop yields, lower the nutrient content of grains and raise food prices, which reduces the quantity and quality of food that poor households can afford. The result is an increase in undernutrition, which weakens immune systems and increases vulnerability to infectious disease.

Children are the most affected. Undernutrition during the first thousand days of life — from conception to age two — causes stunting, which is irreversible and reduces cognitive development, school performance and lifetime earnings. The World Health Organization estimates that climate change will cause an additional 95,000 child deaths per year from undernutrition by 2050, and that the burden will fall disproportionately on South Asia and sub-Saharan Africa.

The CO2 fertilisation effect adds a subtle dimension. Higher carbon dioxide concentrations increase the yield of C3 crops but reduce the concentration of protein, zinc and iron in the grain. Experiments on rice, wheat and soy show declines of 5 to 10 per cent in protein and micronutrients under elevated CO2, which means that a harvest of the same weight may be less nutritious. For populations that depend on a single staple crop, this 'hidden hunger' can matter as much as a decline in calories.

9.6 Mental Health

The mental health effects of climate change are increasingly recognised. Acute events produce post-traumatic stress, anxiety and depression, particularly among people who lose homes, livelihoods or family members. Slow-onset changes — drought, sea level rise, the loss of a familiar landscape — produce a more diffuse distress that researchers describe as solastalgia, the pain of seeing one's home environment change beyond recognition.

The 2019–2020 Australian bushfires were followed by a documented increase in psychological distress, and the 2021 floods in Germany and Belgium produced similar effects. Farmers in drought-affected regions show elevated rates of depression and suicide, and the effect is not confined to the people directly affected: entire communities experience the loss of social networks and economic security that accompanies environmental change.

The mental health burden is undercounted because it is not captured by disaster statistics and because stigma prevents people from seeking help. It is also undertreated: mental health services are scarce in most low- and middle-income countries, and they are rarely included in disaster preparedness plans. The consequence is that the psychological effects of climate change persist long after the physical damage has been repaired.

9.7 Health Systems Under Stress

Health systems are themselves vulnerable to climate change. Hospitals are damaged by floods and storms; power outages disable ventilators and refrigeration; roads are cut, preventing patients from reaching care; and staff are affected by the same events as their patients. The result is that the demand for health care rises at the same time as the capacity to provide it falls.

The 2022 European heatwave illustrated the problem. Hospitals cancelled surgeries because operating theatres could not be kept cool; emergency departments were overwhelmed by heat-related admissions; and ambulance response times increased. In Puerto Rico, Hurricane Maria in 2017 caused a prolonged failure of the electricity system that was associated with an estimated 2,975 excess deaths, many of them from chronic conditions that could not be treated.

Preparedness requires investment in infrastructure, supplies and planning. Hospitals need backup power, cooling, water storage and flood protection. Health systems need surge capacity, stockpiles of medicines and vaccines, and plans for continuity of care during disruptions. They also need surveillance systems that can detect outbreaks early, and a workforce trained to recognise climate-sensitive disease. Most of these measures are inexpensive relative to the cost of failure, but they are rarely prioritised until an emergency occurs.

9.8 Vulnerable Populations

The health burden of climate change is not distributed evenly. Children, pregnant women, older people, people with disabilities, outdoor workers and people living in poverty are at higher risk, and the same groups are less likely to have access to care. The World Health Organization estimates that children under five bear about 88 per cent of the disease burden from climate change, because their immune systems are immature, their bodies are small, and they depend on adults for protection.

Pregnancy is a particularly vulnerable period. Heat exposure is associated with preterm birth, low birth weight and stillbirth, and the effects are strongest among women who work outdoors or live in housing without cooling. Air pollution has similar effects. The 2023 Lancet Countdown reported that heatwaves affected pregnant women in all regions, and that the health systems in most countries were not prepared to protect them.

Disability and age interact with poverty. A person who uses a wheelchair may be unable to evacuate a building without an elevator; a person with dementia may not understand a heat warning; an older person living alone may not be able to open a window or reach a cooling centre. These vulnerabilities are not inevitable; they are the result of design choices and social arrangements that can be changed.

9.9 Surveillance and Early Warning

Early warning is the most effective health adaptation. Heat warning systems that trigger cooling centres, outreach to vulnerable people and changes in hospital staffing can reduce deaths substantially, as the Ahmedabad experience demonstrates. Disease surveillance systems that detect outbreaks early can mobilise vector control and treatment before the epidemic peaks, and climate forecasts can be used to predict where conditions are favourable for transmission.

The integration of climate and health data is improving. The Lancet Countdown publishes annual indicators that track exposure, vulnerability and adaptation across countries, and the World Health Organization has developed a climate and health monitoring framework. Several countries now operate heat-health warning systems, and some are developing malaria early warning systems that use rainfall and temperature forecasts to predict outbreaks.

The gap is in implementation. A warning system is only useful if it reaches the people at risk and if they have the capacity to act on it. In many countries, the warning is issued at the national level but not translated into local action; in others, the data exist but are not shared between health and meteorological agencies. Closing that gap is less a matter of technology than of institutional coordination and trust.

9.10 Co-Benefits and the Health Case for Climate Action

The health argument for climate action is stronger than the environmental argument, because it is immediate and personal. Reducing fossil fuel combustion reduces air pollution, which prevents millions of deaths a year. Increasing walking and cycling reduces obesity and cardiovascular disease while reducing emissions. Improving building insulation reduces energy demand and protects people from heat and cold. Shifting toward plant-rich diets reduces emissions and improves health.

These co-benefits are often larger than the costs of the policies that produce them, particularly in countries with high air pollution. A study of the health benefits of meeting the Paris targets in China estimated that the reduction in air pollution would prevent hundreds of thousands of premature deaths a year, a benefit that is not counted in the conventional cost of decarbonisation.

The health frame also changes the politics. Climate policy is often presented as a cost to be borne for the sake of future generations; health policy is presented as a benefit to be delivered now. The same measures — clean energy, efficient buildings, active transport, healthy diets — can be justified in either language, and the health language often travels further because it speaks to what people experience in their own bodies.

9.11 Case Study: The 2024 Dengue Epidemic

Dengue is a viral infection transmitted by Aedes mosquitoes, and it has been expanding for decades. The virus causes fever, severe headache, joint pain and, in a small fraction of cases, haemorrhagic fever that can be fatal. There is no specific treatment, and the vaccines that exist are only partially effective and are not recommended for everyone. Control depends on reducing mosquito breeding sites and on timely diagnosis and supportive care.

The 2024 epidemic in the Americas was the largest on record. More than 12 million cases were reported, with Brazil accounting for about half. The epidemic followed a year of record heat and heavy rainfall, which expanded the habitat of the mosquito and accelerated the replication of the virus. It also followed the relaxation of vector control programmes during the COVID-19 pandemic, which had left fewer resources for surveillance and spraying.

The health system response varied. Brazil deployed mobile testing teams and expanded hospital capacity; Argentina and Paraguay declared epidemiological alerts; and the Pan American Health Organization coordinated surveillance and laboratory support. But the epidemic exposed the limits of reactive response: by the time cases are detected, the mosquito population has already expanded, and the measures that reduce transmission — eliminating standing water, covering tanks, using repellent — depend on household behaviour that is difficult to sustain.

The case illustrates the connection between climate and health systems. The weather created the conditions for transmission, but the scale of the epidemic depended on the capacity of health systems to detect, treat and control it. Countries with strong surveillance and vector control reported fewer severe cases and deaths than countries with weaker systems, even when the climate conditions were similar.

The lessons are practical. First, vector control must be continuous, not episodic; the mosquito does not wait for an outbreak to be declared. Second, surveillance must integrate climate data, so that health authorities can anticipate where conditions are favourable and act before cases appear. Third, health systems need surge capacity for the diseases that climate change is expanding, including staff, beds, tests and medicines. Fourth, communities must be engaged, because the most effective interventions — eliminating standing water, using bed nets and repellents — happen in homes, not in hospitals.

9.12 Future Scenarios and Predictions: Health in 2050

The historical data show that the health effects of climate change are already measurable. Heat-related mortality among people over 65 has increased by about 70 per cent since the 1990s; the transmission season for dengue has lengthened in many regions; and the area burned by wildfire has increased, exposing more people to smoke. These trends are not projections; they are observations, and they are consistent with the physical changes described in earlier chapters.

Present conditions give the baseline. About 3.5 billion people live in areas suitable for dengue transmission; malaria transmission is expanding in highland regions; and air pollution remains the largest environmental health risk, causing about seven million deaths a year. Health systems in low-income countries are the least prepared, with fewer beds, fewer health workers and less capacity to detect outbreaks.

The assumptions that drive the projections are the emission path, population growth, ageing, urbanisation and the strength of health systems. A world that reduces emissions and invests in health systems can prevent much of the projected burden; a world that does neither will face a burden that grows faster than the climate because the population is ageing and urbanising at the same time. The scenarios below summarise the range.

Table 9.1 Climate-sensitive health risks, 2050

Risk	Climate driver	Change at 2°C	Change at 3°C	Main adaptation
Heat mortality	Heatwaves	+50–100%	+150–300%	Warning; cooling; housing
Malaria	Warming; rainfall	Expanding highlands	Wider range	Bed nets; treatment; surveillance
Dengue	Warming; urban water	+100–200%	+200–400%	Vector control; surveillance
Undernutrition	Crop yields; prices	+50–100k deaths/yr	+100–200k deaths/yr	Social protection; nutrition
Air pollution	Fossil fuels; fires	Falling with policy	Rising without policy	Clean energy; smoke warnings
Displacement health	Floods; drought	+50–100 million exposed	+100–200 million exposed	Shelter; health services

Two conclusions follow. First, the health burden is dominated by indirect pathways — nutrition, infectious disease, air pollution — rather than by the direct effects that make headlines. Second, the burden is preventable with known interventions: clean energy, vector control, water and sanitation, heat warnings, and strong health systems. The gap between what is known and what is done is the central problem of climate and health.

What would change these conclusions? A breakthrough in dengue vaccines or in gene-drive mosquito control could reduce transmission substantially. A rapid expansion of clean energy would reduce air pollution and heat simultaneously. And a strengthening of primary health care in low-income countries would improve the response to every climate-sensitive disease. None of these is a substitute for emission reduction, but each would reduce the human cost of the warming that is already committed.

CHAPTER 10

ECONOMIC IMPACTS AND FINANCIAL RESILIENCE

Theme: the economics of climate damage, insurance and the cost of delay

In August 2017, Hurricane Harvey stalled over Houston and dropped more than 1,500 millimetres of rain in five days. The damage was estimated at US$125 billion, most of it from flooding rather than wind. A few weeks later, Hurricane Irma struck the Caribbean and Florida, and Hurricane Maria struck Puerto Rico, with losses of US$77 billion and US$90 billion respectively. The three storms together produced losses of about US$300 billion, of which only about a third was insured.

The 2017 season illustrates a pattern that is becoming more common. The physical damage is only the first layer of economic loss. Businesses that close do not pay wages; workers who lose wages spend less; local governments that lose tax revenue cut services; insurers that pay claims raise premiums; and households that lose their savings reduce consumption for years. The indirect losses can exceed the direct damage, and they fall hardest on the people with the least cushion.

Climate economics has moved from the margins of the discipline to its centre because the damage is no longer hypothetical. The estimates vary widely, but they converge on a conclusion: the cost of inaction is larger than the cost of action, and the difference grows with every year of delay. The debate is no longer about whether to act but about how to allocate the cost, who bears the risk, and how to finance the transition.

This chapter examines the economic channels through which climate change affects households, firms and states. It distinguishes between direct damage and indirect loss, between stock and flow, and between the risks that can be insured and those that cannot. It closes with the financial instruments that can build resilience — reserve funds, parametric insurance, catastrophe bonds, and the public balance sheets that ultimately backstop them.

10.1 Direct Damage and Indirect Loss

Disaster economics distinguishes between damage, which is the cost of repairing or replacing physical assets, and loss, which is the cost of lost economic activity. A factory that is destroyed represents damage; the production that the factory would have generated during the months it is rebuilt represents loss. The two are often reported together, but they behave differently: damage is a one-time cost, while loss can persist until the factory is rebuilt and the market recovers.

The distinction matters for policy. Compensation that covers damage but not loss leaves businesses unable to pay wages and households unable to replace income, which prolongs the disruption. The 2011 Thailand floods damaged industrial estates that produced a large share of the world's hard drives; the damage was repaired in months, but the global production shortfall lasted longer and raised computer prices for a year. The loss was transmitted through supply chains to firms and consumers who never saw the flood.

Indirect losses are difficult to measure because they are counterfactual: they describe what would have happened without the disaster. Economists estimate them using input-output models, computable general equilibrium models, or econometric comparisons of affected and unaffected regions. The estimates vary, but they consistently find that indirect losses are larger in economies with tighter linkages, weaker social protection and less fiscal space to respond.

10.2 Productivity, Labour and Heat

Heat reduces economic output through labour productivity. The International Labour Organization estimates that heat stress already reduces global working hours by about 2.2 per cent, equivalent to 80 million full-time jobs, and that the loss could reach 2.4 per cent by 2030. Agriculture accounts for about 60 per cent of the lost hours, followed by construction, manufacturing and transport.

The mechanism is physiological and behavioural. Above a certain wet-bulb temperature, workers must slow down, take breaks and drink water; in the absence of those adaptations, heat exhaustion and heatstroke occur. Firms respond by shifting hours, installing cooling, or automating tasks, but each response has a cost. For workers paid by output, the cost is borne directly as lost income; for firms, it appears as lower productivity and higher capital expenditure.

The distributional effect is important. Outdoor workers, agricultural labourers and construction workers are the most exposed, and they are often the least able to insure against lost income. In countries where a large share of employment is informal, heat does not appear in official productivity statistics, because the work is not measured in the first place. The result is that the economic cost of heat is systematically underestimated in the places where it is highest.

10.3 Food Prices and Inflation

Climate shocks to food production are transmitted to the wider economy through prices. When a major producer loses a harvest, global prices rise, and the effect reaches consumers through retail prices and, in poor countries, through the cost of the food basket that dominates household budgets. Central banks have begun to include climate shocks in their inflation models, because food and energy price spikes can destabilise expectations even when core inflation is stable.

The 2022 food price crisis is an example. The war in Ukraine was the immediate trigger, but it arrived on top of a series of climate-related harvest failures in Europe, South Asia and the Horn of Africa. Food prices rose by more than 20 per cent globally, and the increase was largest in countries that import a high share of their calories. The result was a rise in poverty and a decline in dietary quality that persisted after prices fell.

The inflationary effect is not only a problem for households; it is a problem for governments. Food and energy subsidies are politically difficult to remove and fiscally expensive to maintain, and they tend to expand precisely when revenues are falling because the economy is contracting. The result is a pro-cyclical fiscal burden that can turn a climate shock into a debt crisis, particularly in countries with limited access to

international capital markets.

10.4 Sovereign Risk and the Cost of Capital

Climate change affects the cost at which governments and firms borrow. Credit rating agencies have begun to incorporate climate vulnerability into sovereign ratings, and the effect is already visible in the spreads that some countries pay on their debt. A country that is exposed to repeated disasters, dependent on climate-sensitive exports and lacking fiscal buffers is a riskier borrower than one that is not, even if its current debt level is the same.

The mechanism operates through several channels. Disasters reduce tax revenue and increase expenditure at the same time, which worsens the fiscal balance. Repeated disasters can reduce potential growth by destroying capital and discouraging investment. And the expectation of future losses can raise the risk premium that investors demand, which increases the cost of servicing debt and reduces the resources available for adaptation.

The result is a vicious circle for the most vulnerable countries. They face the highest adaptation costs, the highest cost of capital and the least fiscal space, and the combination can make adaptation unaffordable even when the returns are high. International financial institutions have begun to address the problem through concessional finance, debt swaps and pre-arranged emergency lending, but the scale remains small relative to the need.

10.5 Insurance and the Limits of Risk Transfer

Insurance is the mechanism that allows losses to be spread across many policyholders and over time. It works when the risk is measurable, the events are independent, and the premiums cover the expected losses plus a margin. Climate change violates all three conditions: the risk is changing, the events are becoming correlated across regions, and the losses are rising faster than premiums can adjust.

The result is retreat. Insurers have withdrawn from parts of California and Florida, where wildfire and hurricane risk have increased; they have raised premiums in coastal areas; and they have reduced coverage for floods and storm surge. In the United States, the National Flood Insurance Program has been insolvent for years and has been forced to raise premiums, which has made insurance unaffordable for some homeowners and reduced property values.

When insurance fails, the cost shifts to governments and households. Governments provide disaster relief, which is often untargeted and slow; households draw down savings, borrow at high interest, or abandon property. The shift is regressive, because wealthy households can self-insure and poor households cannot. The result is that climate risk becomes a mechanism for transferring wealth from those who cannot bear it to those who can.

10.6 Stranded Assets and Transition Risk

The transition away from fossil fuels creates a different kind of financial risk: assets that were expected to produce revenue may become uneconomic before they are fully depreciated. Coal plants, oil fields, gas pipelines and the infrastructure that supports them could be stranded if demand falls faster than expected or if policy restricts their use. The value of those assets is reflected in the balance sheets of companies, banks and pension funds, which means that a rapid transition could create losses across the financial system.

The risk is not hypothetical. Coal companies in the United States and Europe have written down assets as gas and renewables have displaced coal; oil majors have faced pressure from investors to disclose their climate assumptions; and central banks have begun to stress-test banks against transition scenarios. The magnitude of the risk depends on the speed of the transition and on whether it is orderly or disorderly.

The policy implication is that transition risk and physical risk must be managed together. A disorderly transition, in which policy changes abruptly and assets are written down suddenly, can destabilise financial institutions; a delayed transition, in which physical risk continues to accumulate, can destabilise them even more. The goal is an orderly transition that signals policy direction early, phases out the most exposed assets gradually and protects the workers and communities that depend on them.

10.7 The Cost of Inaction

Estimating the cost of inaction requires comparing a future with climate change to a future without it, and the comparison depends on assumptions about growth, adaptation and discounting. Early integrated assessment models, such as the DICE model developed by William Nordhaus, produced estimates of a few per cent of global GDP by the end of the century, and those estimates were criticised for understating the risk of catastrophic and irreversible impacts.

More recent estimates are larger. A 2024 study in Nature found that a high-emission pathway would reduce global income by about 19 per cent by 2050 relative to a no-climate baseline, with damages of about US$38 trillion a year, and that the losses would fall disproportionately on countries with low historical emissions (Kotz, Levermann and Wenz, 2024). The Swiss Re Institute estimated that climate change could reduce global GDP by 10 per cent by 2050 in a severe scenario, and the World Bank has estimated that poor countries could lose 1.2 to 3.3 per cent of GDP by 2030.

The estimates differ because they use different assumptions, but they share a conclusion: the cost of inaction is larger than the cost of action, and the difference is growing. The economics of climate change is therefore not primarily about whether to act but about how quickly, how fairly and how effectively. The longer the delay, the larger the physical risk and the more abrupt the transition that will be required to avoid it.

10.8 Who Bears the Loss

The economic burden of climate change is distributed unevenly across countries, sectors and households. Low-income countries are more exposed because their economies depend on agriculture, their infrastructure is less resilient, and their fiscal capacity to respond is smaller. Within countries, poor households are more exposed because they live in more hazardous places, work outdoors, and have fewer savings and less insurance.

The distributional effect is compounded by the fact that the countries and households that contributed least to emissions are often the most exposed. This is the basis of the loss and damage debate in international climate negotiations, which led to the creation of a fund at COP27 in 2022. The fund is small relative to the need, and its governance and disbursement remain contested, but it establishes the principle that climate losses are a matter of justice, not only of charity.

The household-level distribution is equally important. A disaster that destroys a poor household's home can push it into poverty permanently, because it has no savings, no insurance and no access to credit. A wealthier household can absorb the same loss and rebuild. The result is that a single event can widen inequality for a generation, and that social protection is as important as physical infrastructure in determining who recovers.

10.9 Financial Instruments for Resilience

Several instruments can transfer or finance climate risk. Sovereign catastrophe bonds pay out when a predefined event occurs, providing rapid liquidity after a disaster. Parametric insurance does the same at a smaller scale, using rainfall or wind speed as a proxy for loss. Contingent credit lines allow governments to borrow quickly when a disaster is declared. Reserve funds accumulate savings in good years for use in bad ones.

Each instrument has strengths and weaknesses. Catastrophe bonds are expensive and require sophisticated legal and financial infrastructure; parametric insurance suffers from basis risk, because the index may not match the actual loss; contingent credit depends on the government's creditworthiness; and reserve funds require fiscal discipline that is difficult to maintain. The most effective approach combines several instruments and layers them so that small events are covered by reserves, medium events by insurance and large events by sovereign or international support.

The deeper issue is that risk transfer does not reduce risk; it only spreads it. A country that insures against floods still experiences the flood, and the premium it pays reduces the resources available for prevention. The most cost-effective financial strategy is therefore to invest in reducing risk first, and to use insurance for the residual risk that cannot be eliminated. That sequencing — prevention before transfer — is the core principle of financial resilience.

10.10 The Macroeconomics of a Warming World

At the macroeconomic level, climate change affects growth through several channels: it destroys capital, reduces productivity, diverts investment from productive uses to reconstruction, and increases uncertainty, which discourages long-term

investment. The combined effect is a reduction in potential output that persists long after the event that caused it.

The effect is not uniform across countries. Economies that depend on agriculture, tourism or outdoor labour are more exposed; economies with diversified production and strong institutions are more resilient. The result is that climate change is likely to widen the gap between rich and poor countries, reversing some of the convergence that globalisation produced in the late twentieth century.

The policy response has three pillars. The first is mitigation, which reduces the magnitude of future damage. The second is adaptation, which reduces vulnerability to the damage that is already committed. The third is financial resilience, which ensures that households, firms and governments can absorb the shocks that cannot be prevented. The three are complementary: mitigation without adaptation leaves people exposed to unavoidable impacts; adaptation without mitigation eventually exceeds its limits; and both without financial resilience leave the poorest to bear the cost.

10.11 Case Study: The 2017 Atlantic Hurricane Season

The 2017 Atlantic hurricane season was one of the most destructive on record. Hurricane Harvey made landfall in Texas in August and stalled, dropping more than 1,500 millimetres of rain over Houston, with losses put at US$125 billion. Hurricane Irma followed in September, devastating several Caribbean islands and inflicting losses of US$77 billion. Hurricane Maria struck Puerto Rico two weeks later, destroying the electricity grid and inflicting damage of about US$90 billion.

The total damage was about US$300 billion, of which only about a third was insured. The gap was concentrated in Puerto Rico and the smaller Caribbean islands, where insurance penetration is low and where the losses were largest relative to the size of the economy. In Puerto Rico, the storm caused an estimated 2,975 excess deaths in the months after landfall, most of them from the failure of the electricity and health systems rather than from the storm itself.

The economic effects persisted for years. In Puerto Rico, the population declined by several per cent as residents moved to the mainland; businesses closed; and the reconstruction of the grid took more than a year and cost billions of dollars. In Texas, the recovery was faster because the state had resources, insurance and access to federal assistance, but the damage to housing was concentrated in low-income neighbourhoods that lacked flood insurance.

The season illustrates the difference between insured and uninsured loss, and between the capacity of rich and poor jurisdictions to recover. It also illustrates the importance of infrastructure: the failure of the Puerto Rican grid turned a hurricane into a public health emergency, and the failure of the Texas power system during a winter storm four years later showed the same pattern in a different season. In both cases, the physical hazard was within the range of what the infrastructure was supposed to withstand; the failure was in the design, maintenance and governance of the system.

The lessons are financial as well as physical. First, insurance penetration is a determinant of recovery, and it is lowest where the need is greatest. Second, infrastructure resilience is a form of financial protection, because it reduces the liability

that governments and households must bear after an event. Third, the poorest households need grants and social protection, not loans, because a loan against future income is not a recovery strategy for someone who has lost their livelihood. Fourth, the cost of reconstruction is rising, and the willingness of donors and insurers to pay is not.

10.12 Future Scenarios and Predictions: The Economic Cost to 2050

The historical record shows that economic losses from climate-related disasters have increased. The total damage from weather-related disasters, adjusted for inflation, has grown by about 5 to 7 per cent a year since the 1970s, faster than exposure alone would predict. Insured losses have grown even faster, because more assets are concentrated in exposed areas and because insurance has expanded into new markets.

Present conditions give the baseline. Global losses from weather-related disasters are estimated at US$200 to US$300 billion a year, of which about half is insured. The uninsured half falls on households, firms and governments, and it is concentrated in low- and middle-income countries. The cost of adaptation is estimated at US$100 to US$300 billion a year by 2030 for developing countries, and the finance available is a fraction of that.

The assumptions that drive the projections are the emission path, the rate of adaptation, and the discount rate used to compare present and future costs. A low discount rate makes future damages look larger and justifies more spending now; a high discount rate makes them look smaller and justifies delay. The choice of discount rate is therefore an ethical choice as much as an economic one, because it determines how much weight is given to people who are not yet born.

Table 10.1 Economic losses under three pathways, 2050

Indicator	Low emissions	Current policies	High emissions	Main driver
Global GDP loss vs no-climate baseline	−3 to −5%	−8 to −12%	−15 to −20%	Labour; agriculture; capital
Annual damage (US$ trillion)	1.5–2.5	4–6	7–10	Extreme events; slow onset
Uninsured losses (share)	40%	50%	60%	Insurance retreat
People pushed into poverty (million)	50–100	100–200	200–400	Food prices; displacement
Adaptation finance needed (US$ bn/yr)	100	200	300+	Infrastructure; social protection

Two conclusions follow. First, the difference between the low and high pathways is measured in percentage points of global GDP, which is to say in trillions of dollars and in hundreds of millions of people pushed into poverty. Second, the losses are not a future that arrives in 2050; they accumulate year by year, and the countries that are least able to adapt are already paying the highest share.

What would change these conclusions? A rapid diffusion of cheap clean energy would reduce both the physical damage and the transition risk, and it is already happening faster than most models assumed. A breakthrough in adaptation finance — through reform of multilateral institutions, debt relief or new levies — would reduce the burden on the poorest countries. And a change in the discount rate used by governments would make long-term damages look larger and justify more investment now. None of these is certain, but the direction of the economics is: delay costs more than action.

CHAPTER 11

MIGRATION AND DISPLACEMENT

Theme: climate migration, displacement and the politics of moving people

Tuvalu is a country of nine coral atolls with a population of about eleven thousand people and a highest point of less than five metres above sea level. At high tide, the capital, Funafuti, is flooded in places; at spring tides, seawater bubbles up through the ground. The government has been negotiating the terms of its own disappearance for more than a decade, and in November 2023 it signed an agreement with Australia, the Falepili Union, that provides for the movement of Tuvaluans to Australia under a special visa arrangement while preserving Tuvalu's sovereignty and maritime boundaries.

Tuvalu is the clearest case, but it is not the largest. Most climate migration is internal, rural-to-urban, and temporary. People move from a village to a city after a drought, from a floodplain to higher ground after a flood, or from a coast to an inland town after salt water contaminates the wells. They move within their own country, often return when conditions improve, and rarely cross an international border. The image of a wave of climate refugees crossing borders is misleading; the reality is millions of small movements, most of them within national boundaries.

The scale is difficult to measure because climate is rarely the only reason for moving. A farmer who leaves a drought-affected village may be moving because of debt, because his children have no school, or because a relative in the city has offered work. Climate change is one pressure among many, and it is often the pressure that tips a decision that was already being considered. That makes attribution difficult and prediction uncertain, but it does not make the phenomenon any less real.

This chapter examines the drivers of climate migration, the difference between voluntary and forced movement, and the destinations that receive people. It pays particular attention to the places that cannot adapt in place — small islands, low deltas and drought-prone rangelands — and to the legal and political frameworks that determine whether movement is managed or chaotic. It closes with the question that receiving communities face: how to absorb people without creating new vulnerability.

11.1 Definitions: Migration, Displacement and Relocation

Migration is a broad term that includes voluntary movement in search of opportunity and forced movement in response to threat. Displacement usually refers to movement caused by a specific event, such as a flood or a conflict, and it implies that people intend to return. Planned relocation refers to the organised movement of a community to a new site, usually with government support. The distinctions matter because they determine which legal protections apply and which forms of assistance are available.

Climate change can produce all three. A flood displaces people temporarily; a drought may push people to migrate permanently; a village that is repeatedly flooded may be relocated by the state. The same household may experience several of these in sequence: displacement to a shelter, return to a damaged home, migration to a city, and

eventually relocation to a new settlement. The categories are useful for policy but they do not describe a linear process.

The legal framework is incomplete. The 1951 Refugee Convention defines a refugee as a person fleeing persecution on the basis of race, religion, nationality, social group or political opinion; it does not cover people fleeing environmental change. There is no binding international instrument that grants climate migrants a right to enter another country or to receive protection. Some countries have created humanitarian visas, and regional agreements have begun to address the gap, but the protection available to a person whose island is disappearing depends on where they happen to be born.

11.2 Rapid-Onset and Slow-Onset Drivers

Rapid-onset events — floods, storms, wildfires — displace people immediately, often in large numbers. The Internal Displacement Monitoring Centre recorded about 26.4 million disaster displacements in 2023, the majority from floods and storms. Most of these displacements are short-lived: people return home within weeks or months, once the water recedes and the damage is repaired. But repeated displacement erodes the assets and social networks that make recovery possible.

Slow-onset changes — sea level rise, desertification, salinisation, glacier retreat — displace people gradually, and the movement is harder to see. A household whose well becomes saline may first buy water, then reduce its herd, then send a son to the city, and finally abandon the farm. By the time the village is recognised as uninhabitable, most of its population may already have left. The slow-onset process is less visible than a flood but often more permanent.

The two interact. A community weakened by repeated drought may be unable to recover from a flood; a coastal city that is subsiding may be overwhelmed by a storm surge that would have been manageable a generation ago. The sequence of shocks matters as much as the magnitude of any single event, which is why vulnerability is best understood as a trajectory rather than a state.

11.3 How Much Migration Is Climate-Related?

Estimating the number of people who move because of climate change is difficult because climate is rarely the sole cause. Researchers use household surveys, satellite data on vegetation and rainfall, and econometric models to identify the effect of climate variables on migration decisions, controlling for income, conflict, land tenure and social networks. The estimates vary widely, but they consistently find that climate shocks increase migration, particularly from rural areas and particularly among households that depend on agriculture.

The World Bank's Groundswell reports projected that 216 million people could move within their own countries by 2050 in six regions under a high-emission scenario, with most of the movement in Sub-Saharan Africa, South Asia and Latin America. The same reports found that most of the movement could be avoided through adaptation — irrigation, crop diversification, social protection and urban planning — which is why they are framed as a call to action rather than a prediction.

The number of people who cross international borders for climate reasons is much smaller, and it is likely to remain so. International migration is expensive, requires documents and networks, and is often deterred by border controls. The people who are most exposed to climate risk are usually the least able to migrate internationally, which is why the most vulnerable often stay in place. This phenomenon — the immobility of the most exposed — is one of the most important and least discussed features of climate migration.

11.4 Pastoralists, Farmers and the Sahel

Pastoralists are among the most climate-exposed groups because their livelihoods depend on grazing that varies with rainfall. In the Sahel, herders have long moved seasonally between dry and wet season pastures, and those movements were governed by customary agreements with farming communities. As rainfall has become more variable and pastures have degraded, the movements have become longer, more frequent and more contested.

The result is conflict over land and water, particularly where the state is weak and where farmers have expanded into corridors that herders once used. The conflict is often described as ethnic or religious, and those identities matter, but the underlying driver is frequently a dispute over access to a shrinking resource. Climate change does not create the conflict, but it intensifies the competition that makes it more likely.

Farmers respond to drought in a different way. They may migrate temporarily to cities, send a family member to work, diversify into non-farm activities, or sell livestock to buy food. The strategies are cumulative, and each one reduces the household's capacity to return to farming. A household that sells its animals and its tools may not be able to restart even if the rains return, which is why drought can produce permanent de-agrarianisation in regions that have experienced it repeatedly.

11.5 Small Island States and Coastal Communities

Small island developing states are the most exposed to sea level rise because their land area is small, their elevation is low, and their populations are concentrated on the coast. Tuvalu, Kiribati, the Marshall Islands and the Maldives have average elevations of a few metres, and their freshwater supplies depend on thin lenses of groundwater that are vulnerable to saltwater intrusion. For these states, adaptation in place has physical limits.

The responses have three tracks. The first is coastal defence: sea walls, reclamation, and raised infrastructure. The second is adaptation within the island: rainwater harvesting, desalination, salt-tolerant crops and the relocation of settlements to higher ground. The third is migration: agreements with other countries that allow citizens to move, work and eventually settle. The Falepili Union between Tuvalu and Australia is the most recent example, and it combines a visa pathway with Australian recognition of Tuvalu's maritime boundaries and a commitment to assist in the event of a major disaster.

The politics of these agreements is delicate. Island states do not want to be seen as giving up, and they do not want to lose their sovereignty or their maritime zones, which are vast relative to their land area. Australia and New Zealand have been willing to accept limited numbers of migrants, but not to recognize a legal category of climate refugee. The result is a patchwork of bilateral arrangements that provides a pathway for some people but not a right for all.

11.6 Floods, Displacement and Return

Floods displace more people than any other climate hazard. The 2022 Pakistan floods displaced about eight million people, many of whom lived in temporary camps for months; the 2010 floods in the same country displaced about 14 million. In Bangladesh, cyclones and floods regularly displace hundreds of thousands, and the people who are displaced are disproportionately poor, landless and living in informal settlements.

Most flood displacement is temporary, but return is not always possible. A household whose home has been destroyed may have nowhere to return to, particularly if it rented or occupied land informally. A household whose land has been eroded by a river may have lost its livelihood permanently. And a household that has been displaced repeatedly may decide not to return, even if the house is repaired, because the risk of another flood is too high.

The quality of displacement matters as much as its duration. People in camps face overcrowding, poor sanitation and limited access to health care and education; people who stay with relatives may be less visible but equally vulnerable. The decision to return is often driven by the availability of work and housing rather than by the level of risk, which means that the poorest are the most likely to return to hazardous places.

11.7 Cities as Destinations

Most climate migrants move to cities, and most cities are not prepared for them. New arrivals often settle in informal areas that are themselves exposed to floods, heat and landslides, and they lack access to water, sanitation, health care and formal employment. The result is that migration transfers vulnerability from the countryside to the city rather than resolving it.

Cities that plan for migration do better. Providing land for affordable housing, extending water and sanitation to informal settlements, and creating work opportunities reduces the risk that newcomers will be pushed into hazardous areas. Some cities have adopted policies that explicitly recognize climate migrants, including access to services and temporary shelter. But most have not, and the arrival of large numbers of people in a short period can overwhelm services and create political tension.

The receiving communities matter as much as the migrants. A city that absorbs people successfully is one where the labour market is growing, the infrastructure is adequate and the social fabric is inclusive. A city that is already stressed by unemployment, housing shortages and weak services may experience conflict. The difference is largely a matter of governance and investment, which means that migration policy is also urban policy.

11.8 Remittances and Transnational Adaptation

Migration is not only a failure of adaptation; it is also a strategy of adaptation. Households that send a member to work elsewhere receive remittances that can be used to invest in irrigation, housing, education and health. In 2023, remittances to low- and middle-income countries reached about US$669 billion, more than foreign direct investment and more than official development assistance. For many countries, remittances are the largest source of external finance.

Remittances can increase resilience. A household that receives regular transfers can buy food during a drought, rebuild after a flood, and keep children in school. They can also finance adaptation: a new roof, a water tank, a solar system, or a small business that is less dependent on rainfall. The effect depends on the reliability of the transfers and on how they are used, but the evidence generally shows that remittances reduce vulnerability.

There is a darker side. Migration can separate families, increase the burden on women who remain behind, and expose migrants to exploitation and dangerous work. The remittances may also create dependence on a single source of income that is itself vulnerable to economic shocks in the destination country. The policy challenge is to maximize the benefits of migration while reducing its costs, which requires safe migration pathways, portable social protection and financial systems that reduce the cost of transfers.

11.9 Legal and Political Frameworks

The legal framework for climate migration is fragmented. The Refugee Convention does not cover environmental displacement; the International Organization for Migration is not a protection agency; and the UN Framework Convention on Climate Change established a task force on displacement but not a right to admission. The result is that most climate migrants depend on ordinary immigration rules, which are designed for other purposes and are often restrictive.

Some progress has been made. The Global Compact for Safe, Orderly and Regular Migration, adopted in 2018, recognizes climate and environmental factors as drivers of migration and commits states to cooperate on admission and protection. The Nansen Initiative and its successor, the Platform on Disaster Displacement, have developed guidance on planned relocation and cross-border displacement. Several countries have created humanitarian visas or temporary protection regimes for people affected by disasters.

The political obstacle is that admission is a sovereign decision, and publics in receiving countries are often hostile to migration. The result is that governments prefer to fund adaptation in countries of origin, which is cheaper and politically easier, rather than to open legal pathways. Adaptation is necessary, but it cannot protect everyone, and the people it cannot protect need somewhere to go. The absence of a legal pathway does not prevent movement; it only makes the movement irregular, dangerous and expensive.

11.10 Planned Relocation

Planned relocation is the organised movement of a community to a new site, usually because the original site has become unsafe. It is the most difficult form of adaptation, because it involves the loss of land, livelihoods, social networks and cultural identity. Relocation projects have a long history of failure when they have been imposed without consultation, compensated inadequately, or sited in places that are themselves vulnerable.

The successful cases share several features. Communities are consulted early and repeatedly; compensation covers land, housing and livelihood; the new site has access to water, employment and services; and the process is paced to allow people to move when they are ready. Fiji's planned relocation of coastal villages, supported by a national framework, is one of the more studied examples; it has been slower than the government intended, but it has generally respected community consent.

The hardest question is who decides. A community may prefer to stay even when the government considers the site unsafe; a government may consider relocation necessary even when the community disagrees. The tension is unavoidable, and it is resolved differently in different political systems. The principle that has emerged from international guidance is that relocation should be a last resort, that it should be voluntary wherever possible, and that it should preserve the rights and dignity of the people who move. Those principles are easier to state than to implement.

11.11 Case Study: Tuvalu and the Falepili Union

Tuvalu consists of nine atolls and reef islands scattered across about 1.3 million square kilometres of ocean, with a total land area of about 26 square kilometres. The population is about 11,000, and the highest point is less than five metres above sea level. The country has no rivers, no significant groundwater beyond thin freshwater lenses, and no high ground to which it could retreat. Its territorial waters, by contrast, are vast, and its exclusive economic zone is its most valuable asset.

The physical pressures are already measurable. Sea level at Funafuti has risen by about 3.4 millimetres a year since records began, close to the global average, and the frequency of flooding at high tide has increased. Saltwater intrusion has contaminated wells and reduced the productivity of taro pits, a staple crop. Coral bleaching has damaged reefs that provide food and coastal protection. The government has built sea walls and raised some roads, but the costs are high relative to the national budget.

The Falepili Union, signed with Australia in November 2023, is the most comprehensive bilateral arrangement of its kind. It provides a special visa pathway for Tuvaluans to live and work in Australia, with an initial cap and growth over time; it commits Australia to assist Tuvalu in the event of a major natural disaster or a security threat; and it recognizes Tuvalu's maritime boundaries and statehood even if the land is lost to the sea. The agreement is reciprocal: Australia gains strategic influence in the Pacific, and Tuvalu gains a pathway for its people.

The agreement has been criticized. Some Tuvaluans argue that it encourages emigration and undermines the incentive to adapt in place; others argue that it is the only realistic option for a country that cannot defend itself against the sea. The government has framed it as an insurance policy rather than a surrender, and it has continued to invest in coastal defence and adaptation. The debate reflects a broader tension in climate policy: whether to invest in places that may become uninhabitable, or to plan for the movement of people before the crisis arrives.

The case has implications beyond Tuvalu. It shows that legal pathways for climate migration can be created through bilateral agreements, even in the absence of a global framework. It shows that the recognition of statehood and maritime boundaries can be separated from the physical territory, which is a legal innovation with implications for other island states. And it shows that the choice between adaptation and migration is not binary: a country can do both, and the balance shifts as the physical risk increases.

11.12 Future Scenarios and Predictions: Migration to 2050

The historical data show that climate shocks already move people. The Internal Displacement Monitoring Centre recorded about 26 million disaster displacements in 2023, and the World Bank estimates that 216 million people could move within their own countries by 2050 under a high-emission scenario. These figures are not predictions of a single wave; they describe millions of individual decisions, most of them short-distance and temporary.

Present conditions give the baseline. About 3.5 billion people live in contexts that are highly vulnerable to climate change, and about 1 billion live in low-elevation coastal zones. The countries that are most exposed are also the ones with the least capacity to manage migration: they have weak registration systems, limited social protection and few legal pathways to other countries.

The assumptions that drive the projections are the emission path, the rate of adaptation, and the openness of receiving countries. A world that invests in adaptation in places of origin can reduce the number of people who are forced to move; a world that opens legal pathways can make the movement safer and more orderly. A world that does neither will face a larger number of people moving irregularly, with higher costs for migrants and for the countries that receive them.

Table 11.1 Projected climate migration by 2050

Region	Internal migrants (million)	Main driver	Adaptation priority	Policy risk
Sub-Saharan Africa	~85	Drought; pastoralism	Water; social protection	Conflict; weak states
South Asia	~40	Floods; heat; sea level	Urban planning; embankments	Informal settlements
Latin America	~17	Drought; storms	Rural livelihoods; insurance	Urban violence
East Asia and Pacific	~48	Sea level; storms	Coastal defence; relocation	Land acquisition
North Africa	~19	Drought; water	Irrigation; employment	Political instability
Small island states	~2–3	Sea level rise	Relocation; legal pathways	Sovereignty; culture

Two conclusions follow. First, the scale of climate migration is large but manageable if it is anticipated; the danger is not the number of people but the absence of systems to receive them. Second, most migration will be internal, which means that the responsibility falls primarily on national and municipal governments, not on international institutions.

What would change these conclusions? A rapid reduction in emissions would reduce the number of people who are forced to move, but it would not eliminate the movement that is already committed by the warming that has occurred. A major expansion of adaptation finance would allow more people to remain in place. And a shift in public opinion in receiving countries would make legal pathways more feasible, which would reduce the human cost of movement even if the number of movers did not change.

CHAPTER 12

CONFLICT, SECURITY AND STABILITY

Theme: climate, conflict and the security of states and communities

The Lake Chad basin was once one of the largest inland water bodies in Africa. In the 1960s it covered about 25,000 square kilometres; today it covers roughly a tenth of that, having shrunk through a combination of reduced rainfall, increased irrigation and population growth. The retreating water left behind a landscape of dry lakebed, competing claims to fishing grounds and pasture, and a population that depends on an ecosystem that no longer supports it. The region is now home to one of the world's longest-running insurgencies and about 10.7 million people in need of humanitarian assistance.

The connection between the shrinking lake and the insurgency is real but not simple. Boko Haram did not emerge because the lake dried; it emerged from a combination of political marginalisation, economic neglect, religious radicalisation and weak institutions. What the environmental change did was to remove the livelihoods that might have provided an alternative, to displace people into camps where recruitment is easier, and to weaken the local economy that might have sustained resistance. Climate change acted as a threat multiplier, not as a cause.

The same pattern appears in other regions. Drought in the Sahel has intensified competition between herders and farmers; water scarcity in the Nile basin has raised tensions between Egypt, Sudan and Ethiopia; food price spikes have preceded riots in dozens of countries. In each case, the environmental stress interacts with political, economic and social conditions that determine whether the outcome is cooperation, protest or violence.

This chapter examines that interaction. It reviews what the research actually shows about climate and conflict — a literature with strong claims and serious disagreements — and then examines the pathways through which environmental stress can contribute to violence. It pays particular attention to the role of institutions, because the same drought that produces famine in one country can produce protest in another and nothing at all in a third. The chapter closes with the question of what security institutions can do, and what they cannot.

12.1 What the Research Shows

The modern literature on climate and conflict began with a series of quantitative studies that found a statistical relationship between temperature and violence. A widely cited meta-analysis by Hsiang, Burke and Miguel (2013) reported that each standard deviation increase in temperature was associated with a 14 per cent increase in intergroup conflict and a 4 per cent increase in interpersonal violence, across studies covering many countries and periods.

The finding was influential and contested. Critics, notably Buhaug (2015), argued that the studies suffered from selection bias, that they aggregated very different forms of violence, and that the effect sizes were small relative to political and economic factors. Later work, including a review in Nature by Mach et al. (2019), concluded that

climate has affected armed conflict risk in some contexts but that the evidence is strongest for low-level violence and weakest for civil war, and that confidence in the overall relationship is limited.

The disagreement is not about whether climate can contribute to conflict; it is about how much, where, and under what conditions. The consensus that has emerged is that climate is a risk multiplier rather than an independent cause. It acts on societies that are already vulnerable, through mechanisms that are political and economic rather than meteorological, and its effect is largest where institutions are weak and where livelihoods depend on climate-sensitive resources.

12.2 Pathways: Livelihoods, Prices and Migration

The first pathway is economic. When a drought destroys a harvest or a flood destroys a factory, incomes fall and unemployment rises. Young men who cannot find work are easier to recruit into armed groups; households that cannot feed themselves are more willing to accept protection from whoever offers it. The effect is not automatic, but it is one of the most consistent findings in the literature: negative income shocks increase the risk of violence.

The second pathway is food prices. When prices rise, urban populations protest, and governments that respond with repression can escalate the protest into conflict. The 2010–2011 food price spike coincided with riots in more than thirty countries and with the political transitions of the Arab Spring, though the causal relationship is debated and varies by country. The mechanism is plausible and well documented: food is the largest expenditure for poor households, and a sudden increase can make the difference between coping and collapse.

The third pathway is migration. The movement of people can create tension in receiving areas, particularly when it is rapid, when it is perceived as unfair, and when the state is unable to provide services. But migration can also be a source of resilience, through remittances and labour. The outcome depends on the capacity of receiving communities and institutions to absorb people, which is why the same inflow can produce integration in one place and conflict in another.

12.3 The Sahel and the Lake Chad Basin

The Sahel is the region where the climate-conflict relationship is most often invoked and most difficult to disentangle. Rainfall has declined since the 1970s, though it has partially recovered in some areas; temperatures have risen faster than the global average; and the population has grown rapidly. Pastoralists and farmers compete for land and water, and the state is weak across much of the region.

The conflict in Mali, Burkina Faso and Niger has killed thousands and displaced millions. The causes are political and military as much as environmental: the collapse of the Libyan state in 2011 flooded the region with weapons; jihadist groups exploited local grievances; and governments responded with strategies that alienated rural populations. Climate stress contributed by undermining livelihoods and by making it harder for the state to deliver services, but it did not create the conflict on its own.

The Lake Chad basin is a similar case. The lake's shrinkage has been driven by reduced rainfall, increased irrigation and population growth, and it has destroyed fishing and farming livelihoods across four countries. The resulting poverty and displacement created a reservoir of recruits for Boko Haram and its offshoots, and the military response has displaced millions more. The lesson is not that the lake caused the insurgency; it is that environmental change can create the conditions in which insurgency becomes possible.

12.4 Syria and the Politics of Attribution

The Syrian conflict is often cited as the first climate war. Between 2006 and 2010, a severe drought hit the country's northeastern agricultural region, destroying crops and livestock and pushing as many as 1.5 million people toward the cities. The drought was the worst in the instrumental record, and studies have found that climate change made it more likely.

The causal chain from drought to war is contested. Some analyses argue that the drought, combined with the Assad government's failure to manage water and agriculture, created a rural crisis that fed the uprising. Others, notably Selby et al. (2017), argue that the drought was one factor among many, that the migration was not as large or as sudden as sometimes claimed, and that the regime's repressive response to peaceful protest was the proximate cause of the war. The second view does not deny the role of climate; it insists on a more careful account of how it operated.

The Syrian case illustrates a general problem in climate-conflict research: the temptation to tell a simple story that connects a weather event to a war. The stories are politically useful because they dramatise the stakes, and they are analytically dangerous because they obscure the political decisions that turn vulnerability into violence. A government that responds to drought with redistribution and relief may face protest; a government that responds with repression may face war.

12.5 Water Disputes and Cooperation

Water is the resource most often associated with climate conflict, and the record is more cooperative than the headlines suggest. The Pacific Institute's Water Conflict Chronology records thousands of water-related incidents, most of them local, and far more treaties than wars. The Indus Waters Treaty survived three wars between India and Pakistan; the Nile Basin Initiative has kept dialogue alive among ten countries with competing claims; and the Senegal River Organisation has managed a shared basin for decades.

That does not mean water is not a source of tension. The Grand Ethiopian Renaissance Dam on the Blue Nile has been the subject of a decade of negotiations among Egypt, Sudan and Ethiopia, and the dispute remains unresolved. Egypt depends on the Nile for almost all of its fresh water and has treated the river as a existential resource; Ethiopia argues that it has the right to develop its own hydropower. The dispute has not become a war, but it has shaped regional politics and delayed cooperation on other issues.

The factors that determine whether water disputes escalate are political, not hydrological. Countries that have institutions for sharing information, dispute-resolution mechanisms and a history of cooperation are more likely to manage scarcity without violence. Countries that lack them, or that are already hostile, are more likely to treat water as a zero-sum resource. The lesson is that water treaties are a form of adaptation, and that their erosion is a security risk.

12.6 Fisheries, Maritime Boundaries and the Arctic

Climate change is opening new areas of the ocean and creating new disputes. As Arctic sea ice retreats, shipping routes, fishing grounds and hydrocarbon reserves become accessible, and the states bordering the Arctic have competing claims to the continental shelf. The Arctic Council has provided a forum for cooperation, but the withdrawal of some members after the war in Ukraine has weakened it.

In the tropics, warming is driving fish stocks toward cooler water and across maritime boundaries. Countries whose fleets have historically fished in their own waters may find that the stocks have moved into the waters of a neighbour, which creates disputes over access and enforcement. The resulting tensions are rarely violent, but they can escalate, particularly where fishing is a major source of employment and where enforcement capacity is weak.

Fisheries also illustrate the role of international institutions. Regional fisheries management organisations set quotas and allocate access, and their effectiveness depends on the willingness of states to comply. Climate change makes their task harder because the stocks they manage are moving, and because the scientific advice on which they rely is uncertain. The result is a governance gap that is filled by unilateral action, which is precisely the outcome that institutions are supposed to prevent.

12.7 State Capacity and Fragility

The most important variable in the climate-conflict relationship is the capacity of the state. A state that can deliver relief, maintain order and provide services during a shock can prevent the spiral from drought to famine to violence. A state that cannot — because it is poor, corrupt, illegitimate or already at war — is where the spiral is most likely to occur.

Fragile states are also the most exposed to climate change. They tend to be agriculture-dependent, to have weak infrastructure and to lack the fiscal capacity to respond to disasters. The combination creates a feedback loop: climate shocks weaken the state, and a weakened state is less able to respond to the next shock. The result is that climate change can contribute to state failure, and state failure can make climate impacts much worse.

The policy implication is that climate adaptation and peacebuilding are the same project in fragile states. Building irrigation, restoring rangelands and providing social protection are adaptation measures, but they are also conflict-prevention measures, because they reduce the competition over resources that fuels violence. The reverse is also true: a peace agreement that does not address access to land and water is unlikely to hold.

12.8 Security Institutions and Climate Risk

Military and intelligence institutions have begun to treat climate change as a security issue. The United States Department of Defense has described climate change as a threat multiplier that will increase demand for military missions, damage installations and destabilise regions. NATO has incorporated climate considerations into its planning, and several countries have appointed climate security advisers.

The military's role is controversial. Some argue that framing climate as a security threat militarises a problem that requires development and diplomacy; others argue that the military is one of the few institutions with the logistics capacity to respond to disasters and that its planning assumptions shape how governments understand risk. The balance depends on the country and on whether security institutions are accountable to civilian authority.

The most useful contribution that security institutions can make is often analytical and logistical rather than kinetic. Early warning systems, scenario planning, disaster response capacity and the protection of critical infrastructure are all within the remit of security institutions and all contribute to resilience. The risk is that a security framing crowds out the development and governance responses that address the root causes of vulnerability.

12.9 Food, Prices and Political Instability

The relationship between food prices and political instability is one of the best-documented in the social sciences. The 2007–2008 price spike was followed by riots in more than thirty countries, and the 2010–2011 spike coincided with the Arab Spring. The mechanism is straightforward: when the price of the staple food rises beyond what households can afford, the result is hunger, anger and, often, protest.

The effect is not deterministic. Some countries with high food prices experienced no unrest, and some unrest occurred in countries with relatively low prices. The difference is explained by the presence of other grievances, the capacity of the state to respond and the willingness of the security forces to repress. Food prices are a trigger, not a cause; they ignite conflicts that are already smouldering.

Climate change makes the trigger more likely by increasing the frequency of harvest failures and by increasing the volatility of prices. It also makes the response more difficult, because governments that are already fiscally stressed may be unable to subsidise food or to provide relief. The result is that the same price shock can be absorbed in one country and produce a crisis in another.

12.10 Prevention, Early Warning and Peacebuilding

Conflict prevention is cheaper than conflict response, and climate adaptation can be a form of prevention. Early warning systems that identify areas at risk of drought or flood can trigger relief before people are forced to migrate or to compete for resources. Dialogue mechanisms that bring herders and farmers together can resolve disputes before they become violent. Land reform that clarifies tenure can reduce the uncertainty that fuels conflict.

The evidence for these interventions is mixed but encouraging. The United Nations Development Programme's climate security projects in the Sahel have supported local mediation and early warning, and some evaluations report reductions in conflict. The challenge is scale: the interventions are small relative to the size of the problem, and they depend on local institutions that are often weak.

The deepest lesson is that climate security is not primarily a military problem. It is a problem of development, governance and justice. The countries that are most vulnerable to climate-driven conflict are those where people lack secure access to land, water and food, and where the state is unable or unwilling to provide them. Addressing those conditions is the most effective security policy available, and it is also the most neglected.

12.11 Case Study: The Lake Chad Basin

Lake Chad sits at the junction of Chad, Niger, Nigeria and Cameroon, and it has supported human settlement for thousands of years. In the 1960s it covered about 25,000 square kilometres, making it one of the largest lakes in Africa. By the 2000s it had shrunk to about 2,500 square kilometres, a decline of roughly 90 per cent, driven by reduced rainfall, increased irrigation and the growing demands of a population that rose from a few million to more than 30 million in the basin.

The shrinking lake destroyed livelihoods. Fishing communities lost their catch; pastoralists lost access to water and pasture; farmers lost the flood-recession land that had produced their crops. The result was a population that was poorer, more mobile and more dependent on whatever work was available. At the same time, the region was marginalised politically and economically, with weak state presence and few public services.

Boko Haram emerged in this context. The group was founded in north-eastern Nigeria in the early 2000s and became violent in 2009 after a government crackdown. Its ideology was religious, but its recruitment drew on grievances that were economic and political: unemployment, corruption, neglect and the collapse of traditional livelihoods. The shrinking lake did not create the group, but it created the conditions in which the group could recruit and operate.

The conflict has killed more than 35,000 people and displaced more than two million. The military response, which includes forces from Nigeria, Chad, Niger and Cameroon, has retaken territory but has not restored the livelihoods that would allow people to return. Humanitarian access is limited, and the region has one of the highest rates of food insecurity in the world. The lake itself continues to fluctuate, and the inter-basin water transfer project that was proposed to restore it has not been built.

The case illustrates the limits of a purely security response. Military action can reduce the capacity of an insurgency, but it cannot restore the livelihoods that make recruitment attractive. A durable solution would require investment in irrigation, fisheries, pastoralism and local governance, combined with a political process that addresses the grievances that the insurgency exploited. Those are development measures, but they are also security measures, which is the central point of this chapter.

12.12 Future Scenarios and Predictions: Conflict Risk to 2050

The historical record does not show a simple trend of climate-driven war. The number of armed conflicts has fluctuated, and the end of the Cold War produced a decline that had little to do with climate. What the record does show is that climate shocks are associated with higher risk of low-level violence, riots and local conflict, particularly in places that are already fragile.

Present conditions give the baseline. About 2 billion people live in countries affected by conflict, and a large share of them are in regions that are highly exposed to climate change. The Sahel, the Horn of Africa, the Middle East and parts of South Asia combine high climate exposure with weak institutions and existing conflicts, which is the combination that the research identifies as most dangerous.

The assumptions that drive the projections are the emission path, the rate of development and the quality of governance. A world that reduces emissions and invests in institutions can reduce the risk; a world that does neither will face a larger number of climate shocks arriving in places that are less able to absorb them. The scenarios below summarise the range, with the caveat that conflict is inherently uncertain and that the confidence in any quantitative projection is low.

Table 12.1 Climate exposure and conflict risk in fragile regions

Region	Climate exposure	Institutional capacity	Conflict risk at 2°C	Priority
Sahel	Very high	Low	High	Land tenure; pastoral corridors
Lake Chad	Very high	Low	Very high	Livelihoods; local governance
Horn of Africa	Very high	Low	High	Drought relief; early warning
Nile basin	High	Medium	Medium	Water agreement; storage
South Asia	High	Medium	Medium	Heat; food prices; migration
Central America	High	Medium	Medium	Drought; rural employment

Two conclusions follow. First, the risk of climate-driven conflict is concentrated in places that are already fragile, which means that prevention must be targeted rather than universal. Second, the most effective prevention is development: secure land rights, functioning water institutions, social protection and accountable local government reduce the competition that turns scarcity into violence.

What would change these conclusions? A major peace agreement in a fragile region could reduce the risk substantially, as could a sustained improvement in governance. A prolonged global recession could reduce the resources available for adaptation and increase the risk. And a rapid reduction in emissions would reduce the frequency of the shocks that trigger conflict, though it would not eliminate the underlying vulnerabilities that make conflict possible.

CHAPTER 13

INFRASTRUCTURE, SHELTER AND ENERGY RESILIENCE

Theme: the built environment, power systems and the design standards of a changing climate

In February 2021, a polar vortex pushed Arctic air into Texas and stayed for a week. Temperatures fell below −10°C across a state whose power system had been designed for summer peaks, not winter lows. Natural gas wells froze, pipelines lost pressure, and thermal plants tripped offline. The grid operator, ERCOT, ordered rolling blackouts to prevent a total collapse; at the peak, more than 4.5 million homes and businesses had no power. Pipes burst in houses and in municipal water systems, and about 14 million people were placed under boil-water notices.

The storm killed at least 210 people, according to the state's own estimate, and caused about US$195 billion in damage, making it the costliest disaster in Texas history. The deaths were not caused by the cold alone. People died of carbon monoxide poisoning from generators and charcoal grills used indoors, of hypothermia in unheated homes, and of dialysis and oxygen failures when clinics lost power. The infrastructure failure turned a weather event into a public health emergency.

Texas is a wealthy jurisdiction with a modern grid, and that is the point. The failure was not caused by poverty or by a lack of engineering capacity; it was caused by design standards that assumed a climate that no longer exists, by a market structure that did not pay for winterisation, and by the interdependence of systems that were each planned separately. The same pattern appears in poorer countries with less capacity to respond, where the consequences are larger and the recovery slower.

This chapter examines the built environment as a climate problem. It looks at housing, power, water, transport, hospitals and schools, and at the standards, financing and maintenance regimes that determine whether they survive a shock. It pays particular attention to the interdependence of systems, because the failure of one often causes the failure of others, and to the question of who pays for resilience when the benefits accrue over decades.

13.1 Infrastructure and the Assumption of a Stationary Climate

Engineering design is based on statistics: the 100-year flood, the 50-year wind, the design temperature that is exceeded on a handful of days each year. Those statistics are derived from historical records and assume that the climate is stationary — that the future will resemble the past. Climate change violates that assumption, which means that infrastructure designed to the old statistics is being asked to perform in conditions it was never intended to withstand.

The consequences are visible in every sector. Roads buckle in heat that exceeds the softening point of asphalt; rail tracks expand and derail; power lines sag and transformers overheat; bridges are scourged by floods larger than the design flow; drainage systems are overwhelmed by rainfall intensities that were once rare. The failures are not caused by poor engineering in the original design; they are caused by the moving baseline against which the design was calibrated.

Adapting infrastructure requires updating the statistics and the standards that depend on them. Some countries have begun to do this: the Netherlands has revised its flood protection standards to account for sea level rise; the United Kingdom has updated its rainfall projections for drainage design; and several cities have raised their design temperatures for roads and rail. But the updates are slow, uneven and often voluntary, and the infrastructure that is already built cannot be redesigned overnight.

13.2 Housing and Shelter

Housing is the first line of protection against climate, and it is also the largest stock of assets that households own. A well-built house protects against heat, cold, wind and rain; a poorly built one exposes its occupants to all of them. The difference is largely determined by building codes, construction quality and maintenance, which are in turn determined by incomes, regulation and the availability of finance.

Informal settlements are the most vulnerable. About one billion people live in slums or informal settlements, often on land that is exposed to floods, landslides or industrial pollution, and often in structures that are not designed to resist wind or heat. Upgrading those settlements — providing tenure, water, sanitation, drainage and safer construction — is one of the most cost-effective adaptation measures available, but it requires political commitment and land reform that go beyond engineering.

In high-income countries, the problem is often the age of the stock and the location of the housing. Buildings constructed before modern codes may lack insulation, cooling, flood protection or fire resistance, and retrofitting them is expensive. Housing in floodplains and on coastlines is exposed to hazards that were underestimated when it was built. The result is that the same storm produces very different losses in different neighbourhoods, and that the differences track income and race.

13.3 Power Systems

Electricity is the backbone of modern infrastructure: hospitals, water treatment, communications, transport and food refrigeration all depend on it. Power systems are also vulnerable to climate in several ways. Heat reduces the efficiency of thermal plants and the capacity of transmission lines; drought reduces hydropower output and the water available for cooling; storms damage lines and substations; and floods inundate equipment that was built at ground level.

The Texas failure in 2021 is the most studied example in a high-income country. The grid was designed for summer peaks, and the winter storm exposed the absence of weatherisation, the dependence on gas supply that froze at the wellhead, and the lack of interconnection with the rest of the country. The failure was compounded by the water system: pumps lost power, pipes burst, and the loss of water pressure created a public health emergency that lasted for days after the power returned.

The solutions are known. Weatherising generators and fuel supply, burying or reinforcing lines, diversifying generation, improving forecasting, and connecting grids so that surplus power can flow from one region to another all reduce risk. The obstacle is cost and incentives: the benefits of resilience accrue over decades and are difficult to attribute to any single decision, while the costs are immediate and visible. Markets that

reward the cheapest electricity often fail to reward reliability, which is why regulation is needed.

13.4 Water and Sanitation Systems

Water systems are climate-sensitive at every stage. Source water depends on rainfall and snowmelt; treatment depends on energy and on water quality; distribution depends on pipes that leak and pumps that need power; and wastewater treatment depends on flows that change with storms and droughts. A failure at any stage can interrupt supply for days or weeks.

Floods are the most visible threat. They inundate treatment plants, contaminate wells and overwhelm sewers, producing a surge in waterborne disease. Droughts are slower but equally damaging: reservoirs fall, groundwater is over-pumped, and utilities are forced to ration. In both cases, the poorest households are the most affected, because they are the least likely to have storage, backup supply or alternatives.

Resilience requires redundancy and diversification. Multiple sources, interconnected networks, backup power, storage and treatment that can operate under variable conditions all reduce the risk of failure. So does demand management: reducing losses in the distribution network, which average about 30 per cent globally, is often cheaper than developing new supply. The case of Windhoek, Namibia, which has recycled wastewater into drinking supply since 1968, shows that diversification can secure a city in an arid region.

13.5 Transport and Supply Chains

Transport networks are exposed to heat, flood, wind and coastal erosion. Roads soften and crack in extreme heat; rail tracks buckle; bridges are scourged by floods; ports are closed by storms; and airports are affected by heat that reduces air density and requires longer runways or reduced payloads. The failures are individually manageable, but they accumulate across a network and can isolate communities for days.

Supply chains depend on the same networks. A flood that closes a road can prevent food from reaching a city; a heatwave that closes a rail line can delay coal deliveries; a storm that closes a port can disrupt imports for weeks. The 2011 Thailand floods showed how a single node in a global supply chain can affect production on other continents, and the 2021 Texas storm showed how a failure of power can cascade into water, health and transport failures.

Resilience measures include raising and waterproofing critical links, building alternate routes, improving drainage, and diversifying suppliers. The most cost-effective measures are often organisational rather than physical: better forecasting, pre-positioned supplies, and contingency plans that have been tested. The problem is that these measures are rarely visible, and therefore rarely funded, until a failure occurs.

13.6 Hospitals, Schools and Critical Facilities

Some buildings must keep functioning during a disaster: hospitals, clinics, emergency services, water treatment plants and shelters. Their failure turns a hazard into a catastrophe, because they are the institutions that would otherwise provide care

and coordination. A hospital that loses power loses ventilators, sterilisation, refrigeration for vaccines and medicines, and the ability to treat the people injured by the event.

The record is poor. Hospitals are often located in floodplains, built to standards that do not account for current hazards, and dependent on the same grid and water systems as everyone else. Hurricane Sandy in 2012 forced the evacuation of several New York hospitals; the 2022 Pakistan floods damaged hundreds of health facilities; and the COVID-19 pandemic showed how quickly health systems can be overwhelmed when demand rises and staff are absent.

Schools are equally important. They are shelters during emergencies, they provide continuity for children, and they are often the buildings with the largest community function. A school that is unusable for months after a flood disrupts education and forces families to choose between work and childcare. Building schools to higher standards, using them as designated shelters, and equipping them with water, power and communications are relatively cheap interventions with large returns.

13.7 Resilient Design and Retrofitting

Resilient design begins with siting: avoiding floodplains, steep slopes and exposed coastlines is cheaper than defending them. It continues with form and materials: raised floors, reflective roofs, shading, cross-ventilation, reinforced connections and flood-resistant materials all reduce vulnerability. And it depends on redundancy: backup power, water storage, multiple exits and alternate routes.

Retrofitting existing buildings is harder and more expensive than building new ones well. Insulation, shading, roof reinforcement, elevated mechanical equipment and flood barriers can be added, but the cost may exceed the value of the building, particularly in low-income housing. The result is that retrofitting is often targeted at public buildings and at households that can afford it, leaving the poorest exposed.

The economics of resilient design are favourable when the full lifecycle is considered. A study by the World Bank and the Global Facility for Disaster Reduction and Recovery found that every dollar invested in resilience saves about four dollars in future losses, and that the ratio is higher for infrastructure than for post-disaster reconstruction. The problem is that the savings accrue to future budgets and future governments, while the costs are borne now, which is why resilience is systematically underfunded.

13.8 Decentralised Systems and Microgrids

Centralised systems are efficient in normal conditions and fragile in extremes. A single failure can affect millions of people; a single point of control can determine whether power flows. Decentralised systems — microgrids, solar with battery storage, local water treatment, community shelters — reduce that fragility by creating islands that can operate independently when the main grid fails.

The technology has become cheaper and more reliable. Solar panels and batteries can now provide continuous power for a clinic or a community centre at a cost that is competitive with diesel generators, and they do not depend on fuel deliveries that

may be disrupted. Microgrids can connect to the main grid when it is available and disconnect when it is not, providing resilience without sacrificing efficiency.

The barriers are regulatory and financial rather than technical. Utilities may resist distributed generation because it reduces their revenue; regulations may not allow local grids to sell power; and financing may not be available for small projects. The communities that would benefit most are often those with the least access to capital, which is why public investment and concessional finance are needed to make decentralised resilience available to the poor.

13.9 Maintenance, Finance and the Build-Neglect-Rebuild Cycle

The most common cause of infrastructure failure is not a design flaw but a maintenance deficit. Bridges corrode, drains silt up, pumps wear out, and levees settle. The deficit accumulates slowly and invisibly, and it is revealed only when an event tests the system. By then, the cost of repair is much higher than the cost of maintenance would have been.

The build-neglect-rebuild cycle is a political economy problem. Building new infrastructure attracts credit and visibility; maintaining it does not. Donors prefer to finance new projects; governments prefer to inaugurate them; and voters do not see the maintenance that did not fail. The result is a systematic bias toward new construction and against the routine spending that keeps existing systems working.

Breaking the cycle requires institutions that make maintenance visible and fundable: asset registers, lifecycle budgets, performance contracts and independent audits. Some countries have created dedicated maintenance funds and require utilities to report on service quality, which has improved performance. The principle is simple: resilience is not a project but a practice, and it must be budgeted every year.

13.10 Codes, Standards and Governance

Building codes and engineering standards are the most powerful instruments for infrastructure resilience, because they determine what is built before any individual decision is made. A code that requires flood-resistant foundations, wind-resistant roofs and heat-resistant materials shapes millions of buildings over decades. A code that is not enforced shapes nothing.

Enforcement is the weak link. In many countries, codes exist but are not applied, particularly in informal construction and in small projects. Inspectors may be underfunded or corrupt; builders may lack the skills to comply; and homeowners may not know the standards. The result is a gap between the code on paper and the building on the ground, which is often widest where the risk is highest.

Governance determines whether standards are updated as the climate changes. That requires institutions that monitor hazards, revise codes, train professionals and enforce compliance, and it requires a political system that is willing to impose costs on developers and homeowners for benefits that may not be visible for years. The countries that do this well tend to have strong professional bodies, well-funded regulators and a public that understands the risk. The countries that do it poorly tend to discover the gap only after a disaster.

13.11 Case Study: The Texas Grid Failure of 2021

Winter Storm Uri arrived in Texas on 13 February 2021 and remained for a week. Temperatures fell to levels that the state had not experienced in decades, and the demand for electricity rose to a winter record. The grid operator had planned for summer peaks, not winter extremes, and it had limited interconnection with the neighbouring grids that could have provided emergency power.

The failures cascaded. Natural gas wells and processing plants froze, reducing fuel supply; thermal plants tripped offline; wind turbines, some of them not weatherised, also failed; and the grid lost more than a third of its generating capacity. ERCOT ordered rolling blackouts, but the outages lasted for days in many areas, and more than 4.5 million customers lost power at the peak. Pipes burst in homes and in municipal systems, and about 14 million people were placed under boil-water notices.

The human cost was concentrated among the vulnerable. At least 210 people died, according to the state, and independent estimates were higher. Many died of carbon monoxide poisoning from generators and grills used indoors; others died of hypothermia, of dialysis that could not be performed, and of oxygen that could not be delivered. The deaths were not evenly distributed: they were higher in low-income neighbourhoods and among people who could not afford to evacuate or to heat their homes.

The economic cost was about US$195 billion, making it the costliest disaster in Texas history. Insured losses were a fraction of that, because much of the damage was uninsured. The political aftermath was intense: the grid operator's leadership resigned, the state legislature required weatherisation of generating facilities, and the federal regulator recommended that the state connect its grid to the rest of the country, a recommendation that has not been implemented.

The case illustrates three lessons. First, infrastructure failures are systemic: the power failure became a water failure, which became a health failure, which became a housing failure. Second, design standards matter: a grid designed for summer failed in winter, not because the engineers were incompetent but because the assumptions were incomplete. Third, the costs of failure are borne disproportionately by the poor, even in a wealthy jurisdiction. The same lessons apply in countries with far less capacity to respond.

13.12 Future Scenarios and Predictions: Infrastructure in 2050

The historical record shows that infrastructure failures are increasing. The number of weather-related power outages in the United States has risen from a few hundred a year in the 1990s to several thousand, and the costs of extreme weather to infrastructure have grown faster than the value of the assets exposed. The trend is not driven by worse engineering; it is driven by more extreme weather and by ageing systems.

Present conditions give the baseline. The American Society of Civil Engineers grades US infrastructure at C−, with a funding gap of trillions of dollars over the next decade. Globally, the Global Infrastructure Hub estimates that the world needs about

US$94 trillion of infrastructure investment by 2040, and that current spending is well below that. Much of the new investment will be in countries that are highly exposed to climate change and have limited capacity to maintain what they build.

The assumptions that drive the projections are the emission path, the rate of retrofitting, and the standards that are adopted. A world that builds to higher standards and maintains what it has can reduce failures even as the climate worsens; a world that continues the build-neglect-rebuild cycle will face rising costs and more frequent cascading failures. The scenarios below summarise the range by sector.

Table 13.1 Infrastructure resilience by sector, 2050

Sector	Main climate stress	Failure mode	Adaptation	Cost signal
Power	Heat; storms; drought	Cascading outages	Weatherisation; microgrids	High upfront; high return
Water	Flood; drought	Contamination; rationing	Storage; reuse; redundancy	Medium; high return
Transport	Heat; flood	Network isolation	Elevation; alternate routes	High; medium return
Hospitals	Heat; flood; outage	Loss of critical care	Backup power; flood proofing	Low; very high return
Housing	Heat; flood; wind	Displacement; death	Codes; retrofit; tenure	Medium; high return
Schools	Heat; flood	Education disruption	Shelter design; cooling	Low; high return

Two conclusions follow. First, the highest returns come from the least glamorous investments: maintenance, backup power, drainage, and building codes. Second, the sectors that matter most for survival — water, health and shelter — are often the least funded relative to their importance.

What would change these conclusions? A large public investment programme could close the maintenance gap, but it would require political decisions that are difficult in countries with constrained budgets. A breakthrough in cheap distributed energy could reduce the vulnerability of power systems, but it would not solve the water and transport problems. And a change in building codes could improve new construction, but it would not address the existing stock. The most important variable is the same as in other chapters: the speed of emission reduction determines how much adaptation is needed, and the speed of adaptation determines how much damage the committed warming causes.

CHAPTER 14

CLIMATE LITERACY AND EDUCATION

Theme: climate literacy, curriculum and the transmission of usable knowledge

In 2019, Italy became the first country to make climate change and sustainable development compulsory subjects in all state schools. The law, which took effect in the 2020–2021 school year, requires about thirty-three hours a year of climate education at every level, from primary school to upper secondary, integrated into existing subjects rather than taught as a separate course. Teachers were trained, materials were developed, and the ministry framed the reform as an obligation to the next generation rather than a matter of political preference.

The Italian case is unusual because most countries have not made climate education compulsory. A UNESCO survey found that only about half of national curricula mention climate change at all, and that where it is mentioned it is often confined to a single subject, taught without connection to local conditions, and assessed in ways that reward memorisation rather than understanding. The result is that many students leave school without the knowledge they need to interpret a weather forecast, evaluate a climate claim or prepare for a hazard.

Climate literacy is not the same as climate science. It includes understanding the basic mechanisms, but it also includes the ability to read a graph, interpret uncertainty, identify a reliable source, and connect global trends to local conditions. It includes practical knowledge — what to do during a heatwave, how to store water, how to read a flood map — and it includes the disposition to act on that knowledge rather than to ignore it.

This chapter examines what climate literacy is, what the evidence says about how much of it exists, and how it can be taught. It pays particular attention to the difference between information and understanding, to the role of teachers, and to the risk that climate education becomes either indoctrination or a superficial ritual. It closes with the question of how to measure whether education is working, which is harder than it sounds.

14.1 What Climate Literacy Means

A climate-literate person understands that the climate is warming, that human activity is the main cause, and that the consequences are already measurable. But literacy goes beyond the headline facts. It includes an understanding of how scientific knowledge is produced, why uncertainty is not the same as ignorance, and how to distinguish a projection from a prediction. It includes the ability to evaluate evidence, which requires some familiarity with data, graphs and statistics.

It also includes practical competence. A household that knows how to prepare for a heatwave, store water, secure a roof or interpret a flood warning is climate-literate in a sense that matters more immediately than the ability to explain radiative forcing. The two forms of literacy are related: understanding the mechanism makes the practical advice coherent, and practical experience makes the mechanism concrete.

Finally, literacy includes the disposition to use knowledge. Many people understand the risks and still do not prepare, because preparation is costly, inconvenient or frightening. Education can address that gap by making preparedness normal, by providing concrete steps, and by connecting individual action to collective outcomes. The behavioural component is the hardest to teach and the most important to measure.

14.2 What the Evidence Shows About Public Understanding

Surveys consistently find that most people in most countries believe the climate is changing, but that the proportion who understand the causes, the consequences and the solutions is much smaller. In the United States, the Yale Climate Opinion Maps report that about 72 per cent of adults think global warming is happening, but only about 61 per cent understand that it is mostly caused by human activities, and the figures vary widely by state and county.

Understanding is uneven across countries and across groups. People with more education, more exposure to science and more trust in institutions tend to know more, but the relationship is not linear, and political identity can override education. In some countries, the gap between what scientists know and what the public believes is widening, not because the science is uncertain but because the issue has become a marker of political identity.

The evidence also shows that information alone does not change behaviour. Campaigns that provide facts without addressing the social context, the perceived cost and the sense of efficacy often fail. What works better is a combination of clear information, social norms, practical support and trusted messengers. That is why education is most effective when it is embedded in communities rather than delivered from a distance.

14.3 Misconceptions and Their Sources

Some misconceptions are simple errors that can be corrected with clear explanation: the difference between weather and climate, the role of the ozone hole, the difference between a prediction and a projection. Others are more resistant because they are connected to identity, ideology or economic interest. The belief that climate change is a hoax, or that it is too late to act, or that individual action is pointless, is not usually the result of a lack of information.

Misinformation spreads through social media, through partisan media and through the normal mechanisms of human communication: vivid stories travel further than statistics, and confident claims travel further than cautious ones. The result is that a single misleading graph can circulate for years after it has been debunked, and that corrections often fail to reach the people who saw the original claim.

Education can help by teaching how to evaluate sources, how to recognise the techniques of manipulation, and how to hold uncertainty without abandoning the consensus. That is a form of media and scientific literacy that is valuable far beyond climate. It is also more durable than memorising a list of facts, because it equips students to evaluate new claims that no one has anticipated.

14.4 Curriculum and Schools

Climate change can be taught in many subjects: geography, biology, physics, chemistry, mathematics, history, civics and economics. The most effective programmes integrate it across subjects rather than confining it to one, because climate is inherently interdisciplinary. A mathematics lesson can use temperature data; a history lesson can examine how societies have responded to environmental shocks; a civics lesson can discuss how decisions are made.

The Italian reform is an example of integration. It does not create a separate subject; it requires teachers to incorporate climate and sustainability into existing lessons, with a target of about thirty-three hours a year. The approach avoids overloading the timetable, but it depends on teachers having the knowledge and confidence to teach the material, which is why training is essential.

Curriculum reform is slow and contested. In some countries, the content of climate education is politically disputed; in others, it is crowded out by exam pressures and by subjects that are tested. The result is that climate education is often present on paper but thin in practice, and that the students who most need it — those in under-resourced schools — are the least likely to receive it.

14.5 Teacher Training and Professional Support

Teachers are the limiting factor in climate education. Most teachers have not been trained in climate science, many are uncertain about the facts, and some are anxious about discussing a politically charged topic. A survey by the United Kingdom's Royal Meteorological Society found that many primary teachers lacked confidence in teaching climate change, even when they believed it was important.

Good training addresses both knowledge and pedagogy. Teachers need a solid understanding of the science, but they also need strategies for handling controversy, for connecting global trends to local experience, and for teaching without inducing despair. The last point is important: young people who learn about climate change without learning about solutions are more likely to experience anxiety and disengagement.

Professional support can come from many sources: universities, science centres, professional associations, and peer networks. The most effective programmes are sustained rather than one-off, and they involve teachers in designing the materials they will use. A workshop that gives teachers a set of slides is less effective than a community of practice that helps them adapt materials to their own students.

14.6 Universities and Research Capacity

Universities produce the research that underpins climate understanding, and they train the professionals who will design infrastructure, manage water, plan cities and deliver health care. The capacity to do that is unevenly distributed: a small number of countries produce most of the research, while many of the countries most exposed to climate change have limited research capacity and few opportunities for graduate training.

The gap matters for adaptation. A country that cannot monitor its own glaciers, aquifers or rainfall cannot plan for their changes; a country that cannot train engineers in climate-resilient design cannot build resilient infrastructure. International collaborations, open data and scholarship programmes can help, but they are not a substitute for domestic capacity.

Universities also have a responsibility to avoid reproducing the conditions that produce vulnerability. Research agendas that are set entirely by external funders may not address local priorities; curricula that are imported wholesale may not fit local contexts. The most effective institutions combine international standards with local relevance, and they build partnerships that transfer capability rather than dependence.

14.7 Informal Learning and Public Communication

Most learning about climate happens outside school. Museums, science centres, documentaries, news media, social media and community organisations all shape what people know and believe. The best of them make the abstract concrete: a science centre that lets visitors see the heat-trapping effect, a museum that tells the story of a local flood, a community group that organises a preparedness workshop.

Communication that works tends to be local, specific and actionable. A message that says the planet is warming is less effective than one that says the local river will flood more often and here is what to do about it. A message that presents only catastrophe induces paralysis; a message that presents only solutions can seem naive. The balance is honest hope: a clear account of the risk, combined with a clear account of what can still be done.

Journalists are important intermediaries. Their training in climate science is often limited, and the pressure to produce balanced coverage can create a false equivalence between the scientific consensus and its critics. Organisations that provide specialised training, data and expert networks have improved coverage in many countries, but the improvement is uneven and vulnerable to commercial pressure.

14.8 Literacy for Adaptation

Adaptation requires a specific kind of literacy: the ability to read a hazard map, interpret a forecast, understand a warning, and act on it. That knowledge is practical, local and often held by institutions rather than individuals. The gap between the institution and the household is where many warnings fail.

Teaching adaptation literacy means teaching people how to prepare for the hazards that are likely in their area, how to interpret official information, and how to help others. It means teaching children what to do if a heatwave arrives while they are at school, and teaching adults how to make a household plan. It also means teaching the limits of individual action, so that people do not blame themselves for risks that require collective responses.

The most effective programmes are embedded in existing institutions: schools, clinics, religious organisations and local government. They use local languages and local examples, and they are repeated regularly so that knowledge does not decay. A single campaign is not enough, because memory fades and because new people arrive in

the community. Preparedness is a practice, not a lesson.

14.9 Education in Emergencies

Climate change disrupts education directly, through the destruction of schools and the displacement of families, and indirectly, through the loss of livelihoods that forces children to work or marry early. The 2022 Pakistan floods damaged thousands of schools and interrupted the education of millions of children; the 2020 cyclones in Bangladesh destroyed schools that had served as shelters; and drought in the Sahel has pushed families to withdraw girls from school.

The interruption has long-term consequences. Children who miss a year of school are less likely to return, particularly girls; the loss of learning reduces future earnings and increases vulnerability to the next shock. Education is therefore both a victim of climate change and a determinant of adaptive capacity, which is why protecting it is a form of adaptation.

The responses include building schools to higher standards, using them as shelters, providing remote learning when schools are closed, and supporting families so that children are not forced to work. The most effective programmes combine physical investment with social protection, because a school that is rebuilt but whose students are hungry or working will not restore learning.

14.10 Measuring Impact and Avoiding Indoctrination

Measuring the impact of climate education is difficult. Test scores can measure knowledge but not behaviour; surveys can measure attitudes but not understanding; and the outcomes that matter most — preparedness, resilience, informed decisions — are difficult to attribute to any single intervention. The result is that many programmes are evaluated by their inputs rather than their effects.

The risk of indoctrination is real and should be taken seriously. Education should teach students how to think, not what to think; it should present the evidence, explain the uncertainties, and give students the tools to reach their own conclusions. That does not mean presenting false balance between the scientific consensus and its critics; it means being honest about what is known, what is uncertain and why.

The best programmes combine rigour with humility. They teach the physics, the evidence and the uncertainties; they invite students to examine their own assumptions; and they connect knowledge to action in ways that respect the students' autonomy. That combination is more demanding than either advocacy or neutrality, and it is the only one that produces citizens who can think for themselves when the facts change.

14.11 Case Study: Italy's Compulsory Climate Curriculum

In November 2019, Italy's education minister announced that climate change and sustainable development would become compulsory in all state schools. The reform took effect in the 2020–2021 school year and requires about thirty-three hours a year of climate education at every level, from primary to upper secondary. The hours are integrated into existing subjects rather than taught as a separate course, and the ministry framed the change as a response to the demands of students and to the obligations of the

state.

The design reflects a deliberate choice. Rather than adding a new subject to an already crowded timetable, the reform asks teachers of geography, science, history, civics and other subjects to incorporate climate content into their lessons. That approach reaches more students and connects climate to the rest of the curriculum, but it depends on teachers having the knowledge and confidence to teach the material.

The implementation has been uneven. Teacher training was expanded, and the ministry provided guidelines and resources, but surveys found that many teachers felt unprepared and that the actual number of hours varied widely between schools. The COVID-19 pandemic interrupted the first year, and the pressure of the conventional curriculum limited the time available for the new content.

The case illustrates both the promise and the limits of top-down curriculum reform. The promise is that a legal requirement can create space for a subject that markets and schools would otherwise neglect, and that it can signal to teachers and parents that climate is a core part of education. The limit is that a requirement on paper does not guarantee learning in practice; it must be supported by training, resources, assessment and time.

For other countries, the lessons are specific. First, integration across subjects is more effective than a separate course, but it requires coordination. Second, teacher training is the binding constraint, and it must be sustained rather than one-off. Third, assessment shapes what is taught, so climate content must be included in examinations if it is to be taken seriously. Fourth, the content must be connected to local conditions, because abstract global trends do not prepare students for the hazards they will face.

14.12 Future Scenarios and Predictions: Education in 2050

The historical record shows that education has expanded dramatically over the past century: literacy rates have risen from about 12 per cent in 1900 to more than 85 per cent today, and the number of students in tertiary education has grown from a few million to more than 200 million. The expansion has been uneven, and climate change is now threatening to reverse some of the gains in the most exposed regions.

Present conditions give the baseline. About 250 million children are out of school, and about 222 million are affected by emergencies and protracted crises. Only about half of national curricula mention climate change, and the share is lower in the countries that are most exposed. The gap between the need for climate literacy and the provision of it is largest where the consequences of ignorance are most severe.

The assumptions that drive the projections are the rate of curriculum reform, the quality of teacher training, and the impact of climate shocks on schooling. A world that invests in education can produce a generation that understands the risks and is capable of adapting; a world that does not will produce a generation that is exposed to hazards it does not understand and cannot prepare for. The scenarios below summarise the range.

Table 14.1 Climate education indicators, 2050

Indicator	Low reform	Current trends	High reform	What it enables
Curricula with climate content	50%	70%	95%	Basic understanding
Trained teachers	20%	50%	85%	Effective teaching
Students in climate-vulnerable areas	1.2 billion	1.1 billion	0.9 billion	Reduced exposure
Schools used as shelters	20%	40%	70%	Community protection
Green skills in workforce	8%	15%	30%	Transition capacity
Learning loss from disasters	High	Medium	Low	Human capital

Two conclusions follow. First, climate literacy is a precondition for adaptation: people cannot prepare for hazards they do not understand, and communities cannot demand protection they do not know they need. Second, the countries that invest least in education are the ones that will face the largest learning losses from climate shocks, which means that the gap between rich and poor will widen unless education is prioritised.

What would change these conclusions? A rapid expansion of teacher training would improve the quality of climate education even without curriculum reform. A shift toward online and community-based learning could reach students in areas where schools are closed, though it would not replace the social function of schooling. And a change in assessment policy could make climate content a core part of what is taught, rather than an optional extra. The most important variable, as in other chapters, is the pace of emission reduction, because it determines how much adaptation the next generation will need to learn.

CHAPTER 15

DATA, STATISTICS AND MISINFORMATION

Theme: climate data, statistical reasoning and the contest over evidence

On 26 September 2024, Hurricane Helene made landfall in Florida's Big Bend as a Category 4 storm and then travelled inland for hundreds of kilometres, dropping extreme rainfall across Georgia, South Carolina, North Carolina and Tennessee. The flooding in the southern Appalachians was catastrophic: whole towns were cut off, the French Broad River in Asheville reached a level not seen in more than a century, and at least 230 people died across six states. Damage was estimated at more than US$78 billion.

Within hours of landfall, a second event was unfolding online. Posts claimed that the federal government had engineered the storm, that relief agencies were confiscating donations, that rescue teams were refusing to help people with certain political views, and that officials were seizing land for lithium mining. None of it was true. The Federal Emergency Management Agency set up a page to rebut the rumours; local officials pleaded for people to stop sharing them; and meteorologists reported threats against their staff.

The misinformation did not stay online. It affected whether people evacuated, whether they applied for aid, and whether they trusted the warnings that arrived before the next storm. In a disaster, information is infrastructure: it determines whether people move, whether they shelter, and whether they accept help. When it fails, the physical response fails with it.

This chapter is about how climate knowledge is produced, how it is misread, and how it is deliberately distorted. It explains what climate data actually are, how scientists measure consensus and attribute extreme events, and which statistical mistakes recur most often. It then turns to the mechanics of misinformation and to what actually works against it. The aim is not to make every reader a statistician but to make every reader harder to fool.

15.1 What a Climate Number Actually Means

The statement that 2024 was the first calendar year more than 1.5 °C above pre-industrial levels is precise only in relation to a baseline, and the baseline is a choice. Pre-industrial usually means 1850–1900, the period when instrument coverage became global enough to estimate a planetary average. Change the baseline and the number changes: relative to 1981–2010, 2024 was about 1.2 °C warmer; relative to the last glacial maximum it is a different quantity altogether.

Climate numbers are usually expressed as anomalies rather than absolute temperatures, because anomalies are more comparable across space and time. A station in Siberia and one in Indonesia have very different absolute temperatures, but their year-to-year changes can be averaged meaningfully. The convention avoids being distorted by the uneven distribution of weather stations, but it also makes the numbers abstract, which is one reason they are easy to misrepresent.

Every climate number also carries an uncertainty range, and the range is not a sign of weakness. It reflects incomplete coverage, instrument error, and the choice of method. The global mean surface temperature estimated by NASA, NOAA, the Met Office and Berkeley Earth agree within a few hundredths of a degree, which is a remarkable convergence given that they use different station networks and different interpolation methods. Convergence of independent methods is the strongest evidence a reader can ask for.

15.2 Where the Data Come From

The instrumental record is assembled from land stations, ocean buoys and ships, weather balloons, aircraft, and satellites. Each has strengths and weaknesses. Land stations give long records but are unevenly distributed, with dense coverage in Europe and North America and sparse coverage in parts of Africa and South America. Ocean measurements cover two-thirds of the planet but were historically taken by merchant ships along trade routes, leaving gaps that models must fill.

Satellites provide global coverage but only since the late 1970s, and they measure different things: microwave sounders estimate lower-troposphere temperature, radar altimeters measure sea level, and radiometers measure sea-surface temperature. Satellite records must be calibrated across successive instruments, and small differences in calibration can produce apparent trends that are artefacts rather than signals — a fact that has been exploited by campaigns against the data.

Reanalysis products such as ERA5 combine observations with a weather model to produce a physically consistent reconstruction of the atmosphere going back to 1940, and proxy records — tree rings, ice cores, corals, sediments — extend the picture back thousands of years. The proxies are less precise but they establish the context: the current warming is far faster than anything in the Holocene, and the concentration of carbon dioxide is higher than at any point in at least 800,000 years, and probably three million.

15.3 Measuring Scientific Consensus

The claim that about 97 per cent of climate scientists agree that human activity is warming the planet comes from several independent studies. Doran and Zimmerman (2009) surveyed Earth scientists and found that 97 per cent of those who actively publish climate research agreed that human activity is a significant contributing factor. Cook et al. (2013) examined nearly 12,000 abstracts and found that among those expressing a position, 97.1 per cent endorsed the consensus, a figure later revised upward when the sample was corrected.

Lynas et al. (2021) reviewed 88,125 climate-related papers published between 2012 and 2020 and found that more than 99.9 per cent of them endorsed the consensus. That is an extraordinary level of agreement for any scientific question, comparable to the consensus on plate tectonics or the germ theory of disease. It does not mean that the science is settled in every detail; it means that the central claim is not seriously disputed by specialists.

The gap between scientific consensus and public perception is itself measured. Surveys in the United States find that the public estimates the share of scientists who agree at around 70 per cent, well below the actual figure, and that this underestimate is associated with lower support for policy. The gap is sometimes called the consensus gap, and it is produced by the same mechanisms that produce other misconceptions: repetition, selective exposure and the prominence of dissenting voices disproportionate to their number.

15.4 Attribution Science

Event attribution asks a counterfactual question: how likely would this heatwave, flood or drought have been in a world without human influence? The method compares large ensembles of climate model simulations with and without historical greenhouse gas emissions, and sometimes combines them with observed trends. It does not say that climate change caused a specific storm, because weather always has multiple causes; it says that the probability or intensity of the event has shifted.

The field has matured quickly. World Weather Attribution, a collaboration of scientists in Europe and the United States, now produces rapid analyses within days or weeks of an event. Studies have found that the 2021 Pacific Northwest heatwave was made at least 150 times more likely by climate change, that the 2022 Pakistan flooding was made more intense by warming, and that the 2024 European heat was made substantially more likely and hotter.

Attribution is powerful because it connects global statistics to local experience, but it is also easy to misrepresent. A finding that an event was made twice as likely does not mean it was twice as severe, and a finding that no human signal was detected in a particular event does not mean climate change is not happening. Absence of attribution for one event is not evidence against the broader trend, just as a single cold week is not evidence against a warming climate.

15.5 The Statistical Fallacies That Recur

The most common fallacy is cherry-picking the baseline. A widely repeated claim that warming stopped in 1998 depends on choosing a year with an exceptionally strong El Niño as the starting point. Begin the analysis three years earlier or later and the trend is clear. The same technique is used to argue that sea ice is recovering, that the 1930s were hotter, or that a particular glacier is growing: select the window that produces the desired slope.

A second fallacy is confusing weather with climate. A cold snap in one region does not disprove a global trend any more than a hot day proves it. The relevant question is whether the frequency and intensity of events are changing over decades and across regions, which requires aggregating data rather than selecting anecdotes. Anecdotes are vivid and statistics are abstract, which is why the fallacy persists.

A third is misreading correlation and causation, usually in the claim that carbon dioxide follows temperature rather than causing it. In the ice-core record, carbon dioxide does rise after temperature at the onset of glacial terminations, because the initial warming is driven by orbital cycles and the ocean then releases carbon dioxide,

which amplifies the warming. That is a feedback, not a refutation; the physics of carbon dioxide absorbing infrared radiation was established in the nineteenth century and is not in dispute.

15.6 Misinformation in Disasters

Disasters create ideal conditions for misinformation: uncertainty is high, official information is slow, emotions are strong, and people are desperate for explanations. Rumours fill the vacuum. After Hurricane Katrina in 2005, reports of widespread violence and hundreds of deaths in the Superdome circulated widely and were later found to be largely false; after the 2011 Tōhoku earthquake, false claims about radiation spread faster than official advisories.

The pattern has intensified with social media. During Hurricane Helene in 2024, false claims about weather modification, land seizure and diverted aid reached millions of people, and officials reported that they interfered with relief. FEMA created a rumour-response page, and the North Carolina authorities warned that misinformation was deterring people from applying for assistance. The effect is measurable: trust in warnings falls, and evacuation and aid uptake decline.

Artificial intelligence has made the problem harder. Generated images of flooded streets or destroyed neighbourhoods can be produced in minutes and are difficult to distinguish from real photographs, especially on a phone screen during an emergency. Detection tools lag behind generation, and the platforms that distribute the content have limited incentive to slow it. The result is that the evidentiary value of a photograph has fallen, which is a problem for journalists, insurers and courts as well as for emergency managers.

15.7 Reading a Graph and Spotting a Trick

Many misleading climate claims are made through graphs rather than words, because a graph appears objective. The common tricks are simple: truncating the vertical axis so that a small change looks dramatic, or extending it so that a large change looks small; choosing a start date that flatters a trend; plotting a short window and drawing a long conclusion; and comparing datasets that measure different things.

A reader can apply four checks. First, what is the baseline or reference period, and why was it chosen? Second, what is the vertical axis, and does it start at zero or at a value that exaggerates the change? Third, what is the time window, and is it long enough to distinguish a trend from noise? Fourth, who produced the data, and is the underlying dataset available? The last question is the most powerful, because manipulated graphs rarely come with accessible data.

Legitimate scientific communication has its own conventions that can look like weakness. Error bars, ranges and conditional language are marks of care, not evasion. A scientist who says an outcome is likely under a scenario is being precise; a commentator who says it is certain is being careless or dishonest. Learning to read hedged language as a sign of rigour is one of the most useful skills a citizen can acquire.

15.8 Prebunking, Debunking and Inoculation

The evidence on correcting misinformation is mixed but not hopeless. Debunking after the fact works partially: it reduces belief among people who see the correction, but it rarely reaches everyone who saw the original claim, and repeating the falsehood can strengthen it among some audiences. The practical implication is to lead with the fact, explain the technique rather than dwell on the claim, and avoid repeating the falsehood more than necessary.

Prebunking — warning people about manipulation techniques before they encounter them — has stronger evidence. Inoculation theory, developed by William McGuire in the 1960s, holds that exposing people to a weakened form of a misleading argument builds resistance to the full version. Field experiments by van der Linden and colleagues found that short videos explaining techniques such as emotional manipulation, false dichotomies and scapegoating improved people's ability to identify misinformation across topics.

The most durable defence is structural: reduce the supply of misinformation, increase the salience of reliable sources, and make accurate information easy to find and act upon. That means funding public broadcasting, requiring platforms to disclose how their algorithms amplify content, supporting local journalism, and ensuring that official warnings are timely, specific and consistent. Individual literacy matters, but it cannot compensate for an information environment designed to confuse.

15.9 Open Data, Reproducibility and Data Gaps

Climate science is unusually open. The major temperature records are published with their methods and code; satellite data are archived and distributed; model outputs are contributed to the Coupled Model Intercomparison Project and made available to researchers worldwide; and the Intergovernmental Panel on Climate Change reports are reviewed by hundreds of scientists and published in full. This openness is the best available answer to claims of conspiracy, because anyone with the skills can reproduce the analysis.

But the data are not evenly distributed. Station coverage is sparse in parts of Africa, South America and Southeast Asia, which means that the regions most vulnerable to climate change often have the weakest observational records. The result is greater uncertainty in local projections precisely where adaptation decisions are most urgent. Programmes to digitise historical records, expand automatic weather stations and support national meteorological services are therefore a form of climate resilience.

Reproducibility is also a social practice. A dataset that is published but undocumented is not usable; a method that is described but not coded is not replicable. The most valuable institutions in this space are those that maintain long records, document their methods and respond to criticism in public. Their reputation is built on being checkable, which is exactly what a conspiracy theory cannot be.

15.10 Communicating Numbers Honestly

Numbers are persuasive but easily misread, so communication requires care. A temperature rise of 1.5 °C sounds small to people who experience 1.5 °C as the difference between a cool and a warm day; the same change in a planetary average is enormous, because it represents a shift in the distribution of heat, not a uniform increase. Explaining the difference between a local temperature and a global average is one of the most useful things a communicator can do.

Uncertainty language matters too. The IPCC uses calibrated terms: virtually certain means 99–100 per cent probability, very likely 90–100 per cent, likely 66–100 per cent, about as likely as not 33–66 per cent. These terms are defined so that readers can interpret them consistently, but they are often read as vague hedging. Translating them into plain probabilities, without stripping the uncertainty, is a skill worth teaching.

Finally, communication should respect the audience. People are not empty vessels into which facts are poured; they have experience, values and trusted sources. A message that begins with local experience, acknowledges what is uncertain, and offers concrete action is more likely to be heard than one that begins with a global statistic and a moral instruction. Honesty about uncertainty is not a weakness in persuasion; it is the foundation of trust, and trust is the resource that matters most when the next warning arrives.

15.11 Case Study: Hurricane Helene and the Information Emergency

Hurricane Helene made landfall on 26 September 2024 as a Category 4 storm near Perry, Florida, then moved rapidly inland. Its wind field was unusually large, and its moisture was drawn from an exceptionally warm Gulf of Mexico. The result was extreme rainfall over the southern Appalachians, a region where steep terrain funnels water into narrow valleys and where communities are often accessible by a single road.

The physical damage was severe. At least 230 people died across six states, making Helene the deadliest mainland hurricane since Katrina in 2005. Asheville and surrounding counties were cut off; the region's water system was destroyed; and damage estimates exceeded US$78 billion. Power and communications failed across a wide area, which meant that many people relied on social media for information precisely when it was least reliable.

The information environment collapsed faster than the roads. Claims that the federal government had created or steered the storm, that relief workers were confiscating supplies, and that officials were seizing land for mining spread across platforms and were amplified by political figures. FEMA published a rumour-response page; state officials urged residents to verify before sharing; and meteorologists reported threats. Surveys later found that belief in the false claims was concentrated among people who distrusted government generally.

The practical consequences were measurable. Officials reported that some residents did not apply for assistance because they believed the aid would be taken or would obligate them; others delayed evacuation before subsequent storms. Aid organisations found that they had to spend time rebutting rumours instead of delivering

supplies. The episode showed that trust is not a soft variable: it determines whether warnings are obeyed and whether help is accepted.

The lessons are specific. Warnings must arrive through trusted local messengers, not only through national agencies. Correction must be fast, concrete and repeated, because rumours spread in the gap between the event and the official response. And platforms must be treated as part of the emergency infrastructure, with obligations that match their influence. A community that has accurate information and trusts it recovers faster than one that has the same physical damage and does not.

15.12 Future Scenarios and Predictions: The Information Environment in 2050

The historical record shows that misinformation is not new; what is new is its speed and scale. Rumours spread by word of mouth during the Lisbon earthquake of 1755; radio amplified propaganda in the twentieth century; and social media compressed the cycle from days to minutes. Each transition increased reach and reduced the time available for verification.

Present conditions define the starting point. Around 5 billion people use social media, and surveys in many countries find that a majority get at least some news from it. Generative artificial intelligence has made synthetic text, images and audio cheap; research by Reuters Institute and others finds that audiences struggle to distinguish them from authentic material. Trust in news media has declined in most wealthy democracies, while trust in scientists remains comparatively high, which creates an opening for scientists who communicate directly.

The assumptions that drive the projections are the pace of regulation, the cost of detection, and the resilience of local institutions. A world that invests in verification infrastructure, media literacy and trusted local reporting can limit the damage; a world that does not will face disasters in which the information environment fails alongside the physical one. The table summarises the range.

Table 15.1 Information environment indicators, 2050

Indicator	Fragmented	Current trends	Resilient	Consequence
Share of news from social platforms	70%	55%	35%	Speed vs verification
Synthetic media detection	Poor	Moderate	Good	Evidentiary trust
Local news outlets	Few	Declining	Stable	Accountability
Trust in official warnings	Low	Moderate	High	Evacuation compliance
Media literacy in schools	20%	45%	80%	Resistance to manipulation
Platform disclosure rules	None	Partial	Strong	Algorithmic transparency

Two conclusions follow. First, misinformation is a disaster-risk multiplier: it does not create the flood, but it determines whether people escape it. Second, the defences that work are local and structural — trusted messengers, funded journalism, open data and literate citizens — rather than a single technological fix.

What would change these conclusions? Regulation that requires platforms to disclose how they rank content could reduce the amplification of falsehood, though enforcement across borders is difficult. Cheap and reliable provenance systems — cryptographic signatures on images and video — could restore part of the evidentiary value of a photograph. And a sustained investment in local journalism could rebuild the institutions that once filtered rumour. The most important variable remains the same as elsewhere: the pace of climate change determines how often the information system will be tested, and the quality of that system determines how many people survive the test.

CHAPTER 16

HOUSEHOLD PREPAREDNESS: WATER, FOOD, SHELTER AND RISK

Theme: household stocks, protective action and the first 72 hours

When Hurricane Maria struck Puerto Rico on 20 September 2017, it did more than damage buildings. It destroyed the island's electricity system, knocked out water treatment and pumping, closed roads and ports, and left 3.4 million people without the systems they relied on. In the following weeks, households that had stored water, food, medication and fuel managed; those that had not queued for hours, or went without.

The official death toll was initially reported as 64. A subsequent study by researchers at George Washington University, published in 2018, estimated 2,975 excess deaths in the months after the storm. Many of those deaths were not caused by wind or water but by the interruption of chronic care: dialysis that could not be performed, insulin that spoiled without refrigeration, oxygen concentrators that stopped when the power failed, and conditions that became fatal because hospitals were overwhelmed.

The Puerto Rico case is often told as a story about infrastructure failure, and it is. But it is also a story about household preparedness, because the households that coped were not necessarily wealthier or luckier. They were the ones that had water in containers, food that did not need cooking, a way to keep medicine cold, a plan for where to go, and neighbours who checked on them.

This chapter is practical. It sets out what a household actually needs to survive the first days and weeks of a disruption, how much water and food that requires, how to shelter from heat, smoke, cold and flood, and how to plan for the things that fail silently: medication, power for medical devices, documents, cash and communication. It also examines why so few households prepare, and what actually changes behaviour.

16.1 Households as the First Responders

In most disasters, the first people to help are not professionals. They are neighbours, family members and passers-by, who pull people from vehicles, share water, clear debris and check on the vulnerable. Research on disaster response consistently finds that informal assistance accounts for a large share of immediate rescue and support, particularly in the first hours when formal services cannot reach the affected area.

The protective action decision model developed by Lindell and Perry (2012) explains why some households act and others do not. People must receive a warning, understand it, believe it is credible and personally relevant, identify a feasible action, and have the resources and social support to carry it out. A failure at any stage produces inaction. This is why warnings that are technically accurate still fail: they arrive in a form that does not connect to what people know, or they recommend actions that people cannot afford.

Preparedness is therefore not only a matter of stockpiles. It is a matter of capability: knowing what to do, having the means to do it, and being connected to others who can help. A household with two weeks of water but no plan for an elderly

neighbour is less prepared than one with three days of water and a functioning neighbourhood network, because the second can adapt to circumstances the first did not anticipate.

16.2 Water: The Binding Constraint

Water is the first thing to fail and the most urgent to replace. A person needs about three to four litres a day for drinking and basic hygiene in normal conditions, and considerably more in heat or during physical work. Emergency guidance commonly recommends one gallon (about 3.8 litres) per person per day, with additional water for pets, cooking and sanitation; a two-week supply for a family of four is therefore roughly 100–150 litres, which is heavier and bulkier than most people expect.

Storage is constrained by space and weight. Water weighs one kilogram per litre, so a two-week supply cannot be carried far and must be stored where it will be used. Households in apartments often have less space than they need; households in informal settlements may have none. That is why community-level storage — tanks at schools, mosques, clinics and community centres — is as important as household storage, and why water points must be planned into buildings from the start.

Treatment matters as much as storage. Boiling water for one minute at sea level kills most pathogens; chlorine tablets or unscented household bleach can disinfect water if dosed correctly; and ceramic or membrane filters remove pathogens and sediment. After the 2021 Texas winter storm, about 14 million people were placed under boil-water notices, many for days, and households that had no way to boil water or no stored supply depended on distribution points that were themselves disrupted by the same storm.

16.3 Food: Calories, Shelf Life and Cooking

A person needs roughly 2,000–2,500 kilocalories a day, and more during cold, hard physical work or recovery. A two-week supply for a family of four is therefore around 250,000 kilocalories, which is a substantial volume: rice, pulses, oil, sugar and canned goods provide the cheapest calories per kilogram, while freeze-dried meals provide convenience at a much higher price. The practical rule is to store what the household already eats, rotate it, and keep the cooking method in mind.

Cooking may be impossible. Power cuts disable electric stoves and microwaves; gas may be shut off; and indoor use of charcoal or portable gas burners creates a carbon monoxide hazard that has killed people in every major outage. Households therefore need food that can be eaten without cooking — canned beans, crackers, nut butters, dried fruit, ready-to-eat meals — as well as a safe way to cook outdoors or in a well-ventilated space if fuel is available.

Particular groups need particular supplies. Infants need formula and clean water; people with diabetes need appropriate food and insulin storage; people with coeliac disease or allergies cannot rely on generic relief supplies. Relief systems often assume a generic adult, which is why households with special needs should plan for themselves and tell a neighbour or relative where their supplies are kept.

16.4 Shelter: Staying, Going and the Space In Between

The first decision in a hazard is whether to stay or to leave. Staying is safer for short, local hazards such as a heatwave, a smoke episode or a flood that will not reach the building; leaving is safer for hazards that will destroy the building, such as a storm surge, a wildfire front or a flood that will submerge the ground floor. The decision depends on the hazard, the building, the route and the person, and it must be made before the event, not during it.

Sheltering in place requires a suitable room. For heat, the coolest room on the lowest floor, shaded and ventilated at night, is best. For smoke, a room that can be sealed with tape and towels, with a portable air cleaner or a homemade filter box, reduces exposure. For winter storms, a room that can be closed off and heated by body warmth and blankets conserves heat. For floods, the highest floor is safer, but only if the building is structurally sound and escape routes remain open.

Evacuation requires a destination, a route and a means of transport. Households without a car are the most vulnerable, and they are disproportionately poor, elderly or disabled. Planning must therefore include a way to leave that does not assume private vehicle ownership: a neighbour with a car, a community transport list, or a designated pickup point. Evacuation that is ordered but not enabled is not a plan.

16.5 Power, Light and Heat

Electricity is the quiet dependency behind almost everything else: refrigeration, communications, medical devices, water pumping and lighting. Outages are becoming longer and more frequent in many countries. In the United States, the average customer experienced about five to six hours of outage a year in the early 2010s; by the early 2020s the figure had risen, and extreme events produced outages measured in weeks. In Puerto Rico after Maria, parts of the island waited months for restoration.

A household does not need to power everything; it needs to power a few things. A small solar panel with a battery can charge phones and run a light; a portable power station can run a refrigerator for part of the day or a medical device; and a generator can run a pump or a heater, but only outdoors and away from windows. The last point is essential: carbon monoxide poisoning is consistently among the leading causes of death in winter storm outages, and it is entirely preventable.

Refrigeration has a clock. A closed refrigerator keeps food safe for about four hours; a full freezer for about 48 hours if the door stays shut. Households can extend that by freezing water bottles in advance, grouping food, and keeping a thermometer inside. Knowing when to discard food is as important as knowing how to store it, because food poisoning during a disaster compounds an already dangerous situation.

16.6 Health, Medication and Disability

For people with chronic conditions, preparedness is a medical matter. A week or two of medication, a copy of prescriptions, a cooler and ice packs for insulin, spare batteries for hearing aids and medical devices, and a written list of conditions and allergies can be the difference between managing at home and needing emergency care.

After Maria, the interruption of dialysis and oxygen supply was a major contributor to excess mortality.

Power-dependent medical equipment requires a plan. Oxygen concentrators, ventilators, suction machines, infusion pumps and refrigerated medicines all depend on electricity, and households should register with their utility or health provider for priority restoration where such schemes exist. Backup power — a battery, a generator, or a planned relocation to a powered site — must be arranged before the outage, because afterwards it may be impossible to obtain.

Disability and frailty shape every other decision. A person who cannot climb stairs cannot shelter on an upper floor; a person who uses a wheelchair cannot evacuate through debris; a person with dementia may not understand a warning. Preparedness must be specific to the person, and it must involve the people around them, because institutional support may not arrive in time.

16.7 Documents, Money and Communication

Disasters destroy the records that prove identity, ownership and entitlement. Birth certificates, passports, title deeds, insurance policies, medical records, prescriptions and bank details should be copied, stored in a waterproof container and kept in a second location or in encrypted digital form. Households that lose these documents struggle to claim insurance, access aid or prove residency for months after the event.

Cash matters because card systems depend on power and connectivity. A modest amount of small-denomination cash allows a household to buy fuel, water or transport when payment systems fail. In countries where mobile money is dominant, a charged phone and a backup power source serve the same function, provided the network stays up — which is why the failure of communications is often the most disorienting part of a disaster.

Communication requires a plan made in advance. Families should agree on an out-of-area contact who can relay messages, a meeting point, and a way to signal that they are safe. During major events, mobile networks congest, so text messages often work when calls do not; offline maps, downloaded documents and pre-written messages reduce dependence on connectivity. The aim is not to stay in constant contact but to be able to reconnect when the network returns.

16.8 Assessing Household Risk

Preparedness begins with knowing what is likely. A household should identify the hazards that affect its area — heat, flood, wildfire, storm, landslide, drought, industrial accident — and the specific exposure of its building and neighbourhood: distance to a river, slope, roof type, floor level, access road, and the location of the nearest shelter and clinic. Local hazard maps, where they exist, are the place to start.

The assessment should include the things that are easy to overlook: how long the household can survive without power, water and transport; who depends on whom; what the evacuation route looks like at night and in rain; and whether insurance covers the likely hazard. Many households discover after a disaster that their policy excludes

flood, or that the sum insured is far below the cost of rebuilding.

Preparedness is unequal. Surveys consistently find that higher-income households are more likely to have supplies, plans and insurance, while lower-income households face the same hazards with fewer resources. That is not a failure of individual responsibility but a structural problem: preparedness costs money and time, and public policy that treats it as a private duty without providing support will produce predictable inequality in survival.

16.9 Beyond Seventy-Two Hours

Official guidance in many countries recommends three days of supplies, on the assumption that outside help will arrive within 72 hours. The assumption is often wrong. After Hurricane Katrina in 2005, parts of New Orleans were without basic services for weeks; after Maria, restoration took months; and in conflict-affected or remote areas, help may take longer still. Three days is a floor, not a target.

A more realistic standard is two weeks of water, food and medication, with the capacity to be self-sufficient for longer if necessary. That is a significant commitment of space and money, and not every household can meet it. The practical approach is layered: a small grab-and-go kit for immediate evacuation, a larger stay-at-home stock for sheltering in place, and a community plan for the cases where neither is enough.

The layers should be tested. A household that has never opened its water containers, run its generator or walked its evacuation route does not know whether the plan works. Annual checks — replacing expired food and medication, testing equipment, updating contacts — turn a stockpile into a capability. Preparedness that is never exercised tends to fail at exactly the moment it is needed.

16.10 Community Preparedness and Mutual Aid

Households do not prepare in isolation. Neighbours, extended families, religious congregations, schools and local associations provide most of the immediate help in a disaster, and their effectiveness depends on how well they know each other before it happens. Programmes such as Map Your Neighborhood in Washington State and Japan's self-help disaster organisations build that familiarity deliberately, by mapping vulnerable residents, agreeing roles and practising responses.

Mutual aid is most effective when it is organised. A neighbourhood that knows who has a generator, who has medical training, who needs help evacuating and who has a vehicle can allocate resources in the first hours, when formal systems are overwhelmed. That requires a list, a communication method and a small amount of practice, not a large budget. The main obstacle is social: people do not know their neighbours, and preparedness programmes that ignore social connection tend to produce supplies without coordination.

Public policy can support this without taking it over. Municipalities can fund community preparedness coordinators, provide training and equipment, and recognise neighbourhood groups in official plans. The most resilient places combine strong household preparedness with strong social networks and functioning local government; none of the three is sufficient alone, and the weakest link determines the outcome.

16.11 Case Study: Hurricane Maria and Household Survival in Puerto Rico

Hurricane Maria made landfall in Puerto Rico on 20 September 2017 as a Category 4 storm, two weeks after Hurricane Irma had passed close to the island. Maria's winds and rainfall destroyed the electricity grid, damaged or destroyed tens of thousands of homes, blocked roads with debris, and disrupted water and communications across the territory of 3.4 million people. Damage was estimated at around US$90 billion.

The grid failure was total. Puerto Rico's electricity system, already weakened by underinvestment and by Irma, collapsed, and restoration proceeded slowly; many households remained without power for months, and some for nearly a year. Water pumping depends on electricity, so the loss of power became a loss of water, and the loss of water became a sanitation and health crisis. Hospitals ran on generators until fuel ran short.

The mortality data tell the story of a slow emergency. The initial official count of 64 deaths was widely criticised, and a study by researchers at George Washington University estimated 2,975 excess deaths in the months following the storm. Many were caused by the interruption of care: dialysis, oxygen, insulin refrigeration and access to medication. Others were caused by heat, unsafe water, carbon monoxide from generators, and the cumulative stress of months without basic services.

Households that coped shared certain features. They had stored water — often in cisterns, which are common on the island — and food that did not require cooking. They had medication, or a plan for obtaining it. They had neighbours who shared generators, fuel and information. And they had a way to cook outdoors without creating a fire or carbon monoxide hazard. Wealth helped, but social connection helped more, especially in rural areas where formal aid arrived last.

The lessons for household preparedness are direct. Water storage must be sized for weeks, not days. Medication and power-dependent equipment require a plan that includes a backup power source and a relocation option. Cooking must be possible without electricity and without indoor combustion. And preparedness must be social, because the households that survived best were embedded in networks that could share resources. A stockpile without neighbours is a fragile plan.

16.12 Future Scenarios and Predictions: Household Preparedness in 2050

The historical record shows a widening gap between the length of disruptions and the length of household supplies. Average outage duration has risen in many countries, displacement from disasters has increased, and the number of people exposed to extreme heat has grown. Meanwhile, survey data suggest that only a minority of households meet basic preparedness standards: in the United States, FEMA surveys have found that around half of adults have taken at least one preparedness action, but far fewer have a two-week supply of water and food.

Present conditions set the baseline. Roughly 2 billion people lack safely managed drinking water at home, and about 3.6 billion lack safely managed sanitation; for them, household storage is not a supplement to a functioning system but a substitute

for one. In wealthy countries, preparedness is uneven: higher-income, older and homeowner households are more likely to have supplies, while renters, younger households and low-income families are less likely, despite facing similar or greater exposure.

The assumptions behind the projections are the frequency of disruption, the cost and availability of preparedness goods, and the strength of community networks. A world that invests in household and community preparedness can reduce mortality even as hazards increase; a world that relies on emergency services alone will see deaths concentrated among those who could not prepare. The table summarises the range.

Table 16.1 Household preparedness by hazard, 2050

Hazard	Lead time	Critical stock	Common failure	Minimum standard
Heatwave	Days	Water; cooling; fans	No cool refuge	2 weeks water; cool room
Flood	Hours	Drinking water; documents	Late evacuation	Go-bag; route; higher floor
Wildfire smoke	Days	Filters; sealed room	No clean air	Air cleaner; N95 masks
Winter storm	Days	Heat; fuel; food	Carbon monoxide	Safe heating; CO alarm
Hurricane	Days	Water; power; medicine	Grid and water loss	2 weeks self-sufficiency
Drought	Months	Stored water; cash	No rationing plan	Storage; conservation plan

Two conclusions follow. First, household preparedness is a form of adaptation that individuals can act on immediately, which makes it psychologically as well as practically valuable. Second, it cannot substitute for public systems: households without safe water or reliable power cannot store their way out of a systemic failure, and policy that expects them to do so transfers risk onto those least able to carry it.

What would change these conclusions? Subsidised preparedness goods and insurance incentives could close the income gap in stockpiles. Building codes that require water storage, safe rooms and backup power in new housing would embed preparedness in the structure rather than in individual effort. And community programmes that map vulnerable residents and organise mutual aid could compensate for the erosion of social connection. The most important variable remains the pace of climate change, which determines how often households will need to rely on their own supplies, and how long they will have to last.

CHAPTER 17

HEALTH, EMERGENCY RESPONSE AND SPECIAL POPULATIONS

Theme: health systems, emergency response and unequal risk

In August 2003, a high-pressure system settled over Europe and stayed for two weeks. Daytime temperatures exceeded 40 °C in parts of France, night-time temperatures stayed high enough to prevent recovery, and the heat persisted long enough to overwhelm the body's ability to regulate itself. Across twelve countries, an estimated 70,000 people died. In France alone, the figure was about 14,800, most of them elderly people living alone.

The deaths were not distributed evenly. They were concentrated among people over 75, among those with cardiovascular and respiratory disease, among people living in top-floor flats under zinc roofs, and among people without air conditioning or social contact. Many died at home, and many were not discovered for days. The crisis was not only meteorological; it was also a failure of care, because the institutions that might have checked on them were closed for the August holiday.

The 2003 heatwave changed how Europe thinks about heat. France created a heat-health warning system, established a register of vulnerable people, and required care homes to have cool rooms. Subsequent heatwaves produced fewer deaths per degree of heat, which suggests that the response, not the hazard, had changed. The 2022 European summer was hotter than 2003 in many places, and the death toll was estimated at 61,700 — high, but far lower per degree than two decades earlier.

This chapter examines the health consequences of climate change and the capacity of health systems to respond. It covers heat as a mass casualty event, the strain on emergency services and hospitals, the particular vulnerability of specific populations, and the long shadow of mental health. It argues that health is where the abstractions of climate change become visible as bodies, and that the distribution of those bodies is a matter of policy, not fate.

17.1 Health Systems as Climate Infrastructure

Health systems are the institution that society turns to when climate change injures people, and they are themselves vulnerable to it. Heat drives demand for ambulance services and emergency departments; floods damage hospitals and cut supply chains; storms disrupt power and water; and outbreaks of infectious disease follow displacement and sanitation failure. A health system that is not prepared for these shocks cannot protect the population it serves.

The World Health Organization has described climate change as the single biggest health threat facing humanity, and the Lancet Countdown tracks the indicators annually. Its 2024 report found that heat-related mortality among people over 65 had risen by 167 per cent compared with the 1990s, that 2023 exposed billions of people to days of dangerous heat, and that health systems in most countries remain underprepared for the risks they face.

Preparedness has four components: surveillance, so that threats are detected early; capacity, so that services can absorb surges; infrastructure, so that facilities keep functioning during hazards; and equity, so that the most vulnerable are reached first. Most health systems are strong on one or two of these and weak on the others, and the weakness is usually equity, because the people at greatest risk are often the least able to demand services.

17.2 Heat as a Mass Casualty Event

Heat kills in ways that are easy to miss. It exacerbates cardiovascular and respiratory disease, kidney injury and diabetes; it increases the risk of preterm birth and stillbirth; it impairs cognition and judgement; and in extreme cases it causes heat stroke, in which core temperature rises faster than the body can cool and organs begin to fail. The deaths are often recorded as heart attacks or kidney failure rather than heat, which is why excess mortality analysis is needed to see the true toll.

The physiological limit is well established. Wet-bulb temperature — a measure that combines heat and humidity — of about 35 °C represents the point at which a healthy adult at rest cannot shed metabolic heat, even with unlimited water and shade. Sherwood and Huber (2010) identified this threshold, and later experimental work by Vecellio et al. (2022) found that even lower wet-bulb temperatures, around 31 °C, can be dangerous for older adults and people with chronic illness.

Mass casualty heat events share a pattern. Mortality rises sharply after two or three consecutive hot days and nights, because the body needs the cool of night to recover; it is concentrated among the elderly, the isolated and the poor; and it is underestimated in real time because deaths are attributed to other causes. The public-health response that works is unglamorous: warnings, cooling centres, checking on vulnerable people, adjusting work hours, and opening public buildings at night.

17.3 Emergency Services and Surge Capacity

Emergency medical services are the first formal layer of response, and they are sensitive to climate. Ambulance response times lengthen in heat, because call volumes rise and crews are slower in protective equipment; floods block roads and strand vehicles; and wildfires force evacuations that overwhelm dispatch. Studies in several cities have found that heat is associated with a measurable increase in emergency calls and in response times.

Hospitals face a parallel problem: a surge in demand coinciding with a reduction in capacity. Staff may be unable to reach work; power and water may fail; elective procedures may be cancelled; and beds may be occupied by people who cannot be discharged because their homes are uninhabitable. COVID-19 demonstrated how quickly a health system can be stretched, and how the stretching falls hardest on the poorest and the oldest.

Surge capacity is built before the event. It requires plans for expanding beds, stockpiles of medicines and oxygen, protocols for triage, and agreements with neighbouring facilities. It also requires staff who are trained and rested, which is why workforce planning is a form of climate adaptation. A hospital with a plan but no staff is

not a surge capacity; it is a document.

17.4 Hospitals and Health Facilities Under Threat

Health facilities are frequently located in hazard-prone areas and frequently fail. Hurricane Sandy in 2012 forced the evacuation of several New York hospitals after power and backup systems failed; Hurricane Katrina in 2005 left Memorial Medical Center in New Orleans stranded without power, water or adequate supplies; and the 2022 Pakistan floods damaged or destroyed more than 2,000 health facilities.

The failure modes are predictable. Flooding enters basements and switchgear; wind damages roofs and windows; heat overwhelms cooling systems; and power outages disable ventilators, imaging, sterilisation and refrigeration. Back-up generators often fail because fuel runs out or because they are located in flooded basements. Water outages disable surgery, dialysis and sanitation, which is why hospitals need on-site water storage and treatment.

Retrofitting hospitals is expensive but cost-effective. Elevating critical equipment, relocating generators above flood level, installing on-site water and power, and hardening building envelopes reduce the risk of evacuation, which is itself dangerous for patients. The World Bank has estimated that making health facilities resilient returns several dollars in avoided losses for every dollar invested, and that the benefit is highest for primary care facilities in exposed regions.

17.5 Special Populations and Unequal Risk

Climate risk is distributed by physiology, occupation and social position. Older adults are less able to regulate temperature and more likely to have chronic disease; children have a higher surface-area-to-mass ratio and are more vulnerable to dehydration; pregnant women face increased risk of preterm birth and complications; and people with disabilities may be unable to evacuate or to access information.

Occupational exposure is a major and often invisible source of risk. Outdoor workers in construction, agriculture, delivery and utilities are exposed to heat for long periods, often with piece-rate pay that discourages rest breaks. Studies in the United States and India have found elevated rates of kidney injury among agricultural workers, and heat standards are weak or absent in many jurisdictions.

Social position determines whether people can protect themselves. Homeless people have no cool room and no way to store water; prisoners are confined in facilities that often lack air conditioning; indigenous communities may be remote from services and burdened by historical dispossession; and low-income households may be unable to afford cooling or to evacuate. Each of these groups requires a specific response, and generic advice will not reach them.

17.6 Mental Health and the Long Shadow

The mental health consequences of disasters outlast the physical ones. Studies after floods, hurricanes and wildfires consistently find elevated rates of post-traumatic stress disorder, depression, anxiety and substance use, often persisting for years. A meta-analysis by Galea and colleagues found that a substantial minority of people

exposed to disasters develop PTSD, with rates highest among those who experienced injury, bereavement or displacement.

Slow-onset events produce their own burden. Drought has been associated with increased suicide rates among rural men in Australia; displacement erodes social networks and identity; and the loss of a home or livelihood can produce a grief that is not recognised by formal systems. Among young people, survey evidence suggests widespread anxiety about climate change, which is sometimes called eco-anxiety and which should be distinguished from pathology: it is a rational response to a real threat.

Mental health services are usually the weakest part of a health system, and they are rarely included in disaster planning. Integrating psychosocial support into emergency response, training community health workers, and providing continuity of care for people with pre-existing conditions are low-cost measures with substantial benefit. The alternative is a population that survives the flood and carries the consequences for decades.

17.7 Surveillance and Early Warning

The first requirement of an effective health response is knowing that something is happening. Heat-health warning systems link meteorological forecasts to health data and trigger actions when thresholds are crossed. The Ahmedabad heat action plan, developed after a 2010 heatwave killed more than 1,300 people, combined warnings, cooling centres, and outreach to vulnerable groups; a subsequent evaluation by Hess et al. (2018) estimated that it had prevented about 1,190 deaths a year.

Syndromic surveillance — monitoring emergency department visits, ambulance calls and pharmacy sales in near real time — can detect outbreaks and heat impacts earlier than formal reporting. Excess mortality analysis, which compares observed deaths with expected deaths, is the gold standard for measuring the true toll of a heatwave or a pandemic. Countries that invested in these systems during COVID-19 are better prepared for climate-related health shocks.

Surveillance must be linked to action. A warning that does not trigger a response is only information. Effective systems specify who does what at each threshold: which agencies open cooling centres, which clinics extend hours, which community workers visit vulnerable residents, and how the public is informed. The plan should be tested before the event, because the first test should not be the emergency itself.

17.8 Outbreaks After Disasters

Disasters create the conditions for infectious disease: displaced populations, contaminated water, crowded shelters, disrupted vaccination and overwhelmed health services. Cholera followed the 2010 Haiti earthquake and killed more than 10,000 people; Yemen's cholera outbreak, exacerbated by conflict and water system collapse, produced more than a million suspected cases; and malaria and dengue surged after the 2022 Pakistan floods.

The risk is not inevitable. Rapid provision of safe water and sanitation, vaccination campaigns, vector control and surveillance can prevent most outbreaks. The failures that produce epidemics are usually failures of coordination and funding rather

than of knowledge. The World Health Organization's response frameworks are well developed; the gap is in implementation, especially in conflict-affected and low-income settings.

Climate change is expanding the range of vector-borne disease. Dengue, carried by Aedes mosquitoes, is spreading into new regions: 2024 saw record outbreaks in the Americas, with more than 12 million suspected cases, and local transmission has been reported in parts of Europe. Malaria is expanding into highland areas that were previously too cool for transmission. These shifts require surveillance and vector control to move faster than the mosquitoes.

17.9 Workforce, Medicines and Supply Chains

A health system is only as strong as its workforce and its supplies. Many countries face chronic shortages of doctors, nurses and community health workers, and the shortages are worst in the regions most exposed to climate change. Migration of health workers from low-income to high-income countries compounds the problem, leaving the poorest systems with the fewest staff.

Supply chains are equally fragile. Medicines, oxygen, vaccines and medical equipment depend on transport, power and refrigeration. A blocked road, a flooded port or a power outage can empty a hospital pharmacy within days. The COVID-19 pandemic showed how quickly global supply chains for oxygen and protective equipment can fail, and how unequally the remaining supplies are distributed.

Resilience requires stockpiles, local manufacturing and diversified suppliers. It also requires cold chains that can survive power cuts, which means solar-powered refrigeration and backup systems at every level. The cost is modest relative to the consequence of a vaccine stock spoiled or an oxygen supply interrupted, and the benefit extends beyond climate to every kind of emergency.

17.10 Universal Health Coverage and Resilient Systems

The most effective protection against climate-related health risk is a health system that people can access without financial ruin. Universal health coverage reduces vulnerability because it ensures that people seek care early, that chronic conditions are managed, and that the costs of disaster-related illness do not push households into poverty. Countries with strong primary care systems consistently show better outcomes in disasters than countries with fragmented, market-based systems.

The World Health Organization's operational framework for climate-resilient health systems identifies six building blocks: leadership and governance, financing, workforce, medical products, information systems, and service delivery. Each must be assessed for climate risk and strengthened accordingly. The framework is useful because it makes resilience a matter of routine management rather than a separate project.

Financing is the binding constraint. The adaptation finance gap for health is large: the World Health Organization has estimated that low- and middle-income countries need several billion dollars a year to strengthen health systems against climate risks, while actual flows are a fraction of that. Closing the gap is not only a matter of

money; it is a matter of priorities, because health is consistently underfunded relative to its contribution to well-being.

17.11 Case Study: The 2003 European Heatwave

In the first two weeks of August 2003, a persistent high-pressure system brought exceptional heat to western and central Europe. Temperatures in France exceeded 40 °C in many places, night-time temperatures remained high, and the heat lasted long enough to exhaust the physiological reserves of vulnerable people. Across twelve countries, an estimated 70,000 people died; France alone recorded about 14,800 excess deaths.

The deaths were concentrated among older people living alone, often in flats with poor insulation and no air conditioning. Many were not discovered for days, because neighbours and family were away on holiday and because the systems that might have checked on them did not exist. Nursing homes were overwhelmed; hospitals were stretched; and morgues ran out of capacity. The event exposed a gap between the meteorological warning and the public health response.

The political reaction was severe. The French health minister resigned, and the government commissioned an inquiry that criticised the absence of a heat plan. In the following years, France created a heat-health warning system, established a register of isolated elderly people, required care homes to provide cool rooms, and ran public information campaigns. Similar systems were developed across Europe.

The systems appear to have worked. When Europe experienced extreme heat in 2022, with temperatures exceeding 40 °C in several countries, the estimated death toll was about 61,700 — still very high, but substantially lower than 2003 relative to the intensity and duration of the heat. Analysis by Ballester et al. (2023) attributed the reduction to improved preparedness, adaptation and public awareness.

The lessons are clear. Heat is a mass casualty event that requires a public health response, not merely a weather warning. The most vulnerable are the isolated elderly, who need active outreach rather than general advice. Preparedness saves lives even when the hazard worsens, which is the central argument for adaptation. And the measure of a response is not the number of alerts issued but the number of people reached and protected.

17.12 Future Scenarios and Predictions: Health Systems in 2050

The historical record shows a clear trend. Heat-related mortality among older adults has risen sharply since the 1990s, the geographic range of vector-borne disease has expanded, and the number of people exposed to extreme heat has increased. At the same time, adaptation has reduced deaths per degree of heat in countries that invested in warning systems, which means the future depends on both the hazard and the response.

Present conditions set the baseline. About half of the world's population lacks access to essential health services, and out-of-pocket spending pushes millions into poverty each year. Health systems in low-income countries face chronic shortages of staff, medicines and infrastructure, and those are precisely the countries where climate risks are growing fastest. In high-income countries, systems are better resourced but ageing, and they face rising demand from an older population.

The assumptions behind the projections are the pace of warming, the level of investment in health systems, and the effectiveness of adaptation. A world that invests in primary care, surveillance and resilient facilities can reduce deaths even as hazards increase; a world that does not will see health gains reversed, particularly in the countries least responsible for emissions. The table summarises the range.

Table 17.1 Health system indicators, 2050

Indicator	Low adaptation	Current trends	High adaptation	Health outcome
Excess heat deaths (annual)	~700,000	~500,000	~250,000	Avoidable mortality
Population in endemic dengue zones	5.5 billion	5 billion	4 billion	Infectious disease
Health facilities climate-resilient	20%	45%	80%	Service continuity
Universal health coverage	60%	75%	95%	Financial protection
Mental health service coverage	15%	30%	60%	Psychosocial burden
Health adaptation finance	Low	Moderate	High	System capacity

Two conclusions follow. First, health is where climate change is most directly experienced, and it is therefore where adaptation has the clearest returns. Second, the distribution of health risk follows existing inequalities, so the countries and communities with the least capacity face the greatest burden.

What would change these conclusions? A rapid expansion of primary care and community health workers would improve resilience even without new infrastructure. Heat action plans that reach isolated elderly people could cut mortality substantially at low cost. And a shift in finance toward health adaptation could close the gap between need and provision. The most important variable remains the pace of emission reduction, because it determines how much adaptation the health system will have to deliver, and how many people will need care that cannot be provided.

CHAPTER 18

SOCIAL CAPITAL, MUTUAL AID AND GOVERNANCE

Theme: social networks, public institutions and the politics of protection

In the early hours of 15 July 2021, heavy rain fell on the Ahr valley in western Germany. Between 100 and 200 millimetres fell in less than 24 hours on ground that was already wet, and the small river rose within hours from a quiet stream to a torrent. In the district of Ahrweiler, 134 people died; across Germany and Belgium, more than 180 died, and damage reached tens of billions of euros.

The warning chain had partially worked. The German meteorological service issued warnings, some local authorities evacuated, and some residents left. But in many places the warnings did not translate into action. People did not believe the river could rise that fast; some local officials did not order evacuation; and Germany had no nationwide system for sending alerts directly to mobile phones, a gap that was closed only later.

The Ahr valley flood is a case about governance rather than meteorology. The rainfall was extreme but not unimaginable; models had warned that such events were becoming more likely. What failed was the chain that connects a forecast to a decision, a decision to a warning, and a warning to a household that acts. Every link in that chain is an institution, and institutions can be designed well or badly.

This chapter examines the social and political infrastructure of survival: the networks that help neighbours, the institutions that issue warnings and organise response, and the governance arrangements that determine whether protection reaches the people who need it. It argues that social capital and public institutions are not alternatives but complements, and that the quality of governance is one of the strongest predictors of how many people die when hazard strikes.

18.1 Social Capital as a Survival Resource

Social capital refers to the networks, norms and trust that enable people to act together. It is not a metaphor: it can be measured, and it predicts outcomes. Sociologists distinguish bonding capital, which links people who are similar and close, bridging capital, which links people across different groups, and linking capital, which connects citizens to institutions with power. Each plays a different role in a disaster.

Bonding capital provides immediate help: the neighbour who checks on an elderly resident, the family that shares water, the congregation that opens its hall. Bridging capital provides access to resources outside the immediate circle: the community group that knows how to apply for aid, the network that can mobilise volunteers. Linking capital determines whether a community can influence the institutions that decide where the levee is built and who receives compensation.

Daniel Aldrich, in a series of studies of disaster recovery in Japan, India and the United States, found that social capital predicted recovery better than income, damage or government spending. Communities with dense networks recovered faster because they could share information, allocate resources and hold institutions accountable. The finding is important because it suggests that resilience is not only bought with money; it

is also built through relationships.

18.2 The Evidence: Neighbourhoods and Survival

The classic study of social capital and heat is Eric Klinenberg's analysis of the 1995 Chicago heatwave, which killed more than 700 people in a week. Klinenberg compared two neighbourhoods with similar demographics and found that death rates differed sharply. In the mostly Latino Little Village, dense social networks, active street life and functioning institutions protected residents; in North Lawndale, depopulation, abandoned buildings and fear kept people isolated and indoors.

The mechanism was social, not physiological. Elderly people in Little Village were visited, were known to their neighbours, and went outside; elderly people in North Lawndale were afraid to open their doors and had no one to check on them. Air conditioning and income mattered, but they did not explain the gap. What explained it was whether people were connected to others who would notice their absence.

Similar findings appear in studies of floods, hurricanes and wildfires. Social isolation is consistently associated with higher mortality, and neighbourhood cohesion with lower mortality. The implication for policy is that programmes which build connection — community centres, neighbourhood associations, volunteer networks, outreach to isolated residents — are forms of disaster preparedness, even when they are not framed that way.

18.3 Mutual Aid and Volunteer Response

When formal systems are overwhelmed, people organise. After Hurricane Katrina, the Cajun Navy — a loose network of volunteers with boats — rescued thousands of people from flooded neighbourhoods. After Hurricane Sandy, Occupy Sandy organised volunteers to deliver supplies to areas that official relief had not reached. After the 2017 Mexico City earthquake, civilian brigades known as topos dug survivors out of rubble before professional teams arrived.

Mutual aid is most effective when it is connected to formal systems rather than replacing them. Volunteers know local conditions and can move faster, but they lack equipment, training and coordination capacity. The best outcomes occur when formal agencies create spaces for volunteers, provide information and logistics, and avoid treating them as a threat to command and control.

The Kerala floods of 2018 illustrate the point. The state government mobilised fishing communities, whose boats and knowledge of local waters were essential for rescuing people from flooded areas. The collaboration worked because the government recognised the fishers as partners and provided fuel, coordination and information. It was not spontaneous harmony; it was organised cooperation.

18.4 The Warning Chain and the Last Mile

A warning is a chain of translations: from observation to forecast, from forecast to decision, from decision to message, from message to household, and from household to action. Each link can fail, and the failure is usually invisible until the event. The Ahr valley flood failed at several links: forecasts were available but not understood,

decisions were delayed, and messages did not reach everyone.

The last mile is the hardest. Even when a warning is issued, it may not reach people who are asleep, who do not speak the language, who have no phone, or who distrust the messenger. Effective systems use multiple channels — sirens, cell broadcast, radio, social media, door-to-door — and repeat the message in plain language with specific instructions. A warning that says a river will flood is less useful than one that says leave this street now and go to this location.

Trust determines whether the message is believed. Communities that have experienced accurate warnings in the past are more likely to respond to the next one; communities that have been ignored or misled are less likely. Building trust requires consistency, transparency about uncertainty, and visible follow-through. It is slow to build and fast to lose, which is why it belongs in the same category as physical infrastructure.

18.5 Institutions, Trust and Risk Governance

The quality of government is one of the strongest determinants of disaster outcomes. Countries with effective bureaucracies, well-funded emergency services and honest procurement suffer fewer deaths and recover faster than countries with the same hazard exposure and weaker institutions. The contrast between the 2010 earthquakes in Haiti and Chile is often cited: Haiti's magnitude 7.0 earthquake killed more than 200,000 people, while Chile's magnitude 8.8 earthquake three weeks later killed about 500, largely because of building standards, preparedness and response capacity.

Trust matters because compliance depends on it. People who trust official warnings are more likely to evacuate, to accept restrictions and to cooperate with relief. Trust is built through competence, honesty and fairness, and it is damaged by failures, contradictions and perceived favouritism. The pandemic illustrated how quickly trust can erode when messaging is inconsistent, and how costly that erosion is.

Risk governance requires coordination across levels and sectors. A heat plan involves health, meteorology, social services, utilities and civil society; a flood plan involves water management, land use, emergency services and infrastructure. The failure mode is fragmentation, where each agency optimises its own mandate and no one is accountable for the whole. Effective governance assigns clear responsibility, provides resources, and requires coordination.

18.6 Decentralisation and Local Capacity

Much of the work of disaster response happens at the local level, where knowledge of terrain, population and relationships is greatest. Decentralisation can improve responsiveness, but only if local governments have the resources, skills and accountability to act. In many countries, decentralisation has transferred responsibilities without transferring revenue, leaving municipalities with mandates they cannot fund.

Indonesia's village law, which gave villages substantial budgets and authority, is one example of an attempt to build local capacity. India's panchayat system and the Philippines' barangay structure serve similar functions. Where local institutions are well functioning, they can organise evacuation, maintain drainage and manage relief; where

they are captured by elites or starved of funds, they can exclude the people who most need help.

The design question is how to build local capacity without creating local capture. That requires transparent budgets, participatory planning, external audit and support from higher levels of government. It also requires time, because institutions develop through practice. A decentralisation that begins five years before a disaster is unlikely to produce capable institutions by the time it arrives.

18.7 Corruption, Capture and Reconstruction

Disasters create large flows of money, and large flows of money create opportunities for corruption. Reconstruction contracts, relief distribution and land deals are all vulnerable. In some cases, the result is that aid does not reach survivors; in others, it reaches them in inferior form, with buildings that fail in the next event.

Elite capture is a subtler problem. Relief and reconstruction can reinforce existing inequalities if the people who control distribution also control access to information, land and credit. Post-disaster land grabs have been documented in several countries, particularly where tenure was informal before the event. Women, renters and informal residents are often the most exposed to these dynamics.

The antidote is transparency. Publishing beneficiary lists, contract awards and spending data allows communities and journalists to hold institutions accountable. Independent audits, community monitoring and grievance mechanisms can catch problems early. None of these is sufficient alone, but together they make corruption more difficult and more visible, which is often enough to change behaviour.

18.8 Disaster Risk Finance and Accountability

Money is decisive in the aftermath of a disaster, and the speed with which it arrives determines whether households recover or fall into debt. Traditional financing relies on post-disaster appeals, which are slow and uncertain. Pre-arranged finance — contingent credit, insurance, catastrophe bonds, regional risk pools — can disburse quickly and predictably.

The Caribbean Catastrophe Risk Insurance Facility, established in 2007, is an example of a regional pool that pays out quickly after a qualifying event. African Risk Capacity and the Pacific Catastrophe Risk Assessment and Financing Initiative serve similar functions. These instruments are not a substitute for national investment in resilience, but they reduce the fiscal shock of a disaster and allow governments to respond without diverting budgets from development.

Accountability is the other side of finance. Governments that receive adaptation and relief funds should be required to report how they were spent, and parliaments, audit institutions and civil society should have the capacity to scrutinise. Without accountability, finance can be absorbed without reducing risk, which is one reason why the correlation between aid received and resilience achieved is weaker than it should be.

18.9 Regional and International Cooperation

Climate hazards do not respect borders. Storms cross coastlines, rivers flow across countries, and displacement spills over frontiers. Regional cooperation can improve early warning, share resources and coordinate response. The ASEAN Coordinating Centre for Humanitarian Assistance on Disaster Management (AHA Centre) is an example: it maintains stockpiles, coordinates regional response and supports member states during emergencies.

International cooperation is most effective when it supports local institutions rather than substituting for them. Externally managed responses often arrive late, cost more, and leave little capacity behind. The Grand Bargain, a set of commitments to improve humanitarian financing, emphasises localisation for this reason, though progress has been slow.

Climate finance is also a form of governance. The commitment by wealthy countries to provide US$100 billion a year in climate finance has been met late and unevenly, and the adaptation share remains far below what vulnerable countries need. The gap is not only a matter of money; it is a matter of trust, because countries that promised and did not deliver have less credibility when they ask for cooperation on emissions.

18.10 Polycentric Governance and Learning

No single institution can manage climate risk. The most effective systems are polycentric: many centres of decision-making, connected by rules and information, each able to act within its domain. Elinor Ostrom's work on governing the commons showed that communities can manage shared resources effectively when they have clear boundaries, rules that fit local conditions, and mechanisms for collective choice.

Polycentric systems are resilient because they do not depend on a single point of failure. If a national agency is slow, a city can act; if a city is captured, a community group can organise. But polycentricity is not anarchy: it requires coordination, shared information and accountability. The challenge is to design systems that allow local initiative while ensuring that the most vulnerable are not left out.

Learning is the final requirement. Institutions that treat each disaster as a lesson, investigate what happened, and change their procedures are more resilient than those that do not. The inquiry into the Ahr valley flood led to the introduction of cell broadcast warnings in Germany; the inquiry into Hurricane Katrina led to reforms in emergency management. Learning requires the political willingness to accept criticism, which is often in short supply when blame is at stake.

18.11 Case Study: The Ahr Valley Flood, Germany 2021

On 14 and 15 July 2021, a slow-moving low-pressure system produced extreme rainfall over western Germany, Belgium and the Netherlands. In the Ahr valley, up to 200 millimetres fell in less than 24 hours, and the river rose within hours. The flood destroyed houses, roads and bridges, cut power and communications, and killed 134 people in the Ahrweiler district alone; across Germany and Belgium the death toll

exceeded 180, and damage was estimated at more than €30 billion.

The meteorological service had issued warnings, and some local authorities acted. But many people did not evacuate, partly because they did not believe the river could rise so fast, partly because the warnings were not specific enough, and partly because no mechanism existed to push an alert directly to every phone in the affected area. Germany introduced cell broadcast warnings only in 2023, after the event.

The failure was distributed across the chain. Forecasts were available but not translated into clear local instructions; some officials delayed evacuation because they feared false alarms; and communication systems failed when infrastructure was destroyed. The result was a disaster that was worse than the rainfall alone would explain, which is the definition of a governance failure in disaster risk.

The response after the event was substantial. Investigations were commissioned, responsibilities were clarified, and warning systems were improved. The flood became a reference point for German disaster policy, and it changed public expectations about warning and preparedness. The lesson was not that Germany lacked capacity; it was that capacity must be organised into a chain that reaches the last household.

For other countries, the lessons are specific. Warnings must be specific and actionable, not merely meteorological. Multiple channels must be used, including ones that do not depend on electricity or internet. Local officials need clear authority and training to order evacuation. And the public must be educated about what warnings mean, so that a warning is not dismissed as routine. Each of these is a governance task, not a technological one.

18.12 Future Scenarios and Predictions: Governance in 2050

The historical record shows that governance quality explains much of the variation in disaster mortality. Countries with strong institutions have reduced deaths from comparable hazards over the past century, while countries with weak institutions have not. The trend is not automatic: it depends on investment, accountability and learning.

Present conditions set the baseline. Trust in government has declined in many democracies, while the demands on government have increased. Local governments, which do most of the operational work, are often underfunded. International climate finance has fallen short of commitments, and adaptation finance reaches a small fraction of what is needed. Meanwhile, the hazards are becoming more severe and more frequent.

The assumptions behind the projections are the trajectory of institutional quality, the level of finance, and the degree of international cooperation. A world that invests in local capacity, transparency and regional cooperation can manage rising risk; a world that does not will see disasters become political crises. The table summarises the range.

Table 18.1 Governance indicators, 2050

Institution	Core function	Common failure	Reform	Indicator
Local government	Evacuation; shelters	Underfunded; unclear authority	Devolved finance; training	Response time
Warning systems	Forecast to household	Last-mile gap	Multi-channel; cell broadcast	Reach and compliance
Emergency services	Rescue; coordination	Fragmentation	Joint command; exercises	Coverage
Social protection	Cash; food; shelter	Exclusion errors	Registries; portable benefits	Coverage of poor
Anti-corruption	Procurement integrity	Elite capture	Open contracts; audits	Audit findings
Regional bodies	Shared stockpiles	Slow coordination	Pre-agreed protocols	Response time

Two conclusions follow. First, governance is a determinant of survival, not a background condition; the difference between a well-organised response and a disorganised one is measured in lives. Second, the reforms that matter most are unglamorous: clear authority, adequate local finance, transparent procurement, and practised procedures.

What would change these conclusions? A sustained investment in local government capacity would improve response even without new technology. Universal social protection would reduce the exclusion that leaves the poorest without help. And stronger regional cooperation could compensate for weak national capacity in small states. The most important variable remains the pace of climate change, which determines how often institutions will be tested, and whether they have time to learn between tests.

CHAPTER 19

RENEWABLE ENERGY AND APPROPRIATE TECHNOLOGIES

Theme: decentralised power, storage and the economics of clean energy

In the chars of northern Bangladesh — islands of silt that appear and disappear with the rivers — grid electricity arrived late, if at all. Households relied on kerosene lamps for light, which produced poor illumination, indoor smoke and a recurring expense that consumed a significant share of household income. Children studied under flickering light; shops closed at dusk; and clinics could not refrigerate vaccines.

Beginning in the late 1990s, a programme run by the Infrastructure Development Company Limited and supported by the World Bank and other donors began financing solar home systems through microcredit. Households paid a small down payment and instalments over two to three years. By the early 2020s, more than four million systems had been installed, reaching an estimated twenty million people, and Bangladesh had one of the largest off-grid solar programmes in the world.

The results were measurable. Kerosene use fell; children gained study hours; shops extended opening; and households reported better air quality and fewer burns. The systems were small — typically 50 to 100 watts — and they did not power industry. But they demonstrated something important: a modular technology, financed appropriately, could deliver a basic service to people whom the grid had not reached and would not reach for years.

This chapter examines renewable energy and appropriate technologies as instruments of resilience. It covers solar and wind, storage and grid flexibility, mini-grids and off-grid systems, clean cooking, efficiency, and the supply chains and politics that determine whether these technologies spread fast enough. The argument is not that technology alone will solve the climate crisis, but that the falling cost of modular clean energy has changed what is possible for households, communities and countries.

19.1 Why Renewables Are a Resilience Technology

Renewable energy is usually discussed as a mitigation strategy — a way to reduce emissions. It is also an adaptation strategy, because it changes the failure modes of energy systems. A coal plant depends on a fuel supply chain, cooling water and a centralised grid; a solar array depends on sunlight. When floods block roads or droughts reduce river flows, the solar array keeps producing, and a system built from many small units fails gradually rather than all at once.

Modularity is the key property. A large central plant is efficient but brittle: one failure can remove a gigawatt from the grid. A distributed system composed of thousands of rooftops, batteries and small generators can lose pieces without losing the whole. That is why hospitals, clinics and community shelters increasingly combine grid connection with rooftop solar and batteries, so that they can operate when the grid fails.

Renewables also reduce exposure to fuel price shocks. Countries that import oil and gas are vulnerable to price spikes caused by conflict, shipping disruption or market manipulation; the 2022 energy crisis, following Russia's invasion of Ukraine, raised

electricity and heating costs across Europe and pushed millions of households into fuel poverty. Solar and wind, once built, have no fuel cost, which makes their operating economics predictable even when commodity markets are not.

19.2 Solar: The Learning Curve

Solar photovoltaic modules have followed the steepest cost decline of any energy technology in history. The price of a module fell by roughly 90 per cent between 2010 and 2023, from more than a dollar per watt to around twenty cents, and installed system costs fell almost as fast. The decline was driven by manufacturing scale, learning by doing, and competition, and it transformed solar from a niche technology into the cheapest source of new electricity in many markets.

Deployment followed the price. Global solar capacity passed one terawatt in 2022 and continued to grow at record rates, with annual additions exceeding 400 gigawatts by 2023 and continuing to rise. China accounted for the largest share of both manufacturing and installation, but growth spread across India, the United States, Europe, Brazil and the Middle East. In many countries, solar now supplies a significant share of daytime electricity.

The technology has limits. Solar produces only during daylight, so it requires storage or complementary generation to meet evening demand; output varies with weather and season; and it requires land or roof space. In high-latitude winters, output can fall sharply. These limits are manageable with storage, transmission and diversified generation, but they mean that solar alone is not a complete system. The integration challenge, not the generation cost, is now the main constraint.

19.3 Wind Power and the Variability Problem

Wind power is the other mature renewable technology. Global capacity passed 1,000 gigawatts in 2023, with the largest installations in China, the United States, Germany, India and Spain. Offshore wind has grown rapidly because turbines can be larger and winds steadier at sea, and because offshore sites are close to coastal demand centres. The cost of offshore wind fell sharply before rising again in 2022–2023 as supply chains tightened and interest rates rose.

Wind shares solar's variability but not its daily pattern. In some regions, wind output is higher at night and in winter, which complements solar; in others, wind droughts — periods of unusually low wind — can last for days or weeks. The 2021 wind drought in Europe reduced output at a time of high demand, contributing to the energy price spike. Systems that rely heavily on wind need storage, interconnection and dispatchable backup to manage these periods.

Wind also has social constraints. Turbines are visible, can be noisy, and affect landscapes and wildlife; opposition has slowed projects in several countries. The constraints are real but manageable with better siting, community benefit sharing and early consultation. The alternative — continuing to rely on fossil fuels — carries larger environmental and social costs, but that comparison is not always made in local planning debates.

19.4 Storage, Flexibility and the Grid

The central challenge of a renewable-heavy grid is matching supply to demand. Electricity cannot be stored cheaply in large quantities for long periods, so system operators must balance generation and consumption in real time. Batteries, pumped hydro, demand response and interconnection are the main tools for doing so.

Lithium-ion battery costs fell by about 85 per cent between 2010 and 2023, and grid-scale storage grew rapidly, from a few gigawatts to tens of gigawatts globally. Batteries are well suited to shifting solar output from midday to evening and to providing fast frequency response, but they are less suited to storing energy across seasons. Pumped hydro, which stores energy as water in an upper reservoir, remains the largest form of grid storage and can provide long-duration capacity where geography allows.

Flexibility also comes from how demand is managed. Time-of-use tariffs, smart thermostats, industrial load shifting and electric vehicle charging can move consumption to times when supply is abundant. Interconnection allows regions with different weather to share power, which reduces the total storage needed. The combination of storage, demand flexibility, interconnection and diversified generation is what makes a high-renewable grid reliable, not any single technology.

19.5 Mini-Grids and Off-Grid Solar

About 700 million people still lack access to electricity, most of them in sub-Saharan Africa and South Asia. Extending the grid to remote areas is expensive and slow, and in some places it may never be economic. Mini-grids — small, local generation and distribution systems, often solar with battery storage — can provide reliable power to communities that the grid does not reach.

The economics have improved dramatically. A solar mini-grid can now deliver electricity at a cost competitive with diesel in many off-grid areas, and far below the cost of kerosene or candles per unit of light. The main barriers are financing, regulation and maintenance: mini-grids require upfront capital, clear rules on tariffs and grid arrival, and local capacity to operate and repair equipment.

The Bangladesh solar home system programme illustrates both the potential and the limits of off-grid approaches. It reached millions of households with basic lighting and phone charging, but the systems were too small for productive use, and as the grid expanded, some became redundant. The next generation of programmes combines larger systems, appliance financing and productive uses — irrigation pumps, cold storage, milling — so that electricity generates income rather than only light.

19.6 Clean Cooking and Household Energy

Around 2.3 billion people still cook with solid fuels — wood, charcoal, dung, crop residue — in open fires or inefficient stoves. The resulting household air pollution is estimated to cause 3.7 million premature deaths a year, mostly among women and children, from respiratory disease, heart disease and stroke. It is one of the largest environmental health burdens in the world, and it is concentrated among the poorest

households.

The transition to clean cooking requires more than a stove. Liquefied petroleum gas requires reliable supply and affordable cylinders; electricity requires reliable power and suitable cookers; improved biomass stoves require fuel processing and maintenance. Households often use a combination of fuels — a practice called stove stacking — because no single option meets all needs. Programmes that assume a single switch tend to fail.

The health and climate benefits are substantial. Clean cooking reduces indoor air pollution, saves time spent collecting fuel, reduces deforestation where wood is harvested unsustainably, and cuts black carbon emissions. The barriers are economic and logistical rather than technical, which is why progress has been slow despite decades of effort and why sustained subsidy, supply chain development and behaviour change are all required.

19.7 Appropriate Technology and Local Capacity

Appropriate technology is technology that fits the context: affordable, repairable, maintainable with local skills, and suited to local conditions. A solar home system that can be repaired by a local technician is more useful than a more efficient system that cannot be fixed. A water pump that can be maintained by the community is more sustainable than a complex system that requires external specialists.

The concept has been criticised as condescending when it implies that poor communities should accept lower technology. The critique is fair when appropriate technology means second-best; it is not fair when it means fit for purpose. The relevant question is not whether a technology is advanced but whether it can be sustained, repaired and afforded over its lifetime, and whether it meets the needs of the people who use it.

Local manufacturing and assembly can improve sustainability by creating jobs, reducing dependence on imports and building repair capacity. Several countries have used industrial policy to develop solar manufacturing, battery assembly and appliance production, with varying success. The most effective programmes combine technology transfer with training and finance, so that local firms can maintain and adapt the equipment rather than merely sell it.

19.8 Critical Minerals and Supply Chains

Clean energy technologies require minerals: lithium, cobalt, nickel, graphite and manganese for batteries; copper and aluminium for grids; rare earths for magnets; and silver and silicon for solar cells. The International Energy Agency estimates that demand for critical minerals could triple by 2030 under announced pledges, and that supply may not keep pace without new investment.

Supply is concentrated. China refines a large share of the world's lithium, cobalt and graphite, and dominates solar manufacturing; the Democratic Republic of the Congo supplies most of the world's cobalt; and Indonesia is a major source of nickel. Concentration creates resilience risks, because a single disruption — a export restriction, a conflict, a natural disaster — can affect global supply.

Diversification, recycling and substitution can reduce the risk. Recycling recovers lithium, cobalt and nickel from batteries and can supply a growing share of demand as the stock of deployed batteries increases. Substitution — sodium-ion batteries, iron-based chemistries, rare-earth-free motors — can reduce dependence on constrained materials. Mining also carries social and environmental costs, including water use, land conflict and labour abuses, which must be addressed if the transition is to be just.

19.9 Efficiency and Demand

The cheapest unit of energy is the one that is not used. Energy efficiency — better insulation, more efficient appliances, LED lighting, heat pumps, efficient motors — reduces demand, lowers bills and cuts emissions, often at negative cost. The International Energy Agency has repeatedly found that efficiency measures could deliver a large share of the emissions reductions needed by 2030, and that many are cheaper than new supply.

Cooling is a critical case. Demand for air conditioning is growing rapidly in hot countries as incomes rise and temperatures increase, and it could become one of the largest sources of electricity demand. Efficient air conditioners, better building design, shading and passive cooling can reduce that demand substantially. The Kigali Amendment to the Montreal Protocol, which phases down hydrofluorocarbons, is expected to avoid significant warming while improving appliance efficiency.

Efficiency policy faces a paradox: it is invisible, distributed and often requires upfront investment for savings that accrue over years. Households and firms systematically underinvest in efficiency because of split incentives — the landlord pays for insulation while the tenant pays the bills — and because of limited attention and capital. Standards, labelling, financing and technical support can overcome these barriers, and they are among the most cost-effective climate policies available.

19.10 The Political Economy of Transition

Energy transitions are political. Fossil fuel subsidies, which totalled more than a trillion dollars in 2022 during the price crisis, distort markets in favour of incumbents. Grid planning, permitting and market rules determine whether renewables can connect. Utility business models designed for centralised generation can resist distributed production. And communities that depend on fossil fuel employment resist changes that threaten their livelihoods.

A just transition requires addressing those interests rather than ignoring them. That means retraining and income support for workers, economic diversification for coal-dependent regions, and consultation with communities affected by new infrastructure. It also means ensuring that the benefits of the transition — cheaper electricity, cleaner air, new jobs — reach the people who bear its costs.

The pace of transition is now the central variable. Renewables are cheaper than new fossil fuel plants in most markets, and the constraints are increasingly grid capacity, permitting, storage and political will rather than technology cost. The countries that build quickly will benefit from lower costs and greater energy security;

the countries that delay will face higher costs, stranded assets and greater exposure to fuel price shocks.

19.11 Case Study: Bangladesh's Solar Home System Programme

In the 1990s, most rural Bangladeshi households had no electricity. Kerosene was the main source of light, and the cost was significant for poor families. The grid reached urban areas and some rural towns, but extending it to remote chars and villages was expensive and slow. The government, with support from the World Bank and other donors, established a programme to finance solar home systems through microcredit institutions.

The model was simple. A household paid a small down payment, typically 10 to 15 per cent of the system cost, and the remainder in instalments over two to three years, financed by microcredit. The systems were small — 50 to 100 watts — sufficient for lights, a phone charger and a small television. Local technicians were trained to install and maintain them, and quality standards were enforced through testing and certification.

The programme grew rapidly. By the early 2020s, more than four million systems had been installed, reaching an estimated twenty million people. Evaluations found that kerosene use fell, study hours increased, shops extended opening hours, and households reported better air quality and fewer burns. The programme also created a network of local technicians and businesses, which built capacity for later renewable energy development.

The limitations were equally instructive. The systems were too small for productive uses such as irrigation, milling or refrigeration, so they did not generate income. As the grid expanded, some households switched to grid electricity and abandoned their systems. And the financing model depended on microcredit, which excluded the poorest households who could not afford the down payment or the instalments.

The lessons for other countries are specific. Off-grid solar can deliver basic services quickly and at scale, but it needs quality standards, local maintenance capacity and financing that reaches the poorest. It works best as a complement to grid expansion rather than a permanent substitute, and it should be designed to support productive use, not only lighting. Bangladesh's programme was not a complete solution, but it demonstrated that modular technology, appropriate finance and local institutions can change millions of lives.

19.12 Future Scenarios and Predictions: Energy in 2050

The historical record shows a rapid shift. Renewables supplied about 30 per cent of global electricity in 2023, up from about 20 per cent a decade earlier, and solar and wind accounted for most new capacity additions. Costs fell faster than almost any forecast anticipated, and deployment repeatedly exceeded projections. The trend is now driven by economics as much as by policy.

Present conditions set the baseline. About 700 million people lack electricity and 2.3 billion lack clean cooking, while global energy demand continues to rise. Fossil fuels still supply about 80 per cent of primary energy, and subsidies continue to favour them. Grids, storage and permitting are the binding constraints in many countries, and critical mineral supply chains are concentrated.

The assumptions behind the projections are the pace of cost decline, the rate of grid investment, and the strength of policy. A world that invests in grids, storage and manufacturing can reach a predominantly renewable electricity system by 2050; a world that does not will continue to depend on fossil fuels and face higher costs and greater volatility. The table summarises the range.

Table 19.1 Clean energy indicators, 2050

Technology	Cost trend	Best use	Constraint	Resilience benefit
Solar PV	−90% since 2010	Daytime power; rooftops	Land; night output	Modular; no fuel
Wind	Falling; volatile	Night and winter power	Siting; variability	No fuel; scalable
Batteries	−85% since 2010	Daily shifting; backup	Duration; minerals	Fast response; islanding
Mini-grids	Falling	Off-grid communities	Finance; regulation	Local control
Clean cooking	Moderate	Household health	Fuel supply; cost	Indoor air; forests
Heat pumps	Moderate	Heating and cooling	Retrofit; grid load	Efficiency; fuel-free
Solar pumps	Falling	Irrigation	Water tables	Drought resilience

Two conclusions follow. First, the energy transition is now primarily an integration and infrastructure challenge rather than a cost challenge: the technologies are cheap enough, but grids, storage, permitting and supply chains determine how fast they spread. Second, decentralised technologies have a specific resilience value that centralised systems cannot match, because they allow communities to keep essential services running when the grid fails.

What would change these conclusions? A breakthrough in long-duration storage would reduce the need for backup generation and make high-renewable grids cheaper. A rapid expansion of grid investment in low-income countries would unlock demand that is currently constrained by unreliable supply. And a sustained decline in battery and mineral costs would make electrification of transport and heating faster. The most important variable remains the pace of policy and investment, because the technology is already available and the cost curve is already falling.

CHAPTER 20

CLIMATE PSYCHOLOGY AND MENTAL HEALTH

Theme: trauma, grief, resilience and the psychology of a changing climate

On the morning of 8 November 2018, a faulty transmission line sparked a fire in the foothills of the Sierra Nevada in northern California. Driven by strong winds, the Camp Fire moved through the town of Paradise in a few hours. Eighty-five people died, most of them elderly or disabled, and nearly 19,000 structures were destroyed. The town of 26,000 people ceased to exist as a functioning community.

The physical recovery began immediately. The psychological one was slower and less visible. Survivors described nightmares, hypervigilance at the smell of smoke, guilt at having escaped while neighbours did not, and a grief for a place that no longer existed. Clinicians in Butte County reported a surge in demand for mental health services that persisted for years after the fire was contained, and surveys of survivors found elevated rates of post-traumatic stress disorder, depression and substance use.

The Camp Fire was followed by the COVID-19 pandemic, which arrived while many survivors were still displaced, and by subsequent wildfire seasons that filled the sky with smoke and retraumatised people who had lost homes. The sequence is characteristic: climate disasters rarely arrive as single events, and the psychological burden accumulates across repeated exposures, disrupted services and uncertain futures.

This chapter examines the psychology of climate change: the trauma of disasters, the stress of slow-onset change, the grief of loss, and the anxiety that many people, especially the young, feel about the future. It also examines what helps: social support, meaning, action and well-designed services. The argument is that mental health is not a soft add-on to climate policy but a core dimension of how societies experience and respond to change.

20.1 The Mental Health Burden of Climate Change

Climate change affects mental health through multiple pathways. Acute events such as floods, fires and storms cause injury, bereavement, displacement and destruction, each of which increases the risk of trauma-related disorders. Slow-onset changes such as drought, sea-level rise and salinisation erode livelihoods and places, producing chronic stress and grief. And the anticipation of future change produces anxiety, anger and despair, particularly among young people.

The burden is large but under-measured. Mental health is rarely included in disaster mortality statistics, and services in most countries are too thin to capture the demand. Where studies exist, they find substantial effects: a meta-analysis by Galea, Nandi and Vlahov (2005) found that a significant minority of people exposed to disasters develop post-traumatic stress disorder, and that rates are higher after events involving injury, bereavement or displacement.

The burden is also unequal. People with pre-existing mental illness are more vulnerable to disruption of care; people living in poverty have fewer resources to cope; and people who are displaced, isolated or discriminated against face higher risks. The result is that climate change compounds existing mental health inequalities rather than

creating a new and separate problem.

20.2 Disaster Trauma

Post-traumatic stress disorder is the most studied consequence of disaster. Its symptoms — intrusive memories, avoidance, negative beliefs, hyperarousal — can persist for years and can impair work, relationships and health. Risk factors include the severity of exposure, the loss of a loved one, injury, displacement, lack of social support and prior trauma.

But PTSD is not the only outcome, and often not the most common. Depression, anxiety, substance use and grief are widespread, and they may not meet diagnostic thresholds while still causing substantial distress. Studies after Hurricane Katrina found elevated rates of psychological distress across the affected population, and research after the 2011 Christchurch earthquakes documented persistent anxiety and sleep disturbance.

The trajectory matters. Most people recover with time and support, but a minority develop chronic conditions. The factors that predict recovery are social: strong networks, stable housing, financial security and access to care. That is why mental health recovery is inseparable from housing, employment and social policy, and why clinical services alone cannot address the burden.

20.3 Slow Onset, Drought and Solastalgia

Slow-onset change produces a different kind of distress. Drought, salinisation, desertification and coastal erosion erode the landscapes and livelihoods that people depend on, often over years. The philosopher Glenn Albrecht coined the term solastalgia in 2005 to describe the distress produced by environmental change while still living in one's home place — a homesickness experienced without leaving home.

The evidence from drought is substantial. Studies in Australia have found associations between drought and suicide, particularly among rural men, with Hanigan et al. (2012) estimating that drought was associated with a small but significant increase in suicide rates among men aged 30 to 49. The mechanism is likely economic and social: loss of income, debt, isolation and the erosion of masculine identities tied to farming.

Slow-onset events are harder to respond to because they lack a clear beginning and end. There is no single disaster to recover from, no reconstruction programme, and no obvious point at which services should be scaled up. The result is that chronic stress accumulates without triggering the institutional response that an acute event would provoke, which is one reason why slow-onset mental health needs are so often neglected.

20.4 Heat and Mental Illness

Heat affects mental health directly and indirectly. Directly, high temperatures are associated with irritability, aggression, impaired cognition and worsening of symptoms in people with psychotic disorders. Indirectly, heat disrupts sleep, increases substance use and reduces the capacity to work, all of which affect mental well-being.

The pharmacological dimension is often overlooked. Many psychiatric medications — antipsychotics, antidepressants, lithium — impair the body's ability to regulate temperature, increasing the risk of heat exhaustion and heat stroke. People taking these medications need specific advice about hydration, cooling and monitoring during heatwaves, and clinicians need to include heat risk in prescribing decisions.

The association with suicide is well documented, though the mechanisms are complex. Studies have found higher suicide rates during hot periods, and some research suggests that heat increases impulsivity and aggression. Carleton (2017) reviewed the evidence and argued that the relationship is robust enough to warrant inclusion in suicide prevention planning, particularly as heatwaves become more frequent and intense.

20.5 Children, Young People and Eco-Anxiety

Young people are exposed to climate change differently from adults: their bodies are more vulnerable to heat and disease, their education is disrupted by disasters, and their futures are shaped by decisions they did not make. They are also more aware of the issue, because it is taught in school and because it saturates social media.

A survey by Hickman et al. (2021), published in the Lancet Planetary Health, polled 10,000 young people aged 16 to 25 in ten countries. It found that 59 per cent were very or extremely worried about climate change, 84 per cent were at least moderately worried, and 45 per cent reported that their feelings about climate change negatively affected their daily life and functioning. Many reported feeling betrayed by governments and abandoned by adults.

The term eco-anxiety is sometimes used to describe these feelings, but it can be misleading. Anxiety is a normal response to a real threat, not a disorder, and pathologising it can obscure the political and social dimensions. What young people need is not treatment for caring but support, agency, accurate information and adults who take their concerns seriously.

20.6 Displacement, Loss and Place

Displacement disrupts the relationships, routines and identities that sustain mental health. People who lose their homes lose not only shelter but also neighbours, schools, churches, familiar streets and the sense of belonging that comes with them. The loss is compounded when displacement is prolonged, when host communities are hostile, and when the possibility of return is uncertain.

Place attachment is a protective factor in normal times and a source of grief when the place is damaged or lost. Research on communities facing relocation from rising seas and eroding coasts has documented anticipatory grief, conflict over whether to stay or leave, and the erosion of social ties that occurs when some households move and others remain. The process of managed retreat, though sometimes necessary, is psychologically costly and must be designed with care.

Cultural loss is a distinct dimension. Indigenous communities facing the loss of ancestral lands, sacred sites and traditional livelihoods experience a form of grief that is not captured by clinical categories. The loss of language, ceremony and

intergenerational knowledge compounds the loss of land, and the resulting distress is collective as well as individual.

20.7 Grief, Denial and Moral Injury

Climate change produces grief for losses that have already occurred and for losses that are anticipated. This grief is sometimes called ecological grief or climate grief, and it is a rational response to real loss rather than a symptom of pathology. Recognising it as grief, rather than as depression or anxiety, can help people make sense of their experience and find ways to live with it.

Denial is a related but different response. It can be literal — refusing to accept the evidence — or interpretive, in the form of minimisation, fatalism or distraction. Psychologists have described the defence mechanisms that climate change evokes: avoidance of threatening information, distancing from the problem, and the belief that someone else will solve it. These mechanisms are normal, but they impede action.

Moral injury is increasingly recognised among people who work on climate: scientists, health workers, emergency responders and officials who feel complicit in a system that is failing. It arises when people are required to act against their values, or when they witness harm they cannot prevent. Addressing it requires institutional support, not only individual resilience, because the injury is produced by the conditions of work rather than by personal weakness.

20.8 What Helps: Support, Meaning and Action

The evidence on what helps is consistent. Social support is the strongest protective factor: people who are connected to family, friends and community recover better than those who are isolated. Practical support — housing, income, childcare, information — is as important as psychological support, because material insecurity drives distress.

Psychological first aid, developed by the Inter-Agency Standing Committee and others, provides a framework for early support: safety, stabilisation, connection, information and referral. It does not require specialist training and can be delivered by community workers, teachers and volunteers. Hobfoll and colleagues identified five essential elements — safety, calming, self-efficacy, connectedness and hope — that should guide early intervention.

Meaning and action also help. People who feel that their actions matter, and who are part of collective efforts, report less despair than those who feel powerless. This is why climate action can be psychologically protective, not only politically useful. The goal is not to deny the severity of the situation but to combine honesty about risk with a sense of agency, which is sometimes called constructive hope.

20.9 Services, Treatment Gap and Integration

Mental health services are the weakest part of most health systems. The World Health Organization estimates that the treatment gap for common mental disorders exceeds 75 per cent in many low- and middle-income countries, and that spending on mental health is a small fraction of total health spending. In disaster settings, the gap

widens because services are disrupted precisely when demand rises.

Closing the gap requires task-shifting: training community health workers, teachers and volunteers to deliver basic psychological support and to recognise when referral is needed. The WHO's mhGAP programme provides protocols for this approach, and evidence from low-income settings shows that it can be effective when supervision and supply are adequate.

Integration is the key principle. Mental health should be part of disaster planning, primary care, school health and social protection, rather than a separate silo activated only after a crisis. That means including psychosocial support in emergency response, screening in primary care, and continuity of care for people with pre-existing conditions. It also means collecting data, because mental health is invisible in most disaster statistics.

20.10 Communicating Without Despair

How climate change is communicated affects how people feel and act. Messages that emphasise only catastrophe can produce paralysis, denial or fatalism; messages that emphasise only solutions can seem naive and can undermine trust when problems persist. The balance is honest hope: a clear account of the risk combined with a clear account of what can be done.

Framing matters. Research on climate communication suggests that messages which connect to values people already hold, which use local and concrete examples, and which offer specific actions are more effective than abstract global warnings. Messages that emphasise shared identity and collective efficacy are more motivating than those that emphasise individual guilt.

For people experiencing distress, the most helpful response is often acknowledgment rather than argument. Telling someone that their grief is understandable, that they are not alone, and that there are things they can do is more supportive than disputing their feelings or offering false reassurance. The aim is not to eliminate anxiety but to help people live with uncertainty while remaining engaged, which is the psychological equivalent of adaptation.

20.11 Case Study: The Camp Fire and Its Aftermath

The Camp Fire began on 8 November 2018 near Pulga, California, and spread rapidly through the town of Paradise. Within hours, much of the town was destroyed. Eighty-five people died, most of them older or disabled residents who could not evacuate quickly, and nearly 19,000 structures were lost. It was the deadliest and most destructive wildfire in California's history.

The immediate response was overwhelmed. Evacuation routes were blocked, communications failed, and survivors were scattered across surrounding counties. Many spent weeks in shelters, motels and the homes of relatives, and many never returned. The town's water system was contaminated by toxic runoff, and rebuilding was slowed by debris removal, insurance disputes and permitting.

The mental health consequences were severe and persistent. Clinicians reported increases in post-traumatic stress, depression, anxiety and substance use, and surveys of survivors found that a substantial minority met criteria for these conditions years after the fire. Children who had been evacuated showed symptoms of trauma, and elderly survivors who had lost their homes and social networks were particularly vulnerable.

The COVID-19 pandemic arrived while many survivors were still displaced, compounding isolation and disrupting services. Subsequent wildfire seasons brought smoke and renewed evacuations, retraumatising people who had already lost homes. The sequence illustrates how repeated exposures accumulate, and how a single event can become a chronic condition when recovery is slow.

The lessons are specific. Mental health services must be scaled up at the same time as shelter and supplies, not weeks later. Support must reach people where they are, including those who do not seek help. Schools, primary care and community organisations must be equipped to recognise distress and provide basic support. And recovery must address housing, income and social connection, because psychological recovery depends on material stability.

20.12 Future Scenarios and Predictions: Mental Health in 2050

The historical record shows that mental health consequences of disasters are large, persistent and under-measured. Studies after major events consistently find elevated rates of post-traumatic stress, depression and anxiety, and the effects can last for years. As climate hazards become more frequent and severe, the cumulative burden is likely to grow, particularly in regions that experience repeated events.

Present conditions set the baseline. Mental health services are underfunded and understaffed in most countries, and the treatment gap exceeds 75 per cent in many low-income settings. Demand is already rising: surveys of young people find widespread anxiety, and disaster-affected communities report sustained increases in need. At the same time, stigma is declining in many places, which may increase demand further.

The assumptions behind the projections are the frequency of hazards, the availability of services, and the strength of social support. A world that invests in community-based mental health, integrates psychosocial support into disaster response, and addresses the material drivers of distress can limit the burden; a world that does not will see mental health systems overwhelmed. The table summarises the range.

Table 20.1 Mental health indicators, 2050

Population	Main stressor	Typical response	Risk factor	Support
Children	Disruption; fear	Anxiety; regression	Lost schooling	Routine; school support
Adolescents	Uncertainty; anger	Eco-anxiety; withdrawal	Social media	Agency; peer groups
Older adults	Heat; loss; isolation	Depression; grief	Living alone	Outreach; cooling; contact
Displaced people	Loss of place	PTSD; grief	Prolonged displacement	Housing; community
First responders	Repeated exposure	Burnout; moral injury	Long shifts	Rotation; support
Farmers	Drought; debt	Depression; suicide risk	Financial loss	Income support; services
People with mental illness	Heat; service disruption	Relapse	Medication effects	Continuity of care

Two conclusions follow. First, mental health is a climate impact in its own right, not a secondary consequence of physical damage, and it requires its own investment and measurement. Second, the most effective interventions are social and material as well as clinical: housing, income, connection and agency protect mental health more reliably than treatment alone.

What would change these conclusions? A large expansion of community mental health workers would close part of the treatment gap at modest cost. Integrating psychosocial support into disaster response would reduce chronicity. And addressing the material drivers of distress — housing, debt, displacement — would reduce the incidence of disorder in the first place. The most important variable remains the pace of climate change, because it determines how many people will experience loss, and how much support they will need.

CHAPTER 21

GOVERNANCE, JUSTICE AND CIVIC ENGAGEMENT

Theme: responsibility, loss and damage and the politics of redress

In the summer of 2022, monsoon rainfall in Pakistan reached roughly three times the national average. The Indus rose, glacial lakes burst, and by September about a third of the country was under water. More than 33 million people were affected, at least 1,700 died, 2 million homes were damaged or destroyed, and losses were estimated at more than US$30 billion.

Pakistan's contribution to cumulative global greenhouse gas emissions is less than one per cent. Its emissions per person are about a tenth of those in the United States. The country that suffered the disaster was not the country that had caused the warming that made it more likely, and it did not have the fiscal capacity to rebuild without external help. That gap between responsibility and harm is the core of climate justice.

The floods coincided with preparations for the twenty-seventh Conference of the Parties in Sharm el-Sheikh, where the question of loss and damage had been on the agenda for years without resolution. At that conference, parties agreed to establish a fund to address loss and damage, a demand that small island states had first made in 1991. It was a historic decision, and it was also an incomplete one: the fund was created, but its scale, contributors and rules were left to be decided.

This chapter examines the governance of climate change through the lens of justice: who is responsible for emissions, who bears the harm, and what institutions exist to redistribute costs and remedy injustice. It covers historical responsibility, loss and damage, litigation, just transition, indigenous and gender dimensions, youth participation and the reform of international institutions. The argument is that climate policy without a theory of justice is not only unfair but ineffective, because agreement on burden-sharing is what makes cooperation possible.

21.1 Why Climate Change Is a Justice Problem

Climate change is a justice problem for three structural reasons. First, the harms are unequal: the countries and communities that contributed least to emissions are often the most exposed to their consequences, because of geography, dependence on agriculture and limited capacity to adapt. Second, the costs of action are unequal: reducing emissions requires changes in production and consumption that fall differently on different groups. Third, the capacity to act is unequal: rich countries and rich households have resources that poor ones do not.

The distribution of emissions makes the point concrete. Oxfam and the Stockholm Environment Institute estimated that the richest 10 per cent of the world's population accounted for about 48 per cent of cumulative emissions between 1990 and 2015, while the poorest half accounted for about 7 per cent. The pattern holds within countries as well as between them: wealthy households drive more, fly more and heat larger homes.

Justice is also a practical condition for agreement. International climate negotiations require states to accept costs that are not immediately repaid, and they will only do so if they believe the distribution is fair. The failure to resolve questions of equity — who cuts, by how much, and who pays — is one reason why negotiations have moved slowly, and why the language of justice has moved from the margins to the centre of the agenda.

21.2 Historical Responsibility and Carbon Budgets

Carbon dioxide accumulates in the atmosphere, so the warming we experience today is the result of cumulative emissions over more than a century. That makes historical responsibility central to any account of fairness. The United States has emitted roughly a quarter of cumulative carbon dioxide since 1750; the European Union and the United Kingdom together about a fifth; China about 14 per cent, though its emissions have risen sharply since 2000. Africa accounts for less than 4 per cent.

The remaining carbon budget — the amount that can still be emitted while keeping warming within a given limit — turns the history into a current constraint. At the time of the Intergovernmental Panel on Climate Change's Sixth Assessment Report, the budget consistent with a 50 per cent chance of limiting warming to 1.5 °C was roughly 500 gigatonnes of carbon dioxide, or about a decade of current emissions. Dividing that budget fairly is a political question, not a scientific one.

Different principles produce different answers. Grandfathering gives each country a share based on current emissions; convergence equalises emissions per person over time; and historical responsibility weights the budget by past contribution. Each principle has defenders and each would produce a very different distribution of effort. The practical outcome has been a hybrid: nationally determined contributions that are self-selected, combined with pledges of finance that have consistently fallen short.

21.3 Loss and Damage

Loss and damage refers to the harms caused by climate change that cannot be avoided through mitigation or adaptation. It includes economic losses such as destroyed infrastructure and lost crops, and non-economic losses such as the loss of territory, culture, biodiversity and human life. For small island states, the non-economic dimension is existential: a country that loses its land loses its statehood.

The demand for compensation dates back to 1991, when Vanuatu proposed an insurance mechanism on behalf of the Alliance of Small Island States. It was resisted for decades, largely because wealthy countries feared open-ended liability. At COP27 in 2022, parties agreed to establish a fund, and at COP28 and COP29 the arrangements were elaborated, with initial pledges in the hundreds of millions of dollars. The sums are small relative to need: estimates of annual loss and damage in developing countries range from US$200 billion to more than US$500 billion by 2030.

The fund is therefore a beginning rather than a solution. Its scale is insufficient, its contributors are a subset of those responsible, and its rules on eligibility are contested. But it establishes a principle that had been denied for thirty years: that climate harm creates obligations beyond voluntary charity. How that principle is

implemented will determine whether it becomes a meaningful mechanism or a symbolic gesture.

21.4 Climate Litigation

Courts have become an important arena for climate governance. The Sabin Center for Climate Change Law tracks more than 2,600 climate-related cases worldwide, and the number is growing. The cases fall into several categories: suits against governments for insufficient action, suits against companies for damages, suits challenging permits for fossil fuel projects, and suits by communities against states for failure to adapt.

The Urgenda case in the Netherlands was a landmark. In 2019, the Dutch Supreme Court upheld a ruling that the state had a legal obligation to reduce emissions by at least 25 per cent below 1990 levels by 2020, based on the right to life and the right to family life under the European Convention on Human Rights. The decision was unprecedented and it inspired similar cases in other jurisdictions.

Cases against companies have also advanced. In 2021, a Dutch court ordered Shell to reduce its emissions by 45 per cent by 2030, a ruling that was later overturned on appeal in 2024 but that signalled that corporate liability is a live question. In the United States, Held v. Montana in 2023 found that the state's failure to consider climate change when permitting fossil fuel projects violated young people's constitutional right to a clean environment. Litigation is slow and uncertain, but it creates leverage, establishes records and shifts expectations.

21.5 Just Transition and Labour

The transition away from fossil fuels will eliminate some jobs and create others, and the distribution of those changes is a matter of justice. Coal miners, oil workers and communities that depend on fossil fuel extraction face the loss of livelihoods that were often dangerous and poorly paid but that provided identity and stability. A transition that ignores them is both unjust and politically fragile.

The International Labour Organization estimates that about 1.2 billion jobs depend on a healthy environment, including agriculture, forestry, fisheries and tourism, and that the transition to a green economy could create millions of new jobs in renewable energy, efficiency and restoration. The question is whether the new jobs are in the same places, require the same skills and offer the same security as the old ones. Often they do not.

Just transition policies include retraining, income support, early retirement, diversification of local economies and guarantees of decent work in new industries. They require consultation with unions and communities, and they require funding. The countries that have made the most progress — Germany's coal commission, Spain's just transition agreements, Scotland's transition training — have done so by treating workers as partners rather than obstacles.

21.6 Indigenous Peoples and Land Rights

Indigenous peoples manage or have tenure over a large share of the world's remaining forests and biodiversity, and their knowledge systems contain detailed observations of environmental change. They are also among the most vulnerable to climate impacts, because their livelihoods, cultures and identities are tied to specific lands and waters that are increasingly exposed to fire, drought, flooding and extraction.

Land rights are central. Where indigenous communities have secure tenure, deforestation rates are lower and forest management is more effective; where tenure is insecure, communities are vulnerable to land grabs, mining, logging and infrastructure projects that are justified as climate action. The principle of free, prior and informed consent is meant to protect communities from decisions taken without their agreement, but it is often ignored in practice.

Climate policy can therefore harm indigenous peoples even when it is well intentioned. Large-scale hydropower, bioenergy plantations, carbon offsets and mineral extraction for clean technologies can displace communities and damage ecosystems. A just transition requires that these projects be assessed for their social impacts, that communities have a genuine say, and that benefits are shared rather than extracted.

21.7 Gender and Intersectionality

Climate change affects women and men differently, and the differences are produced by social roles rather than biology. Women are more likely to be responsible for water collection, food preparation and care work, all of which become harder when resources are scarce. They are more likely to be employed in informal and climate-sensitive sectors, and less likely to have access to land, credit and decision-making.

Disaster mortality shows the pattern. Studies of several events have found higher female mortality where women had lower social and economic status, limited mobility and restricted access to information. The 2003 European heatwave killed more women than men in France, largely because women live longer and were more likely to be widowed and living alone. The pattern is not universal, but it is common enough to require attention in planning.

Intersectionality matters because disadvantages compound. A poor woman with a disability in a rural area faces a different risk profile from a wealthy man in a city, and generic policies will not reach her. Effective adaptation requires disaggregated data, participation by affected groups, and targeted measures such as cash transfers, childcare, accessible shelters and legal aid.

21.8 Young People and Intergenerational Justice

Climate change raises a distinctive question of intergenerational justice: decisions taken now will determine the conditions in which today's children and their descendants live, and those future people have no vote and no voice in the decisions. The ethical problem is not new, but the scale and irreversibility of climate impacts make it unusually acute.

Young people have responded by organising. School strikes, youth-led litigation and advocacy campaigns have changed the political conversation in many countries, and they have done so with a moral clarity that is difficult to dismiss. The movement has also produced a backlash, with some critics arguing that children are being used and others that the protests are disruptive.

Institutional responses are emerging. Some countries have lowered the voting age for certain elections, appointed youth advisers, or created future-generations commissioners. Courts in several jurisdictions have accepted that young people have standing to challenge climate policies. These innovations are imperfect, but they recognise a structural problem: democratic systems discount the future, and climate policy requires institutions that do not.

21.9 Civic Engagement and Democracy

Climate policy is made and constrained by political institutions, and the quality of those institutions determines whether policy is ambitious, durable and fair. Democracies with strong civic participation tend to produce more ambitious climate policy, but they are also vulnerable to backlash when costs are imposed without consultation. Authoritarian systems can impose policy quickly, but they often lack accountability and responsiveness.

Deliberative experiments offer one route to legitimacy. Ireland's Citizens' Assembly, convened in 2016 and 2017, considered climate change among other issues and produced recommendations that informed subsequent legislation. France's Convention Citoyenne pour le Climat, which brought together 150 randomly selected citizens in 2019 and 2020, produced proposals that were partly adopted. The evidence suggests that deliberation can increase public understanding and support, though it does not guarantee implementation.

Civic engagement also includes protest, litigation, journalism, consumer action and local organising. These forms of participation are not substitutes for formal politics, but they shape the environment in which formal politics operates. A healthy democracy requires space for all of them, and a climate policy that relies only on technocratic decisions will struggle to sustain legitimacy when costs become visible.

21.10 Reforming International Institutions

The institutions that govern climate finance were built for a different era. The World Bank and the International Monetary Fund were established in 1944, when most of today's countries did not exist as independent states, and their governance reflects the post-war distribution of power. Reform proposals have circulated for decades, and they have gained urgency as climate finance needs have grown.

The Bridgetown Initiative, launched by Barbados in 2022, proposes a set of reforms: expanding multilateral development bank lending, linking repayments to climate vulnerability, and creating new mechanisms for loss and damage. The initiative has influenced discussions at the International Monetary Fund and the World Bank, and some elements have been adopted, but the scale of change remains modest relative to the need.

Reform also requires attention to debt. Many climate-vulnerable countries spend more on debt service than on health and education, and a disaster can push them into default. Debt swaps, where creditors forgive debt in exchange for conservation or adaptation investment, have been used in a few cases, and they could be expanded. Without fiscal space, even well-designed adaptation plans cannot be implemented, and the gap between promise and practice will continue to widen.

21.11 Case Study: The Loss and Damage Fund

At the 1991 negotiations, Vanuatu proposed an insurance mechanism to compensate small island states for the effects of sea-level rise. The proposal was not adopted, and for the next three decades the question of loss and damage remained on the margins of climate negotiations. Developed countries resisted language that could imply liability, and discussions focused on insurance and risk transfer rather than compensation.

The issue gained momentum after a series of disasters. Hurricane Maria in 2017, Cyclone Idai in 2019, the floods in Pakistan in 2022 and the heatwaves in Europe and Asia created a record of harm that was difficult to attribute to bad luck alone. At COP27 in Sharm el-Sheikh in November 2022, parties agreed to establish a fund for loss and damage, with the details to be negotiated subsequently.

The agreement was historic but limited. It established a principle — that vulnerable countries should receive financial support for climate harms — but it did not establish a scale, a mandatory contribution formula or a timetable. At COP28 in Dubai, the fund was operationalised with initial pledges of a few hundred million dollars, far below the estimated need of hundreds of billions per year.

The fund's significance is partly symbolic. It recognises that climate change creates obligations beyond charity, and it provides a mechanism through which those obligations can be claimed. But a fund of hundreds of millions cannot address a problem of hundreds of billions, and the gap between the two is the central test of whether the international system can deliver justice at the scale required.

The lessons are specific. Principles matter, because they establish the terms of future negotiations. Institutions matter, because they determine whether principles are implemented. And scale matters, because a mechanism that is adequate in principle but tiny in practice can become a substitute for action rather than a route to it. The next phase of the debate will be about whether the fund becomes a genuine instrument of redistribution or a symbol that allows the underlying inequity to persist.

21.12 Future Scenarios and Predictions: Justice in 2050

The historical record shows that the gap between responsibility and harm has widened. Cumulative emissions are concentrated in a small number of countries, while the damages from climate change are concentrated in countries with limited capacity to adapt. The record also shows that institutions respond slowly: the loss and damage demand took three decades to produce a fund, and the fund remains small.

Present conditions set the baseline. Climate finance commitments have been met late and unevenly, adaptation finance remains a fraction of need, and many vulnerable countries spend more on debt service than on climate adaptation. Litigation is increasing, and courts in several jurisdictions have recognised climate obligations, but enforcement remains difficult. Public support for climate action is broad but shallow in many countries, and it declines when costs become visible.

The assumptions behind the projections are the pace of climate change, the strength of international institutions and the level of civic pressure. A world that strengthens redistribution, expands loss and damage finance and protects civic space can reduce injustice even as the climate worsens; a world that does not will see the burden fall on those least responsible. The table summarises the range.

Table 21.1 Climate justice indicators, 2050

Actor	Responsibility	Exposure	Instrument	Gap
High-income states	High historical	Low to moderate	Finance; mitigation	Delivery
Major emerging economies	Rising	Moderate to high	NDCs; adaptation	Finance
Fossil fuel firms	High	Low	Litigation; regulation	Liability
Multilateral banks	Enabling	Indirect	Concessional finance	Scale
Vulnerable states	Low	Very high	Adaptation; loss and damage	Fiscal space
Indigenous peoples	Very low	Very high	Land rights; FPIC	Recognition
Young people	None	Inherited	Participation; litigation	Voice

Two conclusions follow. First, justice is not a moral add-on to climate policy but a condition of its effectiveness, because cooperation depends on a fair distribution of burdens. Second, the instruments that exist are inadequate in scale, and the gap between principle and practice is the central problem of climate governance.

What would change these conclusions? A substantial expansion of loss and damage finance, funded by levies on fossil fuel extraction or shipping, would close part of the gap. Reform of multilateral banks and debt relief would create fiscal space for adaptation. And stronger civic participation would make governments more accountable for their promises. The most important variable remains the pace of climate change, because it determines the scale of harm that must be shared, and whether the institutions of justice can keep pace.

CHAPTER 22

CASE STUDIES: HEAT, FIRE, FLOOD AND URBAN EXTREMES

Theme: comparative disaster cases and the anatomy of urban failure

On 29 June 2021, the village of Lytton in British Columbia recorded 49.6 °C, the highest temperature ever measured in Canada. The previous record had been 45 °C, set in 1937. The heat did not arrive gradually: a high-pressure ridge parked over the Pacific Northwest and pushed temperatures 5 °C above anything in the local record, in a region where most homes have no air conditioning.

The consequences were immediate and severe. Roads buckled, power cables sagged and failed, and hundreds of people died across British Columbia, Washington and Oregon. In British Columbia alone, the coroner attributed about 619 deaths to the heat, most of them older people living alone in homes that became ovens. Two days after the record, a wildfire swept through Lytton and destroyed most of the village.

The heat dome was a case of statistical extremity becoming physical catastrophe. The temperature was not merely hot; it was far outside the range for which the buildings, infrastructure and behaviour of the region were designed. Air conditioning was rare, night-time temperatures stayed high, and emergency services were overwhelmed by calls from people who had no experience of such conditions.

This chapter examines disasters comparatively. It looks at heat, fire, flood and urban extreme events across regions and decades, and asks what they have in common: why some events kill thousands and others dozens, why similar hazards produce different outcomes, and what the differences reveal about exposure, vulnerability, governance and response. The cases are chosen not for their drama but for what they teach.

22.1 How to Read a Disaster Case

A disaster is not the same as a hazard. The hazard is the physical event — the storm, the heat, the flood — and the disaster is the harm it causes, which depends on who is exposed, how vulnerable they are, and how well institutions respond. The same hazard can produce a few dozen deaths in one place and hundreds of thousands in another, and the difference is almost always social and political rather than meteorological.

A useful case analysis separates five elements. Exposure describes who and what is in harm's way. Vulnerability describes the conditions that make harm more likely, such as poverty, age, disability, poor housing or lack of information. Capacity describes the resources available to prepare and respond. Governance describes the decisions that shape exposure and capacity. And timing describes how the event unfolded, because sequences matter: a heatwave followed by a wildfire is worse than either alone.

The cases in this chapter are selected to illustrate variation across those elements. Some show what happens when a rich country is unprepared for an unfamiliar hazard; others show what happens when a poor country is overwhelmed by a familiar one. Read together, they suggest that preparedness is not a function of wealth alone, and

that the most important determinants of survival are often the least visible.

22.2 Heat: Chicago 1995 and Europe 2003

The Chicago heatwave of July 1995 killed more than 700 people in five days. Temperatures reached 41 °C with high humidity, night-time temperatures stayed above 27 °C, and the city's emergency services were overwhelmed. Most of the dead were elderly, poor and socially isolated, and many were found in apartments with windows closed for security.

Eric Klinenberg's study of the event found that death rates varied sharply between neighbourhoods with similar demographics. In areas with dense social networks, active street life and functioning institutions, people checked on each other and mortality was lower; in areas with abandoned buildings, fear and isolation, people remained indoors and died alone. The finding established that heat mortality is a social phenomenon as much as a physical one.

Europe's 2003 heatwave produced a much larger toll across a whole continent: an estimated 70,000 deaths, including about 14,800 in France. The differences between countries were instructive. France had no heat plan, no register of vulnerable people and a health system on holiday; Italy and Spain, with warmer climates and more familiarity with heat, had lower excess mortality relative to the anomaly. The comparison suggests that experience and preparedness matter more than latitude.

22.3 Heat: The Pacific Northwest, 2021

The 2021 Pacific Northwest heat dome was unprecedented in the instrumental record. Lytton reached 49.6 °C, Portland 46.7 °C and Seattle 42.2 °C, and the anomaly — the difference from normal — was larger than in almost any other heat event ever recorded. An attribution study by World Weather Attribution concluded that the event would have been virtually impossible without human-caused climate change, and that it was made at least 150 times more likely.

The human toll was concentrated among people who had no experience of such heat and no means to escape it. In British Columbia, about 619 deaths were attributed to the event, most of them people over 70 living alone in homes without air conditioning. In Oregon, officials reported more than 100 deaths; in Washington, dozens. Many died indoors, which is characteristic of heat events in temperate cities where housing is designed to retain heat.

The event exposed a planning gap. Emergency plans in the region focused on winter storms, earthquakes and wildfires, not heat; cooling centres existed but were not widely known or accessible; and public messaging assumed that people could simply go to a mall or a friend's house. The lesson is that preparedness must be hazard-specific and must reach the people least able to act on generic advice.

22.4 Fire: Black Saturday and the Camp Fire

Australia's Black Saturday fires of 7 February 2009 killed 173 people and destroyed more than 2,000 homes in Victoria. The fires followed a heatwave in which Melbourne recorded three consecutive days above 43 °C, and they were driven by

winds that changed direction without warning. Many of the dead had stayed to defend their homes, following a policy that had encouraged residents to prepare to shelter or defend rather than leave early.

The Royal Commission that followed recommended a shift toward a policy of leaving early on days of extreme fire danger, and toward clearer warning systems. It also found that the “stay or go” policy had been misunderstood by many residents, who believed their homes would be defensible when conditions were beyond the capacity of any fire service. The case illustrates how a policy that is reasonable in moderate conditions can become lethal in extreme ones.

The Camp Fire in California in 2018 produced a different pattern. It moved so fast that evacuation was the only option, and the deaths were concentrated among older and disabled residents who could not leave quickly. The fire destroyed the town of Paradise in hours, and the evacuation routes were overwhelmed. Together, the two cases show that fire safety is not a single policy but a set of decisions that depend on the speed and intensity of the event.

22.5 Fire: Canada 2023 and the Smoke Problem

In 2023, Canada experienced its most severe wildfire season on record. About 18.5 million hectares burned — roughly seven times the ten-year average — and more than 200,000 people were evacuated. The fires burned from coast to coast, and smoke travelled thousands of kilometres, blanketing cities in Canada and the United States, including New York, where the sky turned orange in June.

The smoke was a health event in its own right. Fine particulate matter (PM2.5) from wildfire smoke penetrates deep into the lungs and bloodstream and is associated with respiratory and cardiovascular illness. Studies estimate that wildfire smoke exposure causes thousands of premature deaths a year in North America, and that the health burden is falling disproportionately on low-income and outdoor workers who cannot avoid exposure.

The Canadian case also illustrates the limits of suppression. Fire is a natural part of many forest ecosystems, and decades of fire suppression have increased fuel loads, making fires more intense when they occur. Climate change adds heat and drought that make forests more flammable. The result is a feedback in which suppression, land management and warming interact to produce larger and more destructive fires.

22.6 Flood: Pakistan 2022

Pakistan's 2022 floods were driven by monsoon rainfall about three times the national average, combined with glacial melt from the Himalayas and a heatwave earlier in the year that had dried the soil. By September, about a third of the country was under water, 33 million people were affected, at least 1,700 died and 2 million homes were damaged or destroyed.

The flood exposed structural vulnerabilities. Pakistan's population is concentrated along the Indus and its tributaries, where fertile land and water access have attracted settlement for millennia. Embankments and irrigation works built to control the river constrain its capacity to spread, and when they fail the water moves fast and

deep. Poverty, poor housing and limited insurance meant that households had few buffers.

The response was large but insufficient. National and provincial agencies, the military, UN agencies and local organisations mobilised, and aid reached millions of people. But the scale of the event overwhelmed capacity, and months later many displaced families remained without adequate shelter. The flood became a symbol of the gap between responsibility and harm, because Pakistan's emissions are a small fraction of the global total.

22.7 Flood: Germany and Belgium 2021

The July 2021 floods in western Germany and Belgium were caused by a slow-moving low-pressure system that produced extreme rainfall over saturated ground. In the Ahr valley, the river rose within hours, destroying houses, roads and bridges. More than 180 people died, and damage exceeded €30 billion.

The case is notable because it occurred in a wealthy region with strong institutions and modern forecasting. Warnings were issued, but they did not translate into action. Many residents did not believe the river could rise so fast; some local authorities delayed evacuation; and Germany lacked a nationwide cell broadcast system. The failure was in the last mile of the warning chain rather than in the forecast.

The comparison with Pakistan is instructive. Both events involved extreme rainfall and both produced large losses, but the mechanisms of harm differed: in Germany, the failure was in translation and response; in Pakistan, it was in structural exposure and capacity. The lesson is that investment in forecasting is necessary but not sufficient, and that the weakest link determines the outcome.

22.8 Urban Extremes: Chennai, Cape Town and Seoul

Urban floods reveal the interaction of infrastructure, planning and inequality. Chennai's 2015 floods, driven by record rainfall, were exacerbated by encroachment on wetlands and drainage channels, and by construction that reduced the city's capacity to absorb water. More than 400 people died, and damage was estimated at several billion dollars.

Cape Town's 2018 water crisis was a slow-onset urban emergency. After three years of drought, the city's reservoirs fell to critically low levels, and officials prepared for 'Day Zero', when taps would be turned off and residents would queue for water. The city avoided that outcome through severe restrictions, pressure management and public campaigns, but the crisis exposed the vulnerability of a growing city dependent on a variable water supply.

Seoul's 2022 floods were a reminder that wealthy cities are not immune. Record rainfall in August 2022 flooded basements and semi-basement apartments, killing at least 141 people, many of them poor residents living in the cheapest housing. The event highlighted the way housing markets push low-income households into the most dangerous spaces, and how infrastructure designed for a different climate can fail suddenly.

22.9 Compound and Cascading Events

Many of the most damaging disasters are compound: two or more hazards occur together or in sequence, and their effects interact. A heatwave followed by a wildfire, a drought followed by a flood, a storm followed by a disease outbreak — each combination is worse than the sum of its parts because response capacity is already stretched.

Cascading failures occur when the failure of one system triggers the failure of another. The Texas winter storm of 2021 is an example: the failure of the gas supply caused the failure of power generation, which caused the failure of water pumping, which caused a public health emergency. The Camp Fire showed a similar pattern: power infrastructure ignited the fire, the fire destroyed communications, and the loss of communications hampered evacuation.

Compound events are becoming more likely as the climate changes, because hazards that were once independent now co-occur more often. Planning for them requires a shift from single-hazard thinking to systemic risk assessment, and from sectoral responsibility to coordinated response. The cases in this chapter suggest that most failures are systemic, and that the institutions that manage them are usually organised by sector.

22.10 What the Cases Have in Common

Across very different contexts, the cases share several features. First, deaths are concentrated among people with the least capacity to act: the elderly, the poor, the disabled, the isolated and those in insecure housing. Second, warnings are necessary but not sufficient: the gap between warning and action is where many deaths occur. Third, infrastructure and land-use decisions determine exposure long before an event arrives.

Fourth, governance quality matters more than hazard severity in explaining outcomes. Countries with strong institutions, well-funded services and accountable decision-making consistently suffer fewer deaths from comparable hazards. Fifth, recovery is slower than response, and the long tail of displacement, debt and illness is where much of the human cost accumulates.

The implications are practical. Preparedness must be targeted at the people most at risk, not at the average resident. Warnings must be specific, repeated and delivered through trusted channels. Infrastructure must be designed for the hazards that are coming, not the ones that have passed. And recovery must be planned before the disaster, because the months after an event determine whether a community returns or dissolves.

22.11 Case Study: Lytton and the 2021 Heat Dome

In late June 2021, a ridge of high pressure settled over the Pacific Northwest and produced temperatures unprecedented in the instrumental record. Lytton, a village of about 250 people in the Fraser Canyon, recorded 49.6 °C on 29 June, the highest temperature ever measured in Canada. The previous national record had been 45 °C, set in 1937, and the margin was so large that some scientists initially questioned the

measurement.

The heat was not a normal summer event. It was a statistical outlier of a magnitude that made existing design assumptions obsolete: roads buckled, rail lines were restricted, power cables sagged, and hospitals reported a surge in heat-related admissions. In British Columbia, the coroner attributed about 619 deaths to the heat, most of them people over 70 living alone in homes without air conditioning.

Two days after the record, a wildfire swept through Lytton and destroyed most of the village. The fire was driven by wind and by the desiccation of vegetation during the heat, and it moved faster than residents could evacuate. The combination of heat and fire — a compound event — produced harm that neither hazard would have caused alone.

The response revealed gaps. Cooling centres were available but not widely known, and some were inaccessible to people without transport. Public messaging advised people to seek air conditioning, but many had none and could not travel. Emergency services were overwhelmed by calls, and the coroner's report recommended that heat be treated as a public health emergency with proactive outreach to vulnerable people.

The lessons are specific. Heat plans must be designed for the people most at risk, not for the average resident. Warnings must include concrete actions and locations. Buildings must be adapted for heat, not only for cold. And the possibility of compound events must be built into planning, because the sequence of heat and fire is becoming more common. Lytton shows that a hazard outside the historical range can overwhelm a wealthy, well-governed region when its systems were designed for a climate that no longer exists.

22.12 Future Scenarios and Predictions: Urban Extremes in 2050

The historical record shows that extreme events are becoming more frequent and more intense. Heatwaves that were once rare are now common; fire seasons are longer; rainfall is more concentrated; and coastal flooding is more frequent. The record also shows that impacts are concentrated in cities, where population, infrastructure and inequality intersect.

Present conditions set the baseline. More than half the world's population lives in urban areas, and the share is rising. Many cities are expanding into floodplains, steep slopes and coastal zones, and a large share of urban residents live in informal settlements with limited services. At the same time, urban governments often lack the finance, authority and data they need to manage risk.

The assumptions behind the projections are the pace of warming, the rate of urbanisation and the quality of urban governance. A world that invests in heat plans, drainage, early warning and social protection can reduce urban mortality even as hazards increase; a world that does not will see disasters concentrate in the poorest neighbourhoods. The table summarises the range.

Table 22.1 Comparative disaster cases, 2050

Event	Hazard	Deaths	What failed	Lesson
Chicago 1995	Heat	739	Social isolation	Networks protect
Europe 2003	Heat	~70,000	No heat plans	Preparedness cuts deaths
Black Saturday 2009	Fire	173	Stay-or-go policy	Leave early
Camp Fire 2018	Fire	85	Evacuation capacity	Reach the vulnerable
Lytton 2021	Heat and fire	619 (BC)	Design assumptions	Plan for extremes
Germany 2021	Flood	>180	Last-mile warning	Specific instructions
Pakistan 2022	Flood	1,700+	Exposure and capacity	Structural vulnerability
Seoul 2022	Flood	141	Housing inequality	Protect basement dwellers

Two conclusions follow. First, the same hazard produces very different outcomes depending on exposure, vulnerability and governance, which means that most disaster deaths are preventable in principle. Second, the cases that are most instructive are not the largest but the most diagnostic: they reveal which systems failed and why.

What would change these conclusions? A rapid expansion of heat action plans and cooling infrastructure would reduce deaths from the hazard that is growing fastest. Investment in drainage and floodplain management would reduce urban flood losses. And better data on informal settlements would make visible the people who are currently absent from official risk maps. The most important variable remains the pace of warming, because it determines how far beyond historical experience future extremes will go.

CHAPTER 23

CASE STUDIES: DROUGHT, FOOD, WATER AND SECURITY

Theme: slow-onset emergencies, entitlements and the politics of scarcity

In July 2011, the United Nations declared famine in two regions of southern Somalia. The declaration came after months of warnings from the Famine Early Warning Systems Network, which had projected the crisis in advance and published maps showing the areas at risk. By the time the famine was declared, tens of thousands of people had already died, and the eventual toll was estimated at about 258,000, half of them children under five.

The warning had worked; the response had not. Access to southern Somalia was constrained by conflict with al-Shabaab, which restricted humanitarian operations, and by the policies of donors and agencies that were reluctant to operate in areas controlled by the group. Food was available in some markets, but people could not afford it, and the livestock on which pastoral livelihoods depended had died in the drought.

The Somali famine is a case about entitlements rather than food. Amartya Sen's analysis of famines, first published in 1981, argued that famines occur when people lose the ability to command food — through employment, prices, assets or transfers — rather than when food is physically absent. The insight reframed famine as a problem of political economy, and it explains why famines occur amid plenty and why early warning alone does not prevent them.

This chapter examines slow-onset emergencies: drought, food crises, water scarcity and their intersections with conflict. It looks at the Sahel, Ethiopia, Somalia, Syria, Yemen, Cape Town, São Paulo and California, and at the global food price crises of 2007–08 and 2010–11. The cases show that slow emergencies kill more people than fast ones and receive less attention, because they lack a single dramatic moment and because their victims are usually poor and distant.

23.1 Reading Slow-Onset Cases

Slow-onset emergencies develop over months or years, and they are harder to see than floods or storms. A drought is not a single event but a cumulative deficit of rainfall that depletes soil moisture, groundwater, pasture and reservoirs; its effects appear gradually in crop yields, livestock deaths, food prices, migration and malnutrition. By the time a famine is declared, the process has been underway for a year or more.

The mortality is also slow and dispersed. People die not from a single cause but from the interaction of malnutrition, disease and exhaustion; deaths are recorded as malnutrition, measles, diarrhoea or respiratory infection rather than as drought. The result is that slow emergencies are systematically undercounted, and their scale is recognised only in retrospective surveys.

The cases in this chapter are therefore read with three questions. Who was exposed, and why were they unable to cope? What institutions failed, and at what point? And what would have prevented the outcome — early warning, cash transfers, water investment, conflict resolution or all of them? The answers are rarely technological,

because the knowledge usually exists before the crisis.

23.2 The Sahel Famines

The Sahel has experienced repeated droughts and food crises since the late 1960s. The drought of 1968–1974 desiccated pasture and crops across the region, and the famine that followed killed an estimated 100,000 people. A second major crisis in 1984–1985 affected millions more, and images of starvation in Ethiopia and Sudan shaped global consciousness of famine.

The causes were not simply rainfall. Colonial and post-colonial policies had pushed pastoralists onto marginal land, restricted their movement and undermined traditional coping mechanisms. Export crop production displaced food crops in some areas, and conflict in Chad, Ethiopia and Sudan disrupted markets and relief. The droughts were severe, but the famines were produced by the interaction of drought with poverty, displacement and weak institutions.

The Sahel also shows that adaptation is possible. Since the 1980s, farmer-managed natural regeneration — the protection and pruning of trees on farmland — has restored millions of hectares in Niger, improved soil fertility and increased yields. The practice was developed locally, spread through farmer networks, and supported by policy. It did not prevent all crises, but it reduced vulnerability in the areas where it was adopted.

23.3 Ethiopia 1984–1985 and the Entitlement Debate

The Ethiopian famine of 1984–1985 killed an estimated one million people. It followed two failed rainy seasons, but it was also produced by conflict, resettlement policies and the diversion of resources to war. Food was available in some markets, and the government's response was shaped by its military priorities and by its suspicion of international agencies.

Sen's entitlement approach explained the paradox. People starve when they cannot establish command over food, whether because their assets have collapsed, their employment has disappeared, prices have risen beyond their reach, or transfers have failed. In Ethiopia, pastoralists lost their herds, farmers lost their harvests, and displaced people lost access to both, while markets in other regions continued to function.

The famine also changed humanitarian practice. It led to the creation of the Famine Early Warning Systems Network, the expansion of food aid, and a debate about development rather than relief. The debate continues: agencies now emphasise cash transfers, social protection and resilience, but the underlying problem — that warnings do not automatically produce response — remains.

23.4 Somalia 2011 and 2022

The 2011 Somali famine was declared after months of warning, and it killed about 258,000 people. The 2022 crisis was different in important respects: early action by humanitarian agencies and local organisations, including cash transfers and livestock support, is thought to have prevented a declared famine, although the death toll was still high. A study published in 2023 estimated about 43,000 excess deaths in Somalia in

2022, half of them among children under five.

The comparison is instructive because the rainfall deficits were similar. In 2011, access was restricted, response was delayed and the crisis was allowed to develop; in 2022, agencies and local groups acted earlier, cash reached people faster and malnutrition treatment was scaled up. The difference in outcomes was largely a difference in the speed and flexibility of response.

The lesson is that famine is not a weather event. It is a political and economic outcome that can be prevented by timely action, and the knowledge required to act is usually available before the crisis peaks. What is often missing is the willingness to release funds early, the access to deliver them, and the political attention to sustain the response after the cameras leave.

23.5 Syria: Drought, Displacement and Conflict

Between 2006 and 2010, Syria experienced its worst multi-year drought in the instrumental record. Rainfall fell to a fraction of normal in the northeast, crops failed, and herds were decimated. An estimated 1.5 million people moved from rural areas to cities that were already struggling to absorb Iraqi refugees and to provide jobs and housing.

The drought did not cause the civil war that began in 2011, and analysts caution against simple causal claims. But it contributed to the conditions in which the war unfolded: rural livelihoods collapsed, food prices rose, and the government's failure to respond eroded legitimacy in affected regions. Studies, including work by Kelley et al. (2015) and Selby et al. (2017), debate the weight of the drought relative to political factors, but agree that it interacted with them.

The Syrian case illustrates the concept of a threat multiplier. Climate change does not determine conflict, but it can amplify existing stresses — unemployment, inequality, migration, weak institutions — and reduce the capacity of the state to manage them. The policy implication is not that climate causes war but that climate adaptation and conflict prevention must be designed together.

23.6 Yemen: Water, Conflict and Disease

Yemen is one of the most water-scarce countries in the world, and its aquifers are being depleted faster than they recharge. Sana'a, the capital, has been projected to run out of economically accessible water within decades, and rural areas depend on increasingly deep wells. Agriculture consumes most of the water, including irrigated qat, a cash crop with high water demand.

The conflict that began in 2014 destroyed much of the water infrastructure and disrupted supply chains. By 2017, a cholera outbreak produced more than one million suspected cases, the largest in modern history, driven by contaminated water and overwhelmed sanitation. The outbreak was not caused by drought, but it was made possible by the collapse of water and sanitation systems.

Yemen illustrates the interaction of water scarcity, conflict and disease. Each stressor reduces the capacity to manage the others: war damages infrastructure, scarcity raises prices, disease overwhelms health services, and the resulting instability makes

reconstruction harder. The case shows that water security is not only a matter of supply but of governance, infrastructure and peace.

23.7 Cape Town and São Paulo: Urban Water Crises

Cape Town's 2018 crisis brought a major city close to running out of water. After three years of drought, reservoir levels fell to about 20 per cent, and officials prepared for Day Zero, when taps would be shut off and residents would collect water at distribution points. The city avoided that outcome through severe restrictions, pressure management, leak repair and public campaigns, but the crisis exposed the vulnerability of a growing city dependent on a variable supply.

São Paulo faced a comparable crisis in 2014–2015. The Cantareira reservoir system, which supplies about half the metropolitan area's water, fell below 5 per cent of capacity, and officials considered rationing. The crisis was managed through restrictions, inter-basin transfers and rainfall, but it revealed the risks of relying on a single reservoir system for a metropolitan region of more than 20 million people.

The two cases show that urban water crises are usually produced by a combination of drought, demand growth, leakage and poor planning rather than by rainfall alone. Cities that invest in diverse sources, demand management, leak reduction and public communication can survive severe droughts; cities that do not can come close to collapse even in wealthy countries.

23.8 California: Groundwater and the Long Drought

California's 2012–2016 drought was one of the most severe in the state's recorded history. Reservoir levels fell, snowpack reached record lows, and farmers pumped groundwater to compensate. The result was land subsidence in the Central Valley, where the ground sank by as much as several metres in some areas, damaging canals and roads and permanently reducing aquifer capacity.

The drought exposed a governance failure. Groundwater had been largely unregulated, and pumping was treated as a private right. In 2014, the state passed the Sustainable Groundwater Management Act, which required local agencies to develop plans to bring aquifers into balance by the 2040s. The law was a significant reform, but it was passed after decades of depletion and will take decades to implement.

California also illustrates the interaction of drought and fire. The drought killed millions of trees, and the dead wood fuelled subsequent wildfires. The 2018 Camp Fire and the 2020 fire season were partly products of the drought years. The sequence shows that slow-onset and fast-onset hazards are connected, and that managing one requires attention to the other.

23.9 Food Prices, Export Bans and the 2007–2008 Crisis

In 2007 and 2008, world food prices rose sharply. Wheat prices more than doubled, rice prices tripled, and maize and soybeans reached record highs. The increases were driven by a combination of factors: biofuel demand, rising oil prices, poor harvests in several exporting countries, low stocks, and speculative investment in commodity futures.

Governments responded with export restrictions. India and Vietnam limited rice exports, Argentina taxed soybean exports, and Russia later banned wheat exports after its 2010 drought. The restrictions were rational for each country individually and catastrophic collectively: they removed supply from world markets, pushed prices higher for importers, and created panic buying. More than 30 countries experienced food riots.

The crisis showed how food security depends on trade as well as production. Countries that relied on imports were exposed to price shocks, and countries that restricted exports exported instability. The lesson for climate policy is that adaptation must include trade rules and regional cooperation, because a drought in one breadbasket can become a food crisis on another continent.

23.10 What the Cases Teach

Across the cases, three lessons recur. First, famines and water crises are usually failures of entitlement and governance rather than absolute scarcity. The knowledge required to prevent them exists, and the food or water is often available; what fails is access, affordability and political will.

Second, early warning is necessary but not sufficient. Somalia 2011 had warnings and still experienced famine; Ethiopia 1984–85 had warnings and a government that prioritised war. What changes outcomes is the translation of warning into early action, which requires pre-arranged finance, flexible programming and access.

Third, slow emergencies are cumulative and interconnected. Drought depletes groundwater, groundwater depletion reduces resilience to the next drought, and drought combined with conflict produces famine. Managing them requires long-term investment in water infrastructure, social protection and conflict resolution, not only emergency relief. The cases suggest that the most effective interventions are often the least dramatic: cash transfers, boreholes, pasture management and peace agreements.

23.11 Case Study: The 2011 Somali Famine

The 2011 famine in southern Somalia followed two consecutive failed rainy seasons and a drought that affected the wider Horn of Africa. Pastoralists lost their herds, farmers lost their harvests, and food prices rose sharply. By July, the United Nations declared famine in two regions, and the eventual death toll was estimated at about 258,000, half of them children under five.

The warning system had worked. The Famine Early Warning Systems Network had projected the crisis months in advance and published maps identifying the areas at risk. Humanitarian agencies had plans and resources. But access to southern Somalia was restricted by conflict with al-Shabaab, and agencies were reluctant to operate in areas controlled by the group. The result was a gap between knowledge and action.

The famine was also a failure of livelihoods. Pastoralists depend on livestock for income and food, and when herds died, households lost both. Farmers depended on rainfall, and when crops failed, they had no buffer. People migrated to camps, where overcrowding and poor sanitation increased disease. The mortality was produced by the

interaction of malnutrition, disease and displacement rather than by hunger alone.

The response, when it came, was large but late. Food aid, cash transfers, therapeutic feeding and health services reached millions, and the death rate fell. But the delay cost tens of thousands of lives, and the crisis displaced hundreds of thousands of people into neighbouring countries. The lessons were incorporated into subsequent planning, and the 2022 response was faster and more flexible.

The case illustrates the central argument of this chapter: slow emergencies are political. The rainfall was a trigger, but the famine was produced by conflict, restricted access, failed livelihoods and delayed response. Preventing the next one requires early action as well as early warning, and it requires addressing the conflict and poverty that make drought lethal.

23.12 Future Scenarios and Predictions: Food and Water Security in 2050

The historical record shows that famines have become less frequent but not extinct. The number of people affected by food crises has risen in recent years, driven by conflict, economic shocks and climate extremes. Water scarcity has also increased, with more than two billion people now living in countries under water stress.

Present conditions set the baseline. About 735 million people faced hunger in 2022, and nearly 3 billion could not afford a healthy diet. Irrigation depends on aquifers that are being depleted in many regions, and food systems are exposed to shocks through trade concentration and export restrictions. At the same time, early warning systems, social protection and cash transfers have improved.

The assumptions behind the projections are the pace of warming, the level of investment in water and social protection, and the intensity of conflict. A world that invests in irrigation efficiency, groundwater management, early action and peace can reduce hunger even as the climate changes; a world that does not will see more frequent food and water crises. The table summarises the range.

Table 23.1 Slow-onset cases, 2050

Case	Drivers	Deaths	What failed	Lesson
Sahel 1972–74	Drought; land policy	~100,000	Pastoral access	Protect mobility
Ethiopia 1984–85	Drought; conflict	~1,000,000	Entitlements	Markets and transfers
Somalia 2011	Drought; conflict	~258,000	Access; late response	Early action
Somalia 2022	Drought; prices	~43,000	Funding gaps	Cash at scale
Syria 2006–10	Drought; policy	Displacement	Rural neglect	Adapt before conflict
Yemen 2017	Water; conflict	Cholera	Sanitation collapse	Water as security
Cape Town 2018	Drought; demand	None	Supply diversity	Manage demand early
São Paulo 2014–15	Drought; reliance	None	Single source	Diversify supply

Two conclusions follow. First, food and water crises are preventable with existing knowledge and resources; what fails is political will, access and timing. Second, slow emergencies require long-term investment in water, social protection and peace, because relief alone cannot reverse the depletion of aquifers or the collapse of livelihoods.

What would change these conclusions? Pre-arranged finance for early action would reduce the gap between warning and response. Investment in groundwater recharge and irrigation efficiency would reduce exposure to drought. And conflict resolution would remove the factor that turns a bad harvest into a famine. The most important variable remains the pace of warming, because it determines how often the systems will be tested, and whether they will have time to recover between shocks.

CHAPTER 24

ACTION PLANS AND CHECKLISTS FOR SURVIVAL

Theme: preparedness cycles, priorities and the practice of readiness

On 12 November 1970, the Bhola cyclone struck the coast of East Pakistan, now Bangladesh. The storm surge reached several metres and swept over low-lying islands with little warning. The death toll was estimated at 300,000 to 500,000 people, one of the deadliest tropical cyclones in recorded history. There was no effective warning system, few shelters, and almost no organised evacuation.

Fifty years later, in May 2020, Cyclone Amphan made landfall in a region that had been transformed by preparedness. Bangladesh had built thousands of cyclone shelters, trained volunteers, established a warning system that reached villages by loudspeaker and mobile phone, and practised evacuation every year. More than 2.4 million people were moved to safety before the storm, and the death toll was in the dozens rather than the hundreds of thousands.

The comparison is not an argument that disasters have been solved. Amphan still destroyed homes and livelihoods, and COVID-19 made evacuation harder. But it is an argument that preparedness works, and that the difference between 1970 and 2020 was not the storm but the system: shelters, warnings, volunteers, drills and political commitment sustained over decades.

This final chapter turns the book's analysis into action. It sets out what households, neighbourhoods, cities, businesses and governments can do, in what order, and how to know whether it is working. It is organised by time horizon — the first hour, the first week, the first year — and by level of organisation, from the household to the state. The aim is not a list of everything that could be done but a sequence of what matters most.

24.1 The Preparedness Cycle

Preparedness is a cycle rather than a project: assess the risk, plan the response, act on the plan, check whether it worked, and adapt. The cycle is familiar from safety management and quality improvement, and it applies to households, firms and governments alike. The most common failure is not the absence of a plan but the absence of the check and adapt steps, which is why plans become obsolete as hazards and populations change.

Assessment begins with hazards and exposure: what can happen here, how often, and to whom. It continues with vulnerability and capacity: who will be affected most, what resources exist, and what gaps remain. Good assessments use local knowledge as well as technical data, because residents know which roads flood, which neighbours need help and where water collects.

Planning turns assessment into decisions: who does what, with what resources, and by when. Good plans are short, specific and exercised. A plan that has never been tested is a hypothesis, and the first test should not be the emergency itself. Exercises reveal gaps in communication, authority and logistics that no document can anticipate.

24.2 The First Hour, Day and Week

In the first hour, the priorities are life safety and information. Households should move to the safest place in the building, avoid windows and floodwater, and check on neighbours, especially those living alone. If evacuation is ordered, it should be carried out early and by the designated route, with documents, medication and water taken along. The most common errors are waiting too long, driving through water, and using fuel-burning equipment indoors.

In the first day, the priorities are water, food, shelter, medication and communication. Households should confirm that water is safe, ration it, and begin treating water if supplies are uncertain. They should check refrigeration times, discard food that has spoiled, and ensure that generators are outdoors. They should also register with assistance systems if they need help, because early registration often determines access later.

In the first week, the priorities are stabilisation and documentation. Households should photograph damage before repairs, contact insurers, keep receipts, and apply for assistance. They should watch for health risks — contaminated water, mould, heat, carbon monoxide — and for the mental health effects of loss. Communities should organise information sharing, because official channels are often slow and fragmented.

24.3 The First Month: Recovery and Rights

Recovery is where inequality is reproduced or reduced, and it is the phase that receives the least attention. In the first month, households must navigate insurance, government assistance, landlords, employers and lenders, often while displaced and traumatised. Those with documents, savings, legal knowledge and social networks recover faster; those without may never recover fully.

Practical steps matter. Keep a written log of contacts, claims and payments. Photograph and inventory losses. Read insurance policies for exclusions, especially flood, and appeal decisions that seem wrong. Apply to every programme for which the household may be eligible, and seek help from legal aid, community organisations or disaster case managers where available.

Renters are especially exposed. They may lose their homes without losing an asset, and they often have fewer rights and less access to assistance than owners. Local governments can protect them by enforcing habitability standards, extending eviction protections after disasters, and making assistance portable. Without such measures, recovery programmes can accelerate displacement and gentrification.

24.4 The First Year: Rebuilding Better

The year after a disaster is the window in which reconstruction decisions are made, and those decisions determine vulnerability for decades. Rebuilding in the same place, to the same standard, reproduces the same risk. Rebuilding with elevation, stronger connections, better drainage and cooler materials reduces it, often at modest additional cost.

Households and communities should insist on building back better, and governments should require it as a condition of assistance. That means codes that reflect current and projected hazards, inspections that are actually carried out, and technical assistance for households that cannot afford engineers. It also means addressing land tenure, because people without secure rights cannot invest in improvement.

The year is also the time to strengthen social infrastructure. Neighbourhood networks, volunteer groups and community organisations that formed during the response can be formalised and supported. Their knowledge of who needs help is an asset that no government database can easily replicate, and their existence reduces the cost of the next response.

24.5 City and Municipal Checklist

Cities are where most climate risk is concentrated and where most adaptation decisions are made. A city that wants to be prepared needs a heat action plan with cooling centres and outreach to vulnerable residents; a flood plan with drainage maintenance, land-use controls and evacuation routes; a warning system that reaches every neighbourhood; and a register of people who need assistance.

It also needs the boring infrastructure: maintained drains, functioning pumps, trees and shade, reliable water, backup power for clinics and shelters, and building codes that are enforced. These are not glamorous investments, and they rarely attract political credit, which is why they are chronically underfunded. A city that spends on events but not on drainage is unprepared regardless of what its plan says.

Finally, a city needs institutions that can coordinate. Heat involves health, social services, utilities and transport; floods involve water, planning, emergency services and infrastructure. Somebody must be accountable for the whole, and the coordination must be practised before the event. Cities that have appointed chief resilience officers and exercised joint command have performed better than those that have not.

24.6 National Policy Checklist

National governments set the framework within which everyone else acts. The essentials are known: building codes that reflect current and projected hazards; early warning systems that reach the last mile; social protection that can be scaled up quickly; health systems that can absorb surges; and finance that can be disbursed before a disaster rather than after an appeal.

Governments also control the incentives that shape private decisions. Insurance pricing, subsidies, procurement rules and disclosure requirements can either encourage or discourage risk reduction. A government that subsidises construction in floodplains and then pays for reconstruction is privatising the gain and socialising the loss, which is both inefficient and unjust.

The most important national decision is to treat adaptation as core public infrastructure rather than as a series of projects. That means budget lines, maintenance regimes, performance indicators and audit. It also means accepting that some places cannot be defended indefinitely, and planning for managed retreat where it is unavoidable — a decision that is politically painful and administratively complex, but

less costly than unmanaged collapse.

24.7 Business, Work and Continuity

Businesses are both victims and vectors of climate risk. A factory that floods loses production; a supplier that fails disrupts a global chain; a workplace without heat protection endangers workers. Continuity planning — identifying critical functions, backup sites, supplier alternatives and communication protocols — reduces losses and protects employment.

Workplace heat standards are a growing issue. Outdoor and indoor workers in hot environments need rest breaks, water, shade and adjusted schedules, and the absence of standards is a major cause of preventable illness. Some jurisdictions have adopted heat rules; many have not. Employers that treat heat as a productivity issue rather than a safety issue are both negligent and economically irrational, because heat reduces output as well as harming workers.

Supply chains require particular attention. Firms that depend on a single supplier, a single port or a single region are exposed to disruption, and the 2011 Thailand floods showed how a single event can affect production worldwide. Diversification, inventory buffers and supplier risk assessment are costly, but so is disruption, and the balance has shifted as extreme events have become more frequent.

24.8 Measuring Readiness

What is measured gets managed, and readiness is often measured badly. Plans are counted rather than tested; exercises are conducted for appearance rather than learning; and indicators focus on inputs — workshops held, documents produced — rather than outcomes. A useful readiness measure asks whether people know what to do, whether systems work under stress, and whether the most vulnerable are reached.

Practical indicators include the share of households with two weeks of water and food; the share of vulnerable residents registered for assistance; the time required to issue and deliver a warning; the number of shelter spaces relative to need; the proportion of critical facilities with backup power; and the time taken to restore water and power after an event. Each can be measured before a disaster and compared across places.

Exercises are the most informative test. A drill that reveals a broken communication link, an unclear authority or an inaccessible shelter is valuable, because it exposes problems while they can still be fixed. The organisations that learn fastest are those that treat exercises as experiments rather than performances, and that publish what they find.

24.9 Financing and Sequencing

Resources are limited, so sequence matters. The highest returns usually come from measures that are cheap, fast and protective across many hazards: early warning, heat plans, drainage maintenance, water storage, vaccination and social protection. These are not the most visible investments, which is why they are often skipped in favour of large infrastructure.

The evidence on returns is strong. Studies of early warning systems find benefit-cost ratios of roughly 4:1 to 10:1, and higher in some contexts; investments in resilient infrastructure are estimated to return about 4:1; and heat action plans have been credited with reducing mortality at low cost. The pattern is consistent: preparedness pays, but the benefits accrue later and to people who are not in the room when budgets are decided.

Financing mechanisms should match the timeline. Pre-arranged contingent finance for response, insurance and catastrophe bonds for large losses, budget lines for maintenance, and concessional finance for adaptation in low-income countries each serve a different purpose. The common failure is to rely on post-disaster appeals, which are slow, uncertain and inadequate relative to need.

24.10 What Survival Requires

The book's argument, condensed, is that survival is a function of preparation, cooperation and equity. Preparation reduces the chance that a hazard becomes a catastrophe; cooperation allows people to act together when institutions are overwhelmed; and equity ensures that protection reaches those who need it most. None of the three is sufficient alone, and all three can be built.

The evidence for that claim is in the cases. Bangladesh reduced cyclone deaths by orders of magnitude through shelters, warnings and volunteers. European countries reduced heat deaths after 2003 through plans and outreach. Cities that invested in drainage and warning suffered less than those that did not. The improvements were not produced by better weather; they were produced by better institutions.

The future will test those institutions more severely. Warming is committed, extremes are intensifying, and the window for avoiding the worst outcomes is narrowing. But the choice is not between optimism and despair; it is between preparation and neglect. The book ends with a simple claim: the societies that survive the coming decades will be those that treat foresight as a public good, build institutions that learn, and protect the people who cannot protect themselves.

24.11 Case Study: From Bhola to Amphan

The Bhola cyclone of November 1970 killed an estimated 300,000 to 500,000 people in the low-lying delta of East Pakistan. There was no warning system capable of reaching coastal villages, almost no cyclone shelters, and no organised evacuation. The storm surge arrived at night, and most victims drowned in their homes.

The disaster shaped the institutions that followed. After independence in 1971, Bangladesh invested in cyclone preparedness: a network of volunteers, a warning system linked to radio and loudspeakers, thousands of raised shelters, and annual drills. The Cyclone Preparedness Programme trained tens of thousands of volunteers, many of them women, to warn and evacuate their neighbours.

The results were measurable. Cyclones of comparable intensity in later decades produced far fewer deaths. In 1991, a cyclone killed about 138,000 people, still catastrophic but far fewer than 1970. By 2020, when Cyclone Amphan made landfall during the COVID-19 pandemic, more than 2.4 million people were evacuated, and the

death toll was in the dozens.

The comparison is not a story of complete success. Shelters are unevenly distributed, evacuation is harder for people with livestock or disabilities, and COVID-19 complicated crowding. But the reduction in mortality is one of the clearest examples of preparedness working at scale, and it was achieved by sustained public investment over five decades.

The lessons are transferable. Warning must reach the last mile, and it must be repeated through trusted local messengers. Shelters must be close enough to reach on foot, and accessible to people with disabilities. Evacuation must be practised, because people do not follow unfamiliar instructions under stress. And the system must be maintained between events, because preparedness decays when attention moves elsewhere.

24.12 Future Scenarios and Predictions: A Prepared World in 2050

The historical record shows that preparedness can reduce disaster mortality dramatically, even when hazards intensify. Bangladesh's cyclone deaths fell by orders of magnitude over five decades; heat deaths in Europe fell after 2003; and cities that invested in warning and drainage suffered less than those that did not. The record also shows that progress is uneven and reversible.

Present conditions set the baseline. Preparedness is uneven within and between countries: wealthier households and cities are better prepared, and the people most at risk are often the least protected. Finance for adaptation remains far below need, and maintenance is chronically underfunded. At the same time, the technologies of warning, communication and distributed energy have become cheaper and more capable.

The assumptions behind the projections are the pace of warming, the level of investment and the quality of governance. A world that invests in the measures described in this chapter can reduce deaths even as hazards increase; a world that does not will see the burden fall on those least able to bear it. The table summarises the range.

Table 24.1 Preparedness priorities by horizon, 2050

Horizon	Priority action	Who acts	Cost	Signal of success
First hour	Warn and shelter	Households; local government	Low	People reach safety
First week	Water; food; medicine	Households; agencies	Low to moderate	No preventable deaths
First year	Rebuild better	Owners; government	High	Stronger stock
Decade	Codes; drainage; shelters	Government; utilities	High	Lower losses
Generation	Land use; managed retreat	All levels	Very high	Reduced exposure
Continuous	Maintenance; drills; data	All actors	Moderate	Readiness scores

Two conclusions follow. First, the most important actions are not the most expensive: warnings, drills, heat plans, water storage and social protection deliver large benefits at modest cost. Second, preparedness must be sustained, because systems decay and because hazards change; a plan written once and never exercised is not preparedness.

What would change these conclusions? A major expansion of adaptation finance would allow low-income countries to invest at the scale required. Better data on vulnerable populations would make outreach more effective. And stronger local institutions would allow communities to act before help arrives. The most important variable remains the pace of emission reduction, because it determines how much adaptation will be needed and whether the systems described in this book will be sufficient.

APPENDIX A: LAMPIRAN (CHECKLISTS & PANDUAN)

Master Emergency Checklist

- Item 1: Verify supplies, documents, and communication plans.
- Item 2: Verify supplies, documents, and communication plans.
- Item 3: Verify supplies, documents, and communication plans.
- Item 4: Verify supplies, documents, and communication plans.
- Item 5: Verify supplies, documents, and communication plans.
- Item 6: Verify supplies, documents, and communication plans.
- Item 7: Verify supplies, documents, and communication plans.
- Item 8: Verify supplies, documents, and communication plans.
- Item 9: Verify supplies, documents, and communication plans.
- Item 10: Verify supplies, documents, and communication plans.

Panduan Penulisan dan Debat Akademik

Buku ini dirancang untuk digunakan dalam diskusi akademik. Saat memperdebatkan atau memperluas klaimnya:

- Selalu sitasi sumber primer (paper peer-reviewed, laporan IPCC).
- Bedakan antara observasi, atribusi, dan proyeksi.
- Akui ketidakpastian dan tingkat kepercayaan.
- Pertimbangkan berbagai jalur sosioekonomi (SSP).
- Sertakan dimensi keadilan dan kesetaraan.
- Perbarui klaim seiring munculnya bukti baru.

APPENDIX B: GLOSARIUM

- Adaptation: Adjusting to actual or expected climate impacts.
- Mitigation: Reducing greenhouse gas emissions.
- Resilience: Capacity to absorb disturbance and reorganize.
- Vulnerability: Susceptibility to harm.
- Exposure: Presence of people or assets in hazard-prone areas.
- Sensitivity: Degree to which a system is affected by climate.
- Adaptive capacity: Ability to adjust to climate change.
- Tipping point: Threshold beyond which change becomes self-sustaining.
- Greenhouse effect: Trapping of heat by atmospheric gases.
- Carbon footprint: Total greenhouse gas emissions caused by an entity.
- SSP: Shared Socioeconomic Pathway (scenario framework).
- RCP: Representative Concentration Pathway.
- Committed warming: Warming locked in by past emissions.

APPENDIX C: INDEKS

- Adaptation
- Biodiversity
- Carbon footprint
- Carbon pricing
- Climate finance
- Committed warming
- Conflict
- Drought
- Early warning
- Ecosystem
- Equity
- Exposure
- Flood
- Food security
- Greenhouse effect
- Health
- Heat wave
- Indigenous knowledge
- Infrastructure
- Justice
- Mental health
- Migration
- Mitigation
- Mutual aid
- RCP
- Renewable energy
- Resilience
- SSP
- Sea level rise
- Sensitivity
- Social capital
- Tipping point
- Vulnerability
- Water security

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APPENDIX E: ABOUT THE AUTHOR

Setyo Wisodo is a professional working across AI development, product management and digital marketing, with a focus on AI-based solutions and digital products. Based in Indonesia.

APPENDIX F: EXTENDED ANALYTICAL NOTES

Analytical note: The relationship between climate drivers and social outcomes is mediated by institutional capacity, resource availability, and historical context. No single factor determines the result.

Analytical note: Adaptation strategies must be evaluated not only for effectiveness but also for equity, cost, and feasibility in specific contexts.

Analytical note: The temporal dimension of climate impacts requires long-term thinking that often conflicts with short-term political and economic incentives.

Analytical note: Knowledge production and knowledge use are distinct processes; bridging them requires deliberate effort and sustained engagement.

Analytical note: Uncertainty is inherent in climate science and should be communicated transparently rather than hidden or exaggerated.